PEARSON EDEXCEL INTERNATIONAL A LEVEL BIOLOGY · WBI12

Unit 2: Cells, Development, Biodiversity and Conservation

Revision notes, spec point by spec point (Topics 3 and 4)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

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Topic 3 – Cell Structure, Reproduction and Development

SPEC 3.1 All living organisms are made of cells

WHAT THIS SPEC POINT ASKS

Know that every living organism is made of one or more cells, and that all cells share some basic features.

3.1 Cell theory

- Every living organism is made of **one or more cells**. Unicellular organisms (bacteria, yeast, *Amoeba*) are a single cell; multicellular organisms are made of millions.
- The cell is the **smallest unit** that can carry out all the processes of life (respiration, growth, reproduction, response…).
- New cells only come from **existing cells**, by cell division.

3.1 Features every cell has

Feature	Why every cell needs it
Cell surface membrane	separates the cell from its surroundings and controls what enters and leaves
Cytoplasm	where most metabolic reactions happen
DNA	the genetic material: codes for the cell's proteins and is copied when the cell divides
Ribosomes	the site of protein synthesis (translation)

3.1 Two kinds of cell

	Prokaryotic (bacteria and archaea)	Eukaryotic (animals, plants, fungi, protoctists)
Nucleus	**none**: DNA lies free in the cytoplasm	DNA inside a **nucleus** with a double membrane
DNA	**circular**, not wound round histone proteins; often **plasmids** too	**linear** chromosomes wound round **histones**
Membrane-bound organelles	**none**	yes: mitochondria, ER, Golgi, lysosomes (and chloroplasts in plants)
Ribosomes	smaller, **70S**	larger, **80S** in the cytoplasm
Typical size	about 1–5 μm	about 10–100 μm
Cell wall	always: **peptidoglycan (murein)**	plants: cellulose; fungi: chitin; animals: none

- **Ultrastructure** means the detail of a cell that you can only see with an **electron microscope** (3.3, 3.6).
- The two main groups also show in classification: prokaryotes make up the domains **Bacteria** and **Archaea**; every eukaryote is in the domain **Eukarya** (more in 4.14).

MARK-SCHEME LANGUAGE

- Eukaryote clue: "this organism has a **nucleus**" / "prokaryotic cells do not contain a nucleus". Cell membrane and cytoplasm don't count: every cell has them.
- Domain: "**Eukarya**" (eukaryote / eukaryota accepted). "Prokarya" and "Plantae" are not domains.
- Tick-grids: cell surface membrane, cytoplasm and ribosomes are in **animal, plant and prokaryotic** cells. Cell wall: **plant and prokaryotic** only. Nucleolus: **not** in prokaryotes.

Often asked as: tick-grids of structures in animal, plant and prokaryotic cells; name the domain of an organism; give one reason an organism is not a prokaryote.

COMMON MISTAKE

"Bacteria have a nucleus" or "bacterial DNA is linear". Bacterial DNA is circular and lies free in the cytoplasm.
Naming "plants" or "Archaea" as the domain of a single-celled organism with chloroplasts. A chloroplast is a membrane-bound organelle, so the organism is in the **Eukarya**.

BEYOND THE SPEC, BUT EXAMINED

Circular DNA in eukaryotic cells: mitochondria and chloroplasts carry their own small **circular DNA** and **70S ribosomes**. So "circular DNA" and "ribosomes" are ticked for animal, plant **and** prokaryotic cells (WBI12 Jan 2024). Few candidates knew this.

SPEC 3.2

Cells, tissues, organs and organ systems

WHAT THIS SPEC POINT ASKS

Understand how the cells of a multicellular organism are grouped into tissues, tissues into organs, and organs into organ systems.

3.2 The levels

1. **Specialised cell**: a cell whose structure suits one job (e.g. a red blood cell carrying O_2). Cells become specialised by **differential gene expression** (3.18).
2. **Tissue**: a group of **similar cells** (the same cell type) working together to carry out a particular function, sometimes with material between the cells.
3. **Organ**: a group of **different tissues** working together to carry out one or more functions.
4. **Organ system**: a group of **organs** working together to carry out a major function.

3.2 Examples at each level

Level	Animal example	Plant example
Specialised cell	red blood cell, sperm, ciliated epithelial cell	palisade mesophyll cell, root hair cell, guard cell
Tissue	squamous epithelium, cardiac muscle, blood (a connective tissue), nervous tissue	**xylem, phloem**, palisade mesophyll, spongy mesophyll, sclerenchyma, meristem
Organ	heart (cardiac muscle + connective tissue + epithelium + nervous tissue), small intestine, lung	leaf, stem, root, flower
Organ system	circulatory, digestive, gas exchange, nervous	shoot system, root system

The four main animal tissue types are **epithelial** (lines surfaces), **connective** (blood, bone, cartilage), **muscle** and **nervous**.

MARK-SCHEME LANGUAGE

- Tissue: "group of (similar) cells working together to perform a (specific) function". Cells with a similar structure, function or origin are accepted.
- Organ: "(group of) tissues which work together to perform {a function / functions}". "Same tissues" is not accepted.
- Organ system: "group of organs that work together to perform {one / one or more / specific} functions".
- Organ v tissue: "an organ has many functions **whereas** a tissue has one (or fewer)"; "an organ has several tissues / cell types **whereas** a tissue has one cell type / similar cells". Two separate definitions are not pieced together.
- Plant tissue: xylem, phloem, meristem, sclerenchyma, parenchyma, palisade or spongy mesophyll. "Palisade mesophyll **cell**" is rejected: that's a cell.

Often asked as: one-mark definitions of tissue, organ or organ system; two differences between an organ and a tissue; name a plant tissue.

COMMON MISTAKE

"A tissue is a group of **specialised** cells." Too vague: the mark needs **similar cells / the same type of cell**.
Writing "tissues" for an organ system, or giving the tissue definition for an organ.
Naming a cell, a molecule or an organ when a tissue is asked for.

SPEC 3.3 Eukaryotic cell ultrastructure and organelle functions

WHAT THIS SPEC POINT ASKS

(i) Know the ultrastructure of eukaryotic cells: nucleus, nucleolus, ribosomes, rough and smooth endoplasmic reticulum, mitochondria, centrioles, lysosomes and Golgi apparatus. (ii) Understand what each of these does.

3.3 Organelle → structure → function

Organelle	Structure	Function
Nucleus	largest organelle (about 10–20 μm). **Nuclear envelope** = double membrane with **nuclear pores**. Contains **chromatin** (DNA wound round histones) in the nucleoplasm	holds the genetic material; site of **DNA replication** and **transcription**. Pores let mRNA and ribosome subunits out, and nucleotides and proteins in
Nucleolus	dense, dark region inside the nucleus; **no membrane**; made of rRNA, protein and the DNA that codes for rRNA	makes **rRNA** and assembles **ribosome subunits**
Ribosomes	tiny (about 25 nm), **no membrane**; a large and a small subunit made of rRNA and protein; **80S** in the cytoplasm. Free in the cytoplasm or attached to rER	site of **protein synthesis** (translation). Free ribosomes make proteins used inside the cell; rER ribosomes make proteins for **secretion**, membranes or lysosomes
Rough ER (rER)	flattened membrane sacs (**cisternae**) continuous with the nuclear envelope; outer surface **studded with ribosomes**	proteins made on its ribosomes enter the space inside, **fold** into their 3D shape and are **transported** in vesicles to the Golgi (3.4)
Smooth ER (sER)	membrane tubes and sacs with **no ribosomes**	makes and processes **lipids** (phospholipids, cholesterol) and **steroid hormones**
Golgi apparatus	a stack of flattened, **curved** membrane sacs (cisternae), **not joined** to each other, with **vesicles** budding off the edges; no ribosomes	**modifies** proteins (e.g. adds carbohydrate to make glycoproteins), then **packages** them into vesicles: secretory vesicles for **exocytosis**, or **lysosomes**
Mitochondrion	**double membrane**; inner membrane folded into **cristae** (large surface area); fluid **matrix** with enzymes, its own circular DNA and 70S ribosomes; about 1–10 μm long	site of **aerobic respiration**, which produces **ATP**. Many in very active cells (muscle, sperm midpiece, secretory cells)
Lysosome	small vesicle with a **single membrane**, containing **hydrolytic (digestive) enzymes**; no internal structure; made by the Golgi	**digests** worn-out organelles, pathogens taken in by phagocytes, or whole dead cells. The membrane keeps the enzymes away from the rest of the cell
Centrioles	a pair of short hollow **cylinders** at right angles, each made of **nine triplets of microtubules**; near the nucleus; in animal cells (not most plant cells)	organise microtubules to form the **spindle fibres** in mitosis and meiosis, which **separate chromosomes/chromatids**

The **cell surface membrane** (a phospholipid bilayer with proteins, from Unit 1) surrounds all of this.

Animal v plant cells: plant cells also have a **cellulose cell wall**, **chloroplasts**, a **large permanent vacuole** (with its tonoplast), amyloplasts and plasmodesmata, and usually no centrioles. That is Topic 4 (4.1).

EXAM TIP

"Function" questions want a **verb plus a product**: "aerobic respiration to make ATP", "makes ribosomes", "forms spindle fibres", "contains hydrolytic enzymes that digest pathogens". A description of the structure scores nothing when function is asked.
Drawing a nucleus: **two lines** for the envelope with **gaps** for pores, a dark **nucleolus**, patches of **chromatin**. Draw only the nucleus, not the whole cell.

MARK-SCHEME LANGUAGE

- Mitochondrion: "**aerobic respiration**", "produces **ATP**" (Krebs cycle, oxidative phosphorylation accepted). "Produces energy" is **not** accepted.
- Nucleolus: "produces **ribosomal subunits**" (ribosomes / rRNA accepted). "Contains genetic material" is ignored.
- Nucleus: "contains the genetic material / DNA", "DNA replication", "transcription". Plain "controls the cell" is ignored.
- Centriole: "forms **spindle fibres** in mitosis / meiosis" (forms microtubules accepted) "to separate chromosomes / chromatids". Centromere for centriole is rejected.
- Lysosome: "contains **digestive / hydrolytic** enzymes" or "digests pathogens / organelles". Plain "enzymes" is ignored.
- Drawing the rER: at least two tubules with **ribosomes on the surface**; cisternae and ribosomes labelled. Drawing a nucleus: double membrane, pores as gaps, nucleolus, chromatin. One wrong label caps the label marks.
- Two structures inside the nucleus: "nucleolus" and a named genetic material (DNA, chromatin, chromosomes). The envelope, pores and nucleoplasm aren't credited.
- Prophase events: chromosomes condense, nuclear envelope breaks down, centrioles move to the poles and produce spindle fibres.

Often asked as: name an organelle from a description or EM and give its function; draw and label a nucleus, rER or Golgi; compare a lysosome with an acrosome; suggest why mitochondria sit next to lipid droplets.

COMMON MISTAKE

"Mitochondria produce energy." Energy can't be produced: say "**aerobic respiration**, which produces/releases **ATP**".
Confusing the **nucleolus** with the nucleus, the **centriole** with the centromere, or rER with sER (look for ribosomes). Saying the Golgi "makes proteins": it **modifies and packages** them.
Label lines that stop in the cytoplasm instead of touching the structure.
In a compare question, writing a paragraph about each organelle. Use paired sentences: "both have…; a lysosome… **whereas** an acrosome…".

BEYOND THE SPEC, BUT EXAMINED

- Chloroplasts and mitochondria **both** have a double membrane and both contain ribosomes and circular DNA (WBI12 Jan 2026).
- **Lysosome v acrosome**: both are single-membrane vesicles of hydrolytic enzymes made by the Golgi; a lysosome digests **inside** the cell, the acrosome digests the zona pellucida **outside** it (WBI12 Jun 2023).
- Mitochondria next to **lipid droplets** in an egg: a **short diffusion distance**, so fatty acids reach the mitochondria quickly for respiration to make ATP (WBI12 Jan 2021).
- The number of organelles roughly **doubles in interphase** before a cell divides, so each daughter cell gets a full set (WBI12 Jun 2021).

SPEC 3.4 The rER and Golgi in protein transport

WHAT THIS SPEC POINT ASKS

Understand how the rough ER and the Golgi apparatus move proteins through a cell, including how extracellular enzymes are made and released.

3.4 The secretory pathway, step by step

Follow one protein, e.g. **amylase** made by a pancreas cell:

1. **Nucleus**: the gene is **transcribed** into mRNA, which leaves through a **nuclear pore**.
2. **Ribosomes on the rER**: the mRNA is **translated**; the growing polypeptide passes into the **rER lumen**.
3. **rER**: the polypeptide **folds** into its secondary and tertiary structure (hydrogen, ionic and disulfide bonds form). It is **packaged into a transport vesicle** that buds off the rER.
4. The vesicle moves to the **Golgi** and **fuses** with it, so the protein enters the Golgi.
5. **Golgi**: the protein is **modified** (for example, carbohydrate is added to make a **glycoprotein**), then **sorted and packaged** into **secretory vesicles** that bud off.
6. Secretory vesicles move to the **cell surface membrane**, **fuse** with it and release the enzyme outside the cell by **exocytosis**.

Mitochondria supply the **ATP** for moving vesicles and for exocytosis.

3.4 Extracellular enzymes

- An **extracellular enzyme** works **outside the cell** that made it: digestive enzymes (amylase, trypsin, lipase), enzymes that fungi release to digest wood, the enzymes a Venus flytrap or pitcher plant secretes onto its prey.
- Cells that secrete a lot of protein (pancreas, salivary glands, plasma cells) are packed with **rER, Golgi** and mitochondria.
- The Golgi sends proteins to other places too: **lysosomes** (enzymes kept inside the cell), and proteins built into the **cell surface membrane**.

EVIDENCE: FOLLOWING LABELLED AMINO ACIDS

Pancreas cells are given radioactive (or fluorescent) amino acids for a short time, then samples are taken over time.
The label appears first in the **rER**, then the **Golgi**, then **secretory vesicles**, and finally **outside the cell**.
Explain it: the amino acids are built into polypeptides on the rER ribosomes; the protein moves in vesicles to the Golgi, is modified and packaged, and leaves by exocytosis. Inside the cell the label falls as outside it rises.

EXAM TIP

Use the protein's **name** from the question ("insulin", "the enzyme", "acrosin") at every step. Marks for the link are lost if you write just "protein". If the question gives you the polypeptide, **don't start with transcription**: start where the question starts.

MARK-SCHEME LANGUAGE

- Route (5 marks, insulin): "rER packages insulin in a vesicle"; "vesicle **fuses** with the Golgi apparatus" and insulin is modified; "Golgi packages insulin in (secretory) vesicles"; "vesicles fuse with the cell surface membrane"; "exits by **exocytosis**".
- rER alone: "translation at rER ribosomes / polypeptide enters the rER"; "forms secondary / tertiary / 3D structure" with named bonds; "packaged into vesicles to go to the Golgi".
- Golgi alone: "**modification** of the protein (into the enzyme)"; "packaged into secretory vesicles"; "leaves the cell by exocytosis".
- Glycoprotein: "**carbohydrate added** (in the Golgi)" (glycosylation accepted). "Modified in the Golgi" with no carbohydrate is ignored.
- Drawing a Golgi: at least two **curved cisternae**, no ribosomes or connections, a vesicle, cisternae labelled.
- Tracer data: label "decreases inside the cell / increases outside"; "vesicles fuse with the cell surface membrane and release…". Amino acids themselves leaving is ignored.

Often asked as: describe how an enzyme or hormone is made and secreted (3–5 marks, often in a plant, fungus or sperm context); the role of the rER or Golgi alone; explain tracer data; draw a Golgi.

COMMON MISTAKE

Giving the Golgi's role when the rER was asked, or the reverse. Saying **amino acids** are exocytosed: the finished protein is.
Writing "moves to the Golgi". The mark is for the vesicle **fusing** with the Golgi, and this is the step most often missed.
Writing "proteins" when the question is about **enzymes**; mixing up transcription and translation; writing lipids for proteins.

BEYOND THE SPEC, BUT EXAMINED

- **Glycoproteins** are made in the Golgi by adding carbohydrate chains to a protein, e.g. the Rhesus antigen (WBI12 Oct 2022) and mucus.
- The rER vesicle **fuses** with the Golgi (at its "cis" face, where vesicles arrive); secretory vesicles bud from the opposite "trans" face (WBI12 Jan 2020).
- Enzymes in the egg's **cortical granules** are made by the same rER → Golgi route (WBI12 Oct 2024).
- Role of one organelle in making acrosin: nucleus (transcription), ribosome (translation), rER (folding, packaging), Golgi (modification, packaging) or mitochondria (ATP). The **acrosome** itself is not credited: it only stores the enzyme (WBI12 Jun 2022).

SPEC 3.5

Prokaryotic cells

WHAT THIS SPEC POINT ASKS

(i) Know the ultrastructure of prokaryotic cells: cell wall, capsule, plasmid, flagellum, pili, ribosomes and circular DNA. (ii) Understand what each structure does.

3.5 Structure → function

Structure	What it is	Function
Cell wall	rigid layer of **peptidoglycan (murein)**: polysaccharide chains cross-linked by short peptides	stops the cell **bursting** when water enters by osmosis; keeps its shape
Capsule (slime layer)	layer of polysaccharide **outside** the cell wall	protects against **drying out**, against being engulfed by **phagocytes**, and against antibodies; helps bacteria **stick** to surfaces and to each other
Cell surface membrane	phospholipid bilayer	controls what enters and leaves; some respiration enzymes are on it
Circular DNA	one long **circular** DNA molecule, **not bound to histones**, free in the cytoplasm (the nucleoid region)	carries the genes for almost all the cell's proteins
Plasmids	small extra rings of DNA that copy themselves independently	carry **extra genes**, e.g. for **antibiotic resistance**; can be passed between bacteria
Ribosomes	**70S**: smaller than eukaryotic 80S ribosomes	protein synthesis (translation)
Flagellum	long protein filament attached at the cell membrane, **rotated** by a motor at its base	**movement** (motility) through liquid
Pili	short, thin protein threads on the surface	**attachment** to surfaces and host cells; a **sex pilus** joins two bacteria for **conjugation** (plasmid transfer)

- **Always present**: cell wall, cell surface membrane, cytoplasm, circular DNA, 70S ribosomes.
- **Only some bacteria**: capsule, plasmids, flagella, pili.
- **Never present**: nucleus, nucleolus, mitochondria, ER, Golgi, chloroplasts or any other membrane-bound organelle.

MARK-SCHEME LANGUAGE

- Capsule: "prevents **dehydration** of the cell"; protection from **phagocytes / antibodies / antibiotics** accepted; helps adhesion. Plain "protection" is too vague.
- Pili: "allow bacteria to **adhere** to surfaces"; conjugation / exchange of plasmids accepted. "Sexual reproduction" is ignored.
- Flagellum: "for (cell) **motility** / movement".
- Plasmid: "contains **genes** to make proteins", e.g. antibiotic resistance. Say what it **does**, not just what it is.
- Ribosomes: "**70S** whereas eukaryotic ribosomes are **80S**", or "smaller than eukaryotic ones". You need **both halves** of the comparison.
- Structures that make an enzyme in a bacterium: "chromosome / circular DNA **and** (70S) ribosomes". Both are needed. The nucleus, rER or Golgi are wrong.
- Wall molecule: plant "cellulose"; prokaryote "**peptidoglycan / murein**".
- Bacteria using O_2: "for aerobic respiration / to produce ATP". "Produce energy" is rejected.

Often asked as: state the function of a capsule, pilus, flagellum or plasmid; draw and label two of them on an outline; how prokaryotic ribosomes differ; compare a fungal cell with a bacterium.

COMMON MISTAKE

Drawing the capsule or pili **inside** the cell wall, or a flagellum that doesn't start at the membrane.
Confusing a **plasmid** with **plasmodesmata** (channels between plant cells).
Writing **chitin** for the bacterial wall (chitin is fungal). In a compare question, wrong statements cancel right ones, so leave out anything you're unsure of.

BEYOND THE SPEC, BUT EXAMINED

- **70S v 80S**: "S" is a measure of how fast the particle sinks in a centrifuge, so bigger means larger. Prokaryotes have 70S ribosomes; eukaryotes have 80S in the cytoplasm but 70S inside mitochondria and chloroplasts (WBI12 Jan 2020, Oct 2022).
- **Fungus v bacterium**: a fungal cell has a nucleus, linear DNA, membrane-bound organelles, 80S ribosomes and a **chitin** wall (WBI12 Jan 2025).
- A bacterium living on dry skin doesn't need a flagellum: it isn't in a liquid, so it can't swim anyway (WBI12 Oct 2024).

SPEC 3.6

Recognising organelles in electron micrographs

WHAT THIS SPEC POINT ASKS

Be able to identify the organelles listed in 3.3 in electron microscope (EM) images.

3.6 What to look for

Organelle	How it looks in a TEM	Don't confuse it with
Nucleus	the largest structure; a **double** envelope (two close lines, broken by **pores**); grainy, dark patches of chromatin	a vacuole (single membrane, no contents)
Nucleolus	the darkest, densest round area **inside** the nucleus; no membrane round it	the nucleus itself
Mitochondrion	oval, sausage-shaped or round profiles about 1 μm across; **double** membrane; inner membrane folded into shelves (**cristae**); grainy matrix	a chloroplast (bigger, with stacks of membranes called grana)
rER	parallel flattened sacs whose outer surfaces are **dotted with ribosomes**; often near the nucleus	the Golgi (no dots)
sER	tubes and sacs with **no dots**	rER
Golgi apparatus	a stack of **curved**, flattened sacs, not joined, with small round **vesicles** at the edges; no ribosomes	rER
Ribosomes	tiny dark dots, free in the cytoplasm or on the rER	—
Lysosome	small, round, **dark** body with a **single** membrane and no internal structure	a secretory vesicle (usually paler)
Centrioles	two short cylinders at right angles: one cut lengthwise looks like a striped rectangle; one cut across looks like a **ring of nine** tiny groups	—

3.6 Why the same organelle can look different

- An electron micrograph shows a **thin slice**. A 3D organelle cut **across** looks round; cut **lengthwise** it looks long. Two photos of the same organelle can differ because of the **angle (plane) of the cut**.
- EM images are black and white. Colour in a published EM is **false colour**, added afterwards.
- Fine internal detail (cristae, ribosomes, membranes) means the image came from an **electron microscope**. A 3D surface view means a **scanning** EM (3.7).

MARK-SCHEME LANGUAGE

- Different appearance: "viewed from a different angle / one is a transverse and one a longitudinal section"; "depends on where the cell was cut".
- Which microscope: "**electron** microscope" because of the "**high resolution**", so (ultra)structure can be seen. A 3D image of a pollen grain's surface: **scanning** EM (TEM rejected). "Image is clear" is ignored.
- Chloroplast function: "site of photosynthesis / light absorption". "Produces ATP" alone and "stores starch" are ignored.

Often asked as: name or label structures on an EM; suggest why two images of one organelle differ; state which type of microscope was used and why.

COMMON MISTAKE

Saying two images of the same organelle differ in **magnification, resolution or staining** when they come from the same photo: it's the **plane of the section**.
Choosing a light microscope because the image is 2D or black and white.
Failing to recognise centrioles (often called mitochondria or lysosomes) or a chloroplast.

SPEC 3.7

Magnification, resolution and staining

WHAT THIS SPEC POINT ASKS

(i) Know how light and electron microscopes achieve magnification and resolution. (ii) Understand why specimens are stained.

3.7 Two different ideas

- **Magnification**: how many times bigger the image is than the real object.
- **Resolution**: the ability to **distinguish between two points that are close together** as separate points. The smaller the distance, the higher the resolution and the more detail you see.
- Magnifying beyond the resolution just makes a **bigger blur**. Detail comes from resolution, not magnification.

$$\text{magnification} = \text{image size} \div \text{actual size}$$

Rearranged: **actual size = image size ÷ magnification**; image size = actual size × magnification. Both sizes must be in the **same units**.

Convert	Multiply by
mm → µm	**1000**
µm → nm	1000
cm → µm	**10 000**

3.7 Light v electron microscopes

	Light microscope	Transmission EM (TEM)	Scanning EM (SEM)
Uses	light, focused by glass lenses	a beam of **electrons** passing **through** the specimen, focused by electromagnets	electrons **scanned across the surface**; scattered electrons form the image
Resolution	about **200 nm**	about **0.5 nm** (best)	about 3–10 nm
Useful magnification	up to about ×1500	over ×500 000	lower than TEM (about ×100 000)
Image	colour, 2D	black and white, 2D, **internal** detail	black and white, **3D surface**
Specimen	living or dead	dead, very **thin** section, in a **vacuum**	dead, in a vacuum; needn't be thin

- **Why electrons resolve more**: resolution is limited by **wavelength**. Light has a wavelength of about 400–700 nm, so points closer than about 200 nm blur together. Electrons have a **much shorter wavelength**.
- Drawbacks of EM: specimens must be dead (vacuum), preparation is complex, and preparation can create **artefacts** (structures that weren't there in the living cell).

3.7 Why stain?

- Most cell structures are **colourless and transparent**, so there is little **contrast** between them.
- A stain binds to particular molecules or structures and colours them, so they stand out from their surroundings. Using two stains that bind different things (**differential staining**) separates structures.
- Staining improves contrast, **not** resolution or magnification.

Stain	What it shows
Methylene blue	nuclei (DNA) blue, e.g. cheek cells (CP5)
Acetic orcein / **toluidine blue**	**chromosomes**, so stages of mitosis can be identified (CP6)
Iodine solution	starch blue-black
Phloroglucinol + acid, or safranin	lignin red (xylem, sclerenchyma; safranin also stains nucleic acids)
Nile red	lipids
Eosin	cytoplasm pink

For electron microscopes, specimens are "stained" with **heavy metal** salts, which absorb electrons and show up dark.

WORKED EXAMPLE: MAGNIFICATION FROM A SCALE BAR

A 2 µm scale bar on a micrograph measures **18 mm** with a ruler.
Convert to the same unit: 18 mm × 1000 = 18 000 µm.
Magnification = 18 000 ÷ 2 = **×9000** (write no unit).
A mitochondrion in the same image is 27 mm long. Actual length = 27 000 µm ÷ 9000 = **3 µm** (= 3 × 10^{-3} mm = 3000 nm).

WORKED EXAMPLE: ACTUAL SIZE

A cell in a drawing is **45 mm** across; the magnification is ×1500.
Actual size = 45 000 µm ÷ 1500 = **30 µm**.

MARK-SCHEME LANGUAGE

- Magnification: measure the **scale bar** (not the specimen), convert mm to µm (×1000), divide by the scale bar's value. **No unit**. Usually ±1 mm tolerance; follow the s.f. or d.p. asked.
- Resolution: "the ability to distinguish between two points which are close together".
- Magnification: "how many times bigger an object appears in an image than in real life". The equation alone is ignored.
- Why a structure can't be seen with a light microscope: "the **resolution** (or magnification) of a light microscope is not high enough" **and** "the structure is very small". For the membrane bilayer: "the two layers are very close together".
- Why an EM image shows more: "more structures can be seen as **separate** due to the **higher resolution**". Plain "high resolution" is ignored.
- Why stain: "to increase **contrast**", "to make the **chromosomes / chromatids** visible" (to identify stages of mitosis). "To make cells / organelles visible" is ignored.

Often asked as: calculate a magnification from a scale bar (almost every paper); calculate an actual size; explain why a structure can't be seen with a light microscope; define resolution; why stain.

COMMON MISTAKE

Converting cm to µm by ×1000 (it's ×10 000). Inverting the formula. Putting µm after a magnification. Giving 2 d.p. when 2 s.f. was asked.
Saying an EM shows more because it has **higher magnification**. The reason is **resolution** (shorter wavelength).
Stating the equation without rearranging it when actual size is asked.

BEYOND THE SPEC, BUT EXAMINED

- **Named stains**: acetic orcein (WBI12 Jan 2020); safranin (lignin, nucleic acids) v nile red (lipids) (WBI12 Oct 2025).
- **Sphere volume** $V = \frac{4}{3}\pi r^3$ (formula given): use the **radius**, half the diameter (WBI12 Jun 2025). **SA:V** of a cube: $SA = 6l^2$, $V = l^3$, give it as **x : 1** (WBI12 Oct 2021).
- Unit conversions inside a rate: 15 mm at 0.2 µm s^{-1} = 15 000 ÷ 0.2 = 75 000 s (WBI12 Jan 2024).

SPEC 3.8

Core Practical 5: light microscopy, drawings and graticules

WHAT THIS SPEC POINT ASKS

(i) Use a light microscope to observe suitable animal cells and make labelled drawings. (ii) Use an eyepiece graticule to measure cells, and understand scale.

3.8 The two scales

- The **eyepiece graticule** is a scale of 100 equal divisions inside the eyepiece. It has **no units**: its divisions (eyepiece units, epu) cover a different real length at each objective.
- The **stage micrometer** is a slide with a scale of known length (often 1 mm divided into 100 divisions, so each is **10 µm**; check the slide).
- Calibrating means finding out how many µm **one eyepiece division** is worth at the objective you're using.

CORE PRACTICAL 5: METHOD IN BRIEF

Calibrate

1. Put the stage micrometer on the stage and focus on it with the low-power objective.
2. Turn the eyepiece until its graticule lies alongside the micrometer scale, with the zeros lined up.
3. Find a point further along where the two scales line up again. Note the number of eyepiece divisions and the length on the micrometer.
4. 1 epu = length on micrometer ÷ number of eyepiece divisions.
5. Repeat for **every objective** (each has its own value).

Observe and measure cheek cells

6. Gently scrape the inside of your cheek with a clean cotton bud and smear it on a slide.
7. Add a drop or two of **methylene blue** (stains the nuclei), lower a coverslip at an angle with a mounted needle to avoid air bubbles, and blot off excess stain.
8. Focus with the **low-power** objective first (coarse focus, moving the lens away from the slide), then switch to high power and use the **fine focus** only.
9. Measure several cells in eyepiece units and convert with the calibration value. Calculate a mean.
10. Draw a few cells, label them, and add a title, the magnification and a scale bar.

WORKED EXAMPLE: CALIBRATING AND MEASURING

At ×40 objective, **10 micrometer divisions (100 µm)** line up with **40 eyepiece divisions**.
1 epu = 100 µm ÷ 40 = **2.5 µm**.
A cheek cell measures 24 epu across → 24 × 2.5 = **60 µm**.
At the ×10 objective each eyepiece division covers 4 times more: 1 epu = **10 µm**. That's why you recalibrate when you change objective.

WORKED EXAMPLE: DRAWING A SCALE BAR

Your drawing is at ×800. You want a scale bar for **10 µm**.
Length of bar = 10 µm × 800 = 8000 µm = **8 mm**. Draw an 8 mm line and label it "10 µm".

3.8 Drawing rules

- Sharp HB pencil; **clear, continuous lines**; **no shading** or colouring.
- Draw **only what you see**, large, with proportions correct.
- **Ruled** label lines, no arrowheads, each **touching** the structure, not crossing each other.
- Title, magnification and/or a scale bar. A cell drawing shows individual cells; a **plan diagram** shows only tissue outlines (CP7 in Topic 4).

3.8 Safety, errors and improvements

- Cheek swabs: each used once by one person, then straight into **disinfectant**. Methylene blue stains skin and is an irritant: gloves and eye protection.
- Never point the mirror at direct sunlight.
- Errors: forgetting to recalibrate at a new objective; cells overlapping or folded; judging cell edges. Improve by measuring **many cells** and using a **mean**.

EXAM TIP

CP5 has not been examined as a whole practical in WBI12 (2019–2026). Its skills come up constantly through magnification and scale-bar calculations (3.7). Learn the calibration as a four-line method.

COMMON MISTAKE

Quoting a graticule reading in µm without calibrating. Calling the stage micrometer "the graticule". Shading drawings or using arrows on label lines.

SPEC 3.9 Gene loci and linkage

WHAT THIS SPEC POINT ASKS

(i) Know that a locus is the position of a gene on a chromosome. (ii) Understand linkage of genes on the same chromosome.

3.9 Key words

- **Gene**: a section of DNA that codes for a polypeptide.
- **Allele**: an **alternative form of a gene** (a different base sequence at the same locus).
- **Locus**: the **position of a gene** on a chromosome. Each gene has a fixed locus.
- **Homologous chromosomes**: a pair (one from each parent) with the **same genes at the same loci**, though they may carry different alleles.

3.9 Linkage

- Genes whose loci are on the **same chromosome** are **linked**: they tend to be **inherited together**, because they move into the gametes as one unit in meiosis. They don't assort independently.
- Linked alleles can be separated only by **crossing over** between **non-sister chromatids** of homologous chromosomes in **prophase I** (3.10). Gametes with the new combinations are **recombinants**.
- Crossing over is rare at any one point, so recombinants are always **fewer** than the parental types.
- The **closer** two loci are, the **less likely** a crossover falls between them, so the more tightly they are linked.
- Sex linkage (Unit 1) is a special case of linkage: genes on the X chromosome are inherited together with sex.

WORKED EXAMPLE: GAMETES FROM LINKED GENES

A parent is AaBb. One chromosome carries **A** and **B**; its homologue carries **a** and **b**.
If the genes were on **different** chromosomes: four gamete types, **25% each**.
Linked, with crossing over between them in 10% of gametes:
- **AB 45%** and **ab 45%** (parental types: linkage kept)
- **Ab 5%** and **aB 5%** (recombinants: made by crossing over)

The more the results depart from 25% each, the closer the loci.

MARK-SCHEME LANGUAGE

- Locus: "{location / position} of a {gene / allele} on a chromosome" (chromatid accepted).
- Allele: "an alternative form of a gene".
- Linkage: "the genes are on the **same chromosome**" **and** "have a higher chance of being **inherited together** / not separated by crossing over". Stopping at "same chromosome" loses the second mark.
- Closer loci: "alleles are closer together, **therefore** crossing over between them is less likely".
- Gamete data (not 25% each): "(due to) linkage"; "A and B on the same chromosome"; "AB and ab inherited together"; "Ab and aB formed by **crossing over**" (recombinants).

Often asked as: define locus or allele; explain why two alleles are inherited together more often than others; explain gamete percentages that aren't 25%; decide which genes are linked from chromosome data.

COMMON MISTAKE

Writing about dominance, mutation or independent assortment when linkage is asked.
Crossing over between **different** chromosomes (e.g. chromosome 1 and chromosome 4). It happens only between **homologous** chromosomes.
Confusing locus with chiasma, centromere or centrosome. Using "gene" when you mean "allele".

BEYOND THE SPEC, BUT EXAMINED

- Working out which recombinant combinations a single crossover can make across **three loci** (WBI12 Jun 2021).
- **Linked and unlinked genes together**: two genes on chromosome 1 are linked and inherited together; a third on chromosome 4 assorts independently of both (WBI12 Jun 2022).

SPEC 3.10

Meiosis and genetic variation

WHAT THIS SPEC POINT ASKS

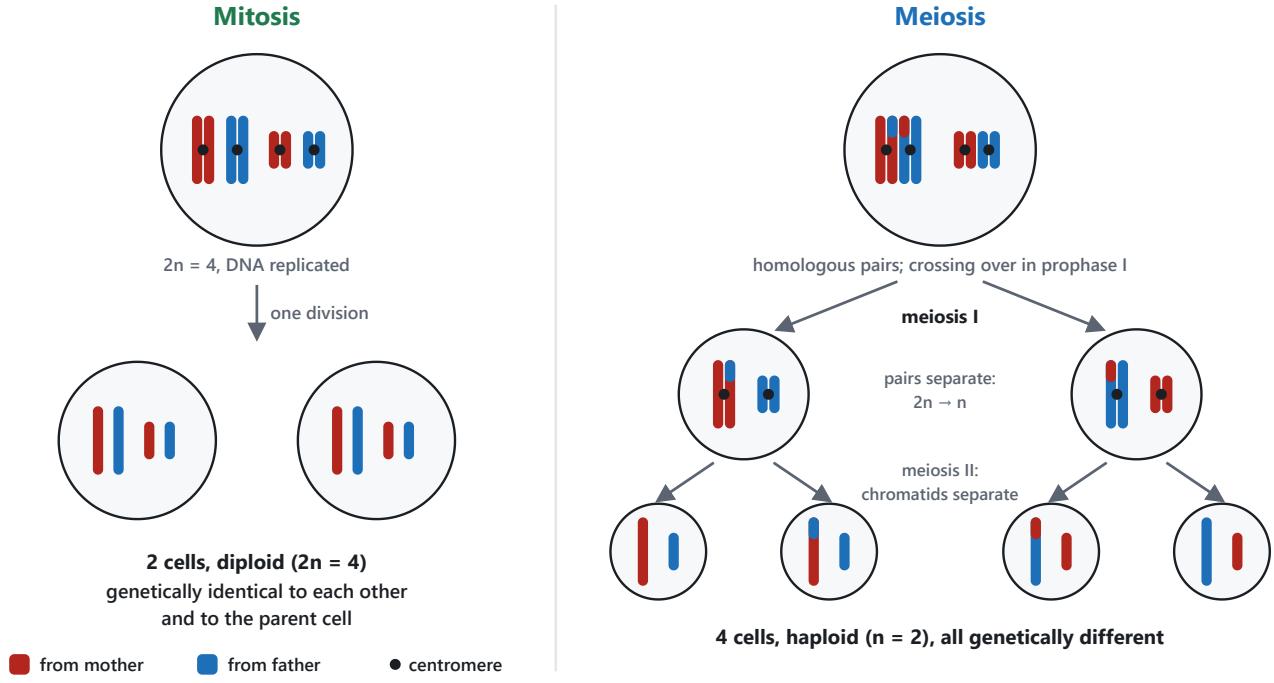
Understand how meiosis makes genetically different gametes, through independent assortment of chromosomes at metaphase I and crossing over between chromatids in prophase I. (You don't need the names of the stages of prophase.)

3.10 Why meiosis

- Meiosis is a **reduction division**: one **diploid** (2n) cell divides **twice** to make **four haploid** (n) cells, each **genetically different**.
- Halving the number means fertilisation (fusing two haploid nuclei) **restores** the diploid number every generation.
- Before meiosis, DNA is replicated (S phase), so each chromosome is two identical **sister chromatids** joined at a **centromere**.

3.10 The two divisions

Stage	Meiosis I: homologous pairs separate	Meiosis II: sister chromatids separate
Prophase	chromosomes condense; homologous chromosomes **pair up** (bivalents); **crossing over** at chiasmata; nuclear envelope breaks down; spindle forms	chromosomes condense again; new spindles form
Metaphase	**pairs** line up on the equator, each pair's orientation random (**independent assortment**)	single chromosomes line up on the equator
Anaphase	**homologous chromosomes** pulled to opposite poles; chromatids stay joined	**centromeres divide**; sister chromatids pulled apart
Telophase	two **haploid** nuclei (each chromosome still two chromatids)	**four haploid** nuclei



A cell with 2n = 4. Mitosis: one division, two identical diploid cells. Meiosis: crossing over (swapped chromatid ends) and independent assortment (which long and short chromosome go together), then two divisions give four haploid cells that all differ.

3.10 Source 1: crossing over (prophase I)

1. Homologous chromosomes pair up closely.
2. **Non-sister chromatids** (one maternal, one paternal) wrap round each other, break at the same points and **rejoin** to the other chromatid. The crossing points are **chiasmata**.
3. Sections, and the **alleles** on them, are **exchanged**. The chromatids now carry **new combinations of alleles** (recombinant chromatids).

3.10 Source 2: independent assortment (metaphase I)

1. Homologous pairs line up on the equator. Whether the maternal or paternal chromosome of each pair faces a given pole is **random**, and each pair lines up independently of the others.
2. So each gamete gets a **different mixture of maternal and paternal chromosomes**, and so a different combination of alleles.
3. Number of combinations = $\mathbf{2^n}$, where n is the **haploid** number.

WORKED EXAMPLE: 2^n

A fruit fly has 2n = 8, so n = 4: 2^4 = **16** combinations of chromosomes in its gametes.
Humans: n = 23, so 2^{23} = **8 388 608** combinations, before crossing over is even counted.
Random fertilisation: any sperm can fuse with any egg, so one couple could make $2^{23} \times 2^{23} \approx \mathbf{7 \times 10^{13}}$ different chromosome combinations in a child.

3.10 Meiosis v mitosis

	Mitosis	Meiosis
Number of divisions	one	**two**
Cells made	**two**	**four**
Chromosome number	stays the same (2n → 2n)	**halved** (2n → n)
Genetic make-up	daughters **identical** to each other and to the parent	daughters **all different**
Homologous chromosomes pair?	no	**yes**, in prophase I
Crossing over	no	**yes**, prophase I
Metaphase	chromosomes line up singly	**pairs** line up (metaphase I); singles (metaphase II)
Where	body cells: growth, repair, asexual reproduction	making **gametes** (testes, ovaries; anthers, ovules)

EXAM TIP

Read the question's wording. "Offspring" means add **random fertilisation**. "Separation of chromosomes" means **independent assortment only**: crossing over earns nothing there. Always say "prophase **I**" and "metaphase **I**".

MARK-SCHEME LANGUAGE

- "Crossing over results in different **allele** combinations"; "independent assortment results in different combinations of chromosomes / alleles in the gametes". "Gene" for allele is rejected.
- Crossing over: "sections of chromatids **exchanged** at **chiasmata**" "between **non-sister** (maternal and paternal) chromatids". Sister chromatids is rejected.
- Offspring variation: independent assortment, "a mixture of maternal and paternal chromosomes / different combination of alleles", and "**random fertilisation**" (which gametes fuse is random).
- Egg v body cell: "23 chromosomes / haploid"; "so fertilisation forms a **diploid** zygote"; different "due to crossing over / independent assortment".
- DNA graph for meiosis: same shape as mitosis to start, then "a **further division**" so DNA falls in **two steps** (4 → 2 → 1 a.u.).
- Metaphase of mitosis v meiosis: similarities "chromosomes line up on the equator", "centromeres attach to spindle fibres"; differences "two metaphases in meiosis", "**homologous pairs** line up in metaphase I but single chromosomes in mitosis", "independent assortment in metaphase I". Full marks need a similarity.

Often asked as: explain how meiosis produces variation; complete a diagram after crossing over; draw the gametes from a given cell; sketch or explain a DNA-content graph; compare metaphase in mitosis and meiosis.

COMMON MISTAKE

Writing "fusion of egg and sperm **cells**". It is the **nuclei** that fuse.
Drawing a mitosis graph for meiosis, or dropping straight from 4 to 1 a.u.
Forgetting random fertilisation when the question says "offspring".
Ignoring the drawing or chromosome number in the question (e.g. "26 chromosomes": half come from each parent).

BEYOND THE SPEC, BUT EXAMINED

- **Non-disjunction**: a pair of chromosomes (meiosis I) or chromatids (meiosis II) fails to separate, so one gamete gets an **extra** chromosome and another gets **none**. In Patau syndrome the zygote has three copies of chromosome 13 (WBI12 Oct 2024).
- **Hybrid sterility**: chromosomes from two different species can't **pair up** in prophase I, so meiosis fails and no gametes form. "An odd number of chromosomes" isn't credited (WBI12 Jun 2022).
- An error in crossing over **inside a gene** can create a new allele, or duplicate a gene (WBI12 Jan 2023, Oct 2025).

SPEC 3.11

Mammalian gametes

WHAT THIS SPEC POINT ASKS

Understand how sperm and egg cells are specialised for their functions, including the acrosome in sperm and the zona pellucida in the egg.

3.11 Sperm: built to swim and to get in

Feature	How it helps
Haploid nucleus (in the head)	fuses with the haploid egg nucleus to make a **diploid** zygote
Acrosome: a vesicle of **hydrolytic enzymes** (e.g. **acrosin**) over the front of the head, made by the Golgi	the enzymes **digest a path through the zona pellucida**
Receptors / proteins on the head membrane	**bind** to the zona pellucida
Midpiece packed with **mitochondria**	**aerobic respiration** releases **ATP** for the flagellum
Flagellum (tail) of microtubules	beats to make the sperm **swim** to the egg
Small, streamlined, very little cytoplasm	less **resistance**, so it swims fast; huge numbers can be made

3.11 Egg (secondary oocyte): built to feed the embryo and let in only one sperm

Feature	How it helps
Haploid nucleus	fuses with a sperm nucleus to restore the diploid number
Very large, with lots of **cytoplasm**: **lipid droplets**, protein, mitochondria, ribosomes	nutrients and ATP for the early embryo until it implants
Zona pellucida: a thick layer of **glycoprotein** just **outside** the cell surface membrane	sperm **bind** to receptors in it (which sets off the acrosome reaction); after fertilisation it **hardens** to keep other sperm out
Cortical granules: vesicles of enzymes just under the membrane	released in the **cortical reaction** (3.12) to harden the zona
Follicle cells (outside the zona)	protect the egg

- A human egg is about 0.1 mm across, visible to the naked eye; a sperm head is only about 5 µm long.
- Many sperm, few eggs: most sperm never reach the egg, and they **compete** to fertilise it.

MARK-SCHEME LANGUAGE

- Mitochondria: "**aerobic respiration**", "provide **ATP**", "to move the flagellum". "Produce / make energy" is not accepted.
- Flagellum: "rotates / moves the sperm"; streamlined shape "**reduces resistance**".
- Longer flagellum or faster sperm: "swim faster", "**increased chance of fertilising** the egg", "competition with sperm from other males". "First to reach the egg" alone is ignored.
- Sperm head: "**haploid** nucleus, so a diploid zygote forms"; "acrosome contains **enzymes** to **digest** the zona pellucida".
- Egg: "haploid"; "**lipid** droplets as a source of energy / ATP"; "cortical granules / zona pellucida, which prevent polyspermy"; "more cytoplasm / named organelles". "Food store", "make energy" and "jelly coat" are ignored.
- Sperm drawing: acrosome, nucleus, **midpiece with mitochondria**, flagellum.

Often asked as: explain the role of the mitochondria or flagellum; draw and label a sperm; suggest why a species has longer sperm; explain how the egg is adapted; why the egg is so much larger than the sperm.

COMMON MISTAKE

"Produce energy." Use "release / provide ATP".
Saying "reach the egg" when the mark needs "**fertilise** the egg". Offering a haploid nucleus as an adaptation for **swimming**.
Placing the zona pellucida **inside** the cell membrane. Describing a feature without saying how it helps.

BEYOND THE SPEC, BUT EXAMINED

- Sperm contain DNA in both the **nucleus** and the **mitochondria**; the acrosome's enzymes are coded by **nuclear** genes (WBI12 Jun 2019, Oct 2019).
- **Acrosin** is the enzyme in the acrosome (WBI12 Jan 2024).
- Eggs release chemicals that **attract** sperm, and have surface glycoproteins that bind them (WBI12 Oct 2021).
- Lipid droplets in large fish eggs supply energy for the developing embryo (WBI12 Jan 2023).

SPEC 3.12

Fertilisation in mammals

WHAT THIS SPEC POINT ASKS

Know the process of fertilisation in mammals: the acrosome reaction, the cortical reaction and the fusion of nuclei.

3.12 The sequence

1. Sperm are deposited near the cervix and swim through the **uterus** into the **oviduct**, where the egg is. Chemicals released by the egg attract them.
2. A sperm passes between the follicle cells and reaches the **zona pellucida**.
3. **Acrosome reaction**: the sperm head binds to receptor glycoproteins on the zona. The **acrosome** fuses with the sperm's membrane and releases its **hydrolytic enzymes** (acrosin). The enzymes **digest a path** through the zona pellucida, and the sperm swims through.
4. The sperm's **cell membrane fuses** with the egg's cell surface membrane, and the **sperm nucleus enters** the egg.
5. **Cortical reaction**: sperm entry triggers a rise in Ca^{2+} in the egg. **Cortical granules** move to the egg membrane, **fuse** with it and release their **enzymes** (exocytosis) into the zona. The zona pellucida **hardens** and its sperm receptors are altered, so **no other sperm can bind or enter**. This prevents **polyspermy** and keeps the zygote **diploid**.
6. The egg (a secondary oocyte) now **completes meiosis II**.
7. The **haploid sperm nucleus and haploid egg nucleus fuse**, forming a **diploid zygote** (2n = 46 in humans). This fusion of nuclei is fertilisation. The zygote then divides by **mitosis** (3.17).

3.12 Don't mix up the two reactions

	Acrosome reaction	Cortical reaction
Whose?	the **sperm**	the **egg**
Trigger	sperm binding to the zona pellucida	a sperm entering the egg
Vesicles involved	the acrosome	cortical granules
Effect on the zona	**digests** a path through it	**hardens** it
Purpose	lets a sperm reach the egg membrane	stops **more** sperm entering (blocks polyspermy)

MARK-SCHEME LANGUAGE

- Cortical reaction (3 marks): "cortical granules **fuse** with the (egg cell surface) membrane and **release enzymes**"; "zona pellucida **hardens** / thickens"; "to prevent **polyspermy** / more than one sperm entering" (so the zygote stays diploid). "Chemicals" is ignored.
- Hardening also "modifies the sperm receptors, preventing other sperm binding".
- Acrosome reaction: "acrosome **releases enzymes / acrosin**"; "enzymes **digest** the zona pellucida"; "sperm membrane **fuses** with the egg membrane (so the nucleus enters)". "Break down / penetrate" without digest is not credited.
- After entry: "**fusion of two haploid nuclei** to form a diploid nucleus".
- An egg with three nuclei: one egg nucleus + **two sperm nuclei** (polyspermy, e.g. because the zona was damaged). "Three sperm entered" is rejected.
- Low acrosin activity: the zona is digested more slowly, so fewer eggs are fertilised.

Often asked as: explain the cortical reaction; describe the events after a sperm enters; explain an egg with extra nuclei; interpret data on acrosin, ZP3 or polyspermy; evaluate an IVF technique (6-mark).

COMMON MISTAKE

Mixing up the acrosome reaction and the cortical reaction.
Leaving out the first step of the cortical reaction: the granules **fuse with the membrane and release enzymes**. "Toughened outer layer" and "fertilisation membrane" aren't precise enough.
Describing sperm entry when the question only asks what happens **after** it.
Mixing up a mammal's egg with a plant ovule (there's no triploid endosperm in mammals).

BEYOND THE SPEC, BUT EXAMINED

- **ZP3** is the zona glycoprotein that sperm bind to. If ZP3 is missing or changed in shape, the sperm can't bind and the acrosome reaction isn't triggered (WBI12 Jun 2025).
- A **fast electrical block**: the egg's membrane potential changes within seconds of sperm entry, before the zona hardens (sea urchin data, WBI12 Jan 2026).
- **ICSI** (injecting one sperm directly into the egg) bypasses the zona, so it helps sperm with little acrosin or poor swimming; used in IVF of endangered species (WBI12 Jan 2024).
- Protein fibres (Rab27a) carry cortical granules to the membrane; without them the cortical reaction fails (WBI12 Oct 2025).

SPEC 3.13

Fertilisation in flowering plants

WHAT THIS SPEC POINT ASKS

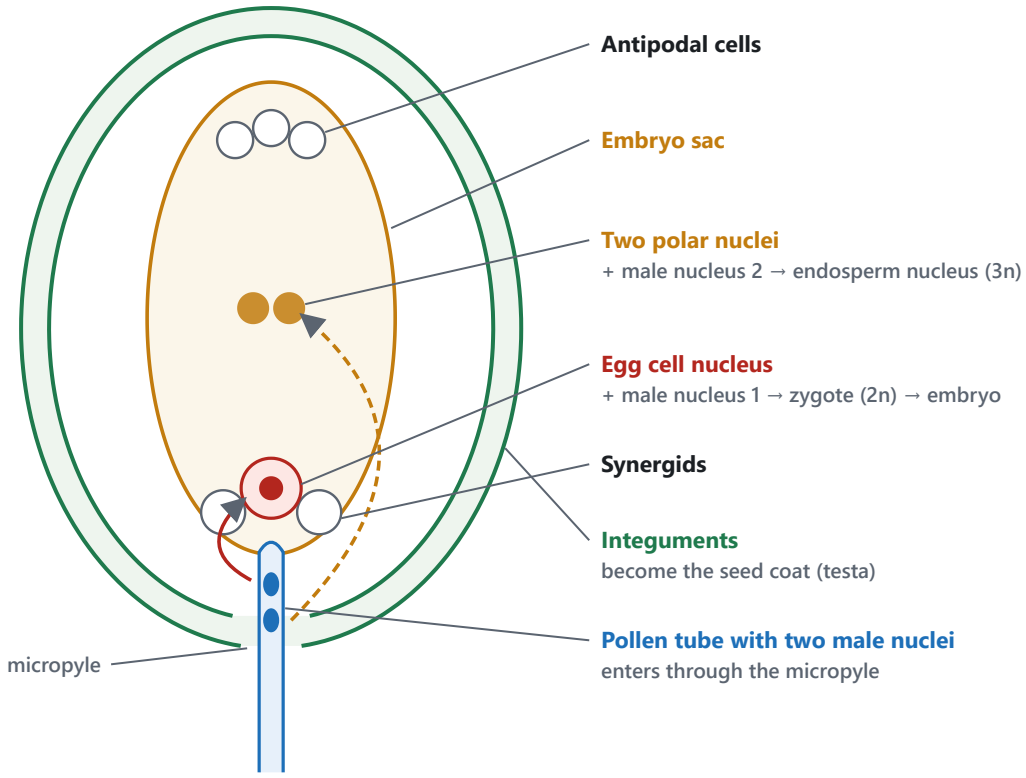
Know how fertilisation happens in flowering plants, from the growth of the pollen tube to the fusion of nuclei.

3.13 Pollination comes first, then fertilisation

- **Pollination**: transfer of pollen from an **anther** to a **stigma**. It isn't fertilisation, in the same way that mating isn't fertilisation in a mammal.
- A pollen grain is **haploid** and contains **two nuclei**: a **tube nucleus** and a **generative nucleus**. There are no sperm cells.
- The **ovule** (inside the ovary) contains the **embryo sac**: a haploid **egg cell** near the **micropyle**, **two polar nuclei** in the centre, plus synergids and antipodal cells.

3.13 The steps

1. The pollen grain lands on a compatible stigma, absorbs water and sugar, and **germinates**: a **pollen tube** grows out of it.
2. The tube grows **down through the style** towards the ovary. It secretes **hydrolytic (digestive) enzymes** that digest a path through the style tissue, and absorbs the products for growth.
3. The **tube nucleus** sits at the growing tip and **controls the tube's growth** (it holds the genes for the digestive enzymes). The tube is guided by chemicals from the ovule (chemotropism).
4. The **generative nucleus** divides by **mitosis** to form **two male nuclei** (haploid). They travel down the tube behind the tube nucleus.
5. The tube enters the ovule through the **micropyle**. The **tube nucleus breaks down**, the tip bursts, and the two male nuclei enter the embryo sac.
6. **Double fertilisation**:
    - one male nucleus **fuses with the egg nucleus** → **diploid zygote** (2n), which develops into the **embryo**;
    - the other male nucleus **fuses with the two polar nuclei** → **triploid endosperm nucleus** (3n), which divides to form the **endosperm**, the food store for the embryo.



Schematic ovule (not to scale). The pollen tube enters through the micropyle and releases two male nuclei: one fuses with the egg nucleus (zygote, 2n), the other with the two polar nuclei (endosperm, 3n).

Fusion	Product	Chromosome sets	Becomes
male nucleus + **egg nucleus**	zygote	**2n** (diploid)	embryo (the new plant)
male nucleus + **two polar nuclei**	endosperm nucleus	**3n** (triploid)	endosperm (food store in the seed)

After fertilisation the ovule becomes the **seed** (integuments → seed coat, or testa) and the ovary becomes the **fruit**.

RECOMMENDED PRACTICAL: FACTORS AFFECTING POLLEN TUBE GROWTH

1. Make **sucrose solutions** of different concentrations (e.g. 0, 5, 10, 15, 20%), or keep one concentration and vary temperature.
2. Put a drop of each on a cavity slide and dust **pollen** from the same flower species onto it. Add a coverslip.
3. Keep the slides at a constant **temperature** (e.g. 25 °C) in a humid box.
4. After a fixed time (e.g. 30 minutes), measure the **length** of several pollen tubes with a calibrated **eyepiece graticule**, or count the percentage of grains that have germinated.
5. Repeat at intervals to find the **rate** of growth; use a **mean** of many tubes.

Controls: same species and age of flower, volume of solution, temperature, time. Sucrose supplies sugar for respiration and sets the water potential; too high a concentration draws water **out** of the grain.

MARK-SCHEME LANGUAGE

- Pollen tube: "(tube) grows (through the style) to the ovary / ovule / embryo sac"; "(by releasing) **digestive / hydrolytic enzymes**"; "**carries** the male nuclei / generative nucleus".
- Generative nucleus: "**divides by mitosis** to form two male (haploid) nuclei". "The generative nucleus fertilises / fuses" is **rejected**.
- Double fertilisation: "one male nucleus fuses with the **egg cell nucleus** to form the zygote"; "the other male nucleus fuses with the **two polar nuclei** to form the endosperm".
- Endosperm "**3n / triploid**"; embryo cells **2n**.
- Tube nucleus: "**controls the growth** of the pollen tube"; "contains the genes for / controls production of the digestive enzymes". "The tube nucleus fertilises" is ignored.
- Data on a drug: "as the concentration increases, the length of the pollen tube decreases", plus a manipulated figure for "determine".

Often asked as: explain the role of the pollen tube (often 5 marks); the function of the tube nucleus or generative nucleus; why embryo and endosperm cells contain different amounts of DNA; identify structures on an ovule diagram; interpret pollen-tube data.

COMMON MISTAKE

Saying the **generative nucleus** fertilises the egg. It must **divide by mitosis first**; the two **male nuclei** do the fusing.
"Degenerative nucleus", "the pollen grain moves down the tube", "two sperm cells".
Writing "egg" for "egg **nucleus**", or leaving out the **polar nuclei**. Calling the endosperm haploid.
Confusing the tube nucleus with a male nucleus.

BEYOND THE SPEC, BUT EXAMINED

- The name **double fertilisation**, the **polar nuclei**, the **micropyle** and the **embryo sac** are all expected, though the spec doesn't list them (WBI12 Oct 2019, Oct 2020, Jan 2021).
- Why the DNA differs: embryo cells have **two** sets of chromosomes (one male nucleus + egg), endosperm cells have **three** (one male nucleus + two polar nuclei) (WBI12 Jan 2024).
- Calcium ions are needed for pollen tube growth; a drug that blocks Ca^{2+} uptake (verapamil) shortens the tubes (WBI12 Oct 2019).
- Strategies that prevent self-fertilisation increase genetic variation but lower the chance of seeds forming (WBI12 Jun 2023, 6-mark).

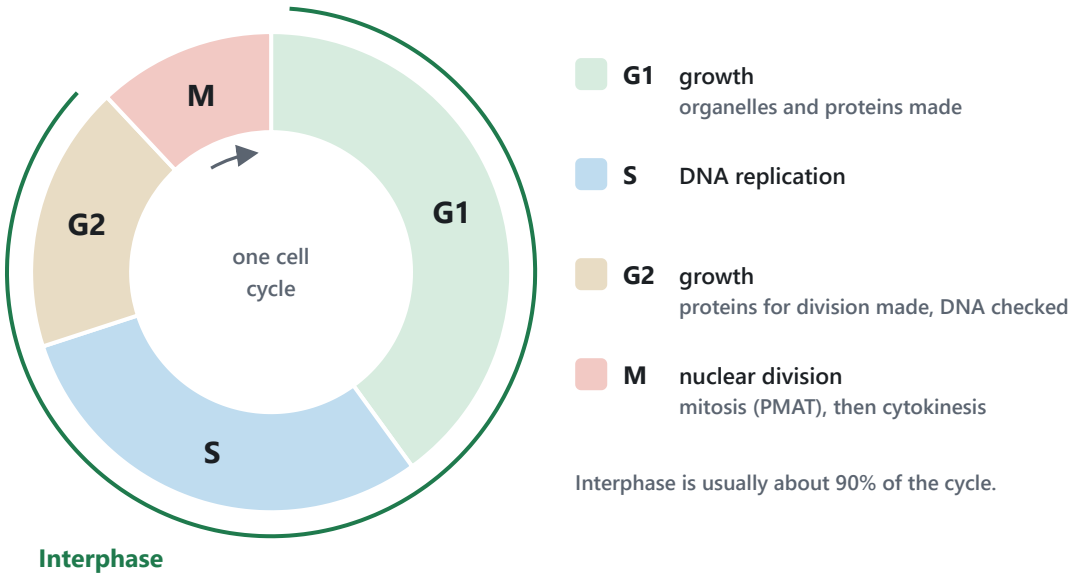
SPEC 3.14

Mitosis and the cell cycle

WHAT THIS SPEC POINT ASKS

Understand how mitosis and the cell cycle produce genetically identical daughter cells for growth and asexual reproduction.

3.14 The cell cycle



The cell cycle. Interphase (G1, S, G2) takes most of the time; mitosis and cytokinesis are short. The length of each phase varies between cell types.

- **Interphase** (not part of mitosis): the cell is very active.
  - **G1**: the cell grows, makes proteins (including enzymes for DNA replication) and **makes new organelles**.
  - **S**: **DNA replicates** (semi-conservatively), so each chromosome becomes two identical **sister chromatids**.
  - **G2**: more growth, proteins for division made; the cell checks the DNA has been copied correctly.
- **Mitosis**: the nucleus divides (prophase, metaphase, anaphase, telophase).
- **Cytokinesis**: the cytoplasm divides, making two cells.

3.14 The stages of mitosis (PMAT)

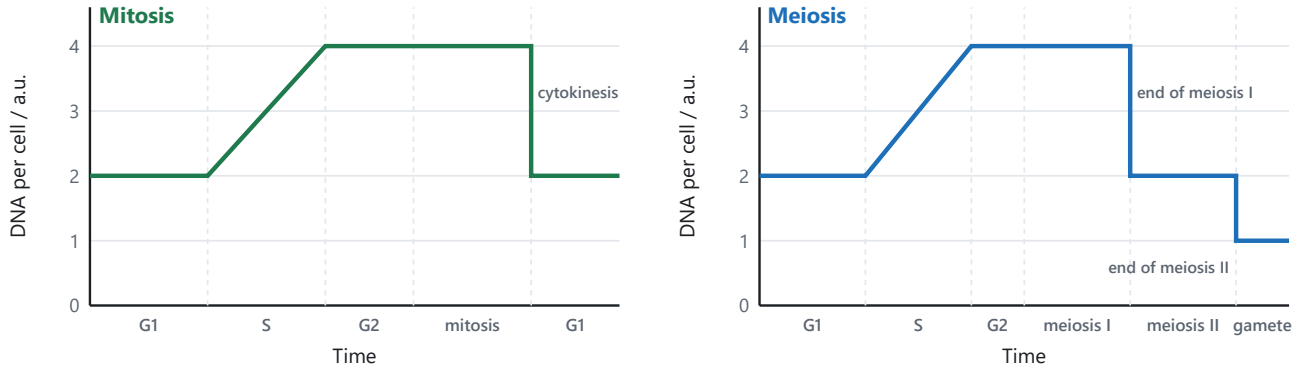
Stage	What happens
Prophase	chromosomes **condense** (shorten and thicken) and become visible as two chromatids joined at a **centromere**; the **nuclear envelope** and nucleolus break down; **centrioles** move to opposite poles and the **spindle** forms
Metaphase	chromosomes line up on the **equator** of the spindle; each **centromere attaches to spindle fibres**
Anaphase	**centromeres divide**; spindle fibres **shorten** and pull the **sister chromatids** to opposite poles, centromere first
Telophase	chromatids (now chromosomes) reach the poles and **uncoil**; a **nuclear envelope** re-forms round each set; spindle breaks down
Cytokinesis	animal cells: the membrane pinches in (a **cleavage furrow**). Plant cells: a **cell plate** forms across the middle and becomes the new cell walls

3.14 Why the daughter cells are identical

1. In S phase, every chromosome is **replicated** exactly, so the two sister chromatids are identical.
2. In anaphase, the sister chromatids of **every** chromosome are separated, one to each pole.

3. So each daughter cell receives **one identical copy of every chromosome**: same number (diploid stays diploid), same alleles.

3.14 DNA content through the cycle



DNA per cell doubles during S phase, stays constant through G2 and mitosis, then halves at cytokinesis. In meiosis it halves twice: after meiosis I and after meiosis II.

3.14 What mitosis is for

- **Growth**: a zygote → embryo → adult; meristems in plants.
- **Repair and replacement**: healing wounds, replacing skin and gut lining cells, blood cells.
- **Asexual reproduction**: one parent produces genetically identical offspring (clones): budding in yeast and *Hydra*, runners in strawberries, tubers in potatoes. (Bacteria divide by **binary fission**, which isn't mitosis.)

WORKED EXAMPLE: TIME IN A PHASE FROM A PIE CHART

A cell cycle lasts **24 hours**. On a pie chart, interphase is 300°, mitosis 45° and cytokinesis 15°.
1° = 24 × 60 ÷ 360 = **4 minutes**.
Mitosis: 45 × 4 = 180 min = **3 hours**. Cytokinesis: 15 × 4 = **60 minutes**.
Give the unit the question asks for (hours or minutes).

WORKED EXAMPLE: CELLS AFTER REPEATED DIVISIONS

One cell divides by mitosis 8 times, every cell dividing each time.
Cells = 2^8 = **256** (not 2 × 8 = 16).

MARK-SCHEME LANGUAGE

- DNA graph (4 marks): "DNA content remains **constant** during G1 / G2 / mitosis"; "DNA content **doubles** due to DNA **replication** (S phase)"; "the cell **divides**"; "producing two diploid, genetically identical daughter cells". "Interphase" alone isn't an explanation.
- Identical cells: "**derived from the zygote**" (in an embryo); "DNA / chromosomes are **replicated**"; "each cell receives one identical copy of each chromosome". "Mitosis" alone isn't enough.
- Spindle: "centromeres attach to spindle fibres"; "**separation of (sister) chromatids**"; each daughter cell gets identical genetic material.
- Drug that stops spindle fibres shortening: "(sister) chromatids **cannot be separated**" in **anaphase**.
- Role of the cell cycle: "synthesis of **organelles**", "increase in cell **size**", "**replication of DNA**". It is not a list of mitosis stages.
- Drawing anaphase for 2n = 6: spindle fibres, centrioles, **six** chromosomes at **each** pole, no nuclear envelope. Metaphase: each chromosome drawn as **two chromatids joined at a centromere**, on the equator; drawing a nuclear envelope loses the mark.
- Plant cell just after mitosis: nuclear envelopes and **cell plate** (new wall) forming.

Often asked as: order or identify stages; draw a stage for a given 2n; explain a DNA-content graph; role of the spindle or centromere; times from a pie chart; evaluate anti-cancer drug data (6-mark).

COMMON MISTAKE

Describing a graph when "explain" is asked; missing the final mark for **why DNA halves** (the cell divides).
Drawing 3 chromosomes per pole in anaphase when 2n = 6 (it's 6 at each pole). Drawing anaphase inside a nucleus.
Leaving out DNA **replication** when explaining why daughter cells are identical.
Describing the stages of mitosis when the role of the whole **cell cycle** is asked.

BEYOND THE SPEC, BUT EXAMINED

- **Cancer** is uncontrolled mitosis. Drugs that stop spindle fibres working hold cells in metaphase, so they **raise** the mitotic index of a sample while stopping division (WBI12 Oct 2022).
- **Epigenetic marks** are copied when DNA replicates, so they pass to both daughter cells (WBI12 Oct 2024; see 3.20).
- Cells of the **morula and blastocyst** divide by mitosis, not meiosis (WBI12 Jun 2024).
- Error bars on drug data: non-overlapping bars suggest a **significant** difference (WBI12 Jun 2019).

SPEC 3.15 Core Practical 6: root tip squash

WHAT THIS SPEC POINT ASKS

Prepare and stain a root tip squash and use it to observe the stages of mitosis.

3.15 Principle

- In a root, cells divide only in the **meristem**, just behind the root cap (the first 1–5 mm of the tip). That's where you'll see cells in mitosis.
- The tissue must be **separated into a layer one cell thick**, and the **chromosomes stained**, so you can see them clearly.

CORE PRACTICAL 6: METHOD IN BRIEF

1. Grow roots from a garlic clove or onion with its base in water for a few days.
2. Cut off the last **5–10 mm** of a few root tips.
3. Put them in warm (about 55–60 °C) **1 mol dm^{-3} hydrochloric acid** for about **5 minutes**.
4. Rinse the tips in cold water and blot them dry.
5. Put a tip on a slide, cut off and keep only the end 1–2 mm (the meristem), and add a few drops of **stain** (acetic orcein or toluidine blue). Leave a few minutes (some methods warm it).
6. Break the tip up with a mounted needle, add a coverslip, and **squash** gently by pressing down through filter paper. Don't twist the coverslip.
7. Find the meristem cells at **low power** (small, square, in rows, no large vacuoles), then view at **×400**.
8. Identify and draw cells in each stage; count cells in each stage and in interphase for the mitotic index (3.16).

Step	Why
Use the **root tip**	the meristem is where cells are dividing
Warm **HCl**	**breaks down the middle lamella** (pectin) holding cells together, so the cells separate and can be squashed flat
Rinse after the acid	removes acid, which would stop the stain working
Stain (acetic orcein / toluidine blue)	binds to DNA so the **chromosomes** are visible and the stages can be told apart (acetic orcein also fixes the cells, stopping division)
Squash	spreads the cells into a **single layer**, so light passes through and cells don't overlap
Low power first	to find the meristem; then high power to see chromosomes

3.15 Recognising the stages in a squash

- **Interphase**: a clear, round nucleus with a nucleolus; no visible chromosomes.
- **Prophase**: threads of chromosomes visible inside the nucleus.
- **Metaphase**: chromosomes in a line across the middle of the cell.
- **Anaphase**: two groups of V-shaped chromatids moving to opposite ends.
- **Telophase**: two dense groups at opposite ends; a new cell wall may be forming between them.

3.15 Safety and errors

- Acetic orcein is **corrosive** and stains: eye protection and gloves. HCl is an irritant at this concentration: eye protection. Hot water bath: scald risk.
- Errors: squashing too hard (chromosomes destroyed) or too lightly (cells overlap); missing the meristem; it can be hard to tell late interphase from early prophase. Count **many fields of view** and repeat with several roots.

MARK-SCHEME LANGUAGE

- Why stain: "to make the **chromosomes / chromatids** visible", "to distinguish the stages of mitosis". "To see which cells are dividing" and "to make organelles visible" are ignored.
- Using a photo: "count the number of cells in mitosis and the total number of cells", then divide (× 100).
- Choosing a stain: the slide stained for **nucleic acids** (e.g. safranin) shows chromosomes; one stained for lipids (nile red) doesn't. Both the choice and the reason are needed.

Often asked as: the reason for the staining step (the most common question); how to find the mitotic index from a slide or photo. The full method hasn't been asked in WBI12 2019–2026, but learn it.

COMMON MISTAKE

"The stain makes the cells visible." More than half the candidates lost this mark (Jan 2020). Say **chromosomes**.
Saying the acid "softens the cell walls". It breaks down the **middle lamella** between cells, so they separate.

SPEC 3.16

Calculating mitotic indices

WHAT THIS SPEC POINT ASKS

Be able to calculate a mitotic index.

$$\text{mitotic index} = \text{number of cells in mitosis} \div \text{total number of cells counted}$$

- "Cells in mitosis" = cells in prophase + metaphase + anaphase + telophase (chromosomes visible).
- **Total** = cells in mitosis **+ cells in interphase**. Multiply by 100 for a percentage.
- A **high** mitotic index means cells are dividing fast: a growing region (meristem), or a **tumour**. Doctors use it to judge how aggressive a cancer is.

WORKED EXAMPLE: MITOTIC INDEX

In one field of view: interphase 112, prophase 14, metaphase 4, anaphase 3, telophase 4.
Cells in mitosis = 14 + 4 + 3 + 4 = **25**. Total = 112 + 25 = **137**.
Mitotic index = 25 ÷ 137 = **0.18** (or **18.2%**).

WORKED EXAMPLE: HOW LONG IS EACH STAGE?

The proportion of cells in a stage ≈ the proportion of the cycle that stage takes.
Using the counts above with a 20-hour cycle, anaphase = 3 ÷ 137 × 20 × 60 = **26 minutes**.
Anaphase is the shortest stage, so the fewest cells are caught in it.

WORKED EXAMPLE: WORKING BACKWARDS

A tissue sample has a mitotic index of **0.24**, and 60 cells are in mitosis.
Total cells = 60 ÷ 0.24 = **250**. Cells in interphase = 250 − 60 = **190**.

3.16 Getting a reliable value

- Count at least 100 cells, in **several fields of view**, from several root tips; calculate a mean.
- Use a clear rule for cells cut by the edge of the field (e.g. count those on two edges only).

- Count cells at the **same distance from the tip** when comparing treatments.

MARK-SCHEME LANGUAGE

- "Number of cells in mitosis ÷ total number of cells (× 100)". 0.22 and 22% are both accepted.
- Method: "count the number of cells in mitosis (visible chromosomes)"; "divide by the **total** number of cells"; "× 100".
- A treatment that stops chromosomes condensing: "fewer cells in mitosis", "**lower** mitotic index", "higher survival".
- Higher mitotic index → more advanced cancer stage and lower survival.

Often asked as: calculate an index from counts (show working); rearrange to find the total or interphase cells; describe how to find the index from a slide; predict survival from an index.

COMMON MISTAKE

Dividing by the number of **interphase** cells instead of the **total**. This is the most common error every year.
Adding up the total wrongly; getting a number of cells larger than the total; forgetting × 100 when a percentage is asked.

SPEC 3.17 Stem cells, potency, the morula and the blastocyst

WHAT THIS SPEC POINT ASKS

(i) Understand the terms stem cell, totipotent, pluripotent, morula and blastocyst. (ii) Discuss how society uses scientific knowledge to make decisions about using stem cells in medical therapies.

3.17 Key terms

- **Stem cell**: an **unspecialised** cell that can **divide by mitosis** to make more stem cells, and can **differentiate** into specialised cell types.
- **Totipotent**: can differentiate into **any** cell type, including the **extra-embryonic** cells of the placenta and membranes. Only the zygote and the cells of the first few divisions (up to the morula).
- **Pluripotent**: can differentiate into **almost any cell type in the body**, but **not** the placenta. The inner cell mass of the blastocyst.
- **Multipotent** (adult stem cells, e.g. bone marrow): only a limited range of related cells, e.g. the different blood cells.

3.17 Early development (human)

Stage	Roughly when	What it is	Potency
Zygote	day 0	one diploid cell after fertilisation	totipotent
Morula	day 3–4	a **solid ball** of about 16 cells, all made by mitosis from the zygote	**totipotent**
Blastocyst	day 5	a **hollow ball** of about 100 cells: an outer layer (**trophoblast**, which forms the placenta), a fluid-filled **cavity**, and an **inner cell mass** (which forms the embryo)	inner cell mass **pluripotent**

As cells divide, some genes are **switched off** permanently (3.18), so potency is lost step by step.

3.17 Stem cells in medicine

- **Bone marrow transplants**: donor stem cells replace a patient's blood cells after leukaemia treatment.
- Research and trials: replacing **heart muscle** after a heart attack, nerve cells in Parkinson's disease or spinal injury, **β cells** in type 1 diabetes, retinal cells in macular degeneration.

Source	For	Against
Embryonic (inner cell mass of spare IVF embryos)	pluripotent: can make almost any cell type; grow readily	the **embryo is destroyed**; some believe life begins at fertilisation; donor cells may be **rejected**
Adult (bone marrow, etc.)	no embryo destroyed; can be the patient's **own**, so no rejection; simple to collect	multipotent: limited range of cells; few in number
Induced pluripotent (iPS) (adult cells, e.g. skin, reprogrammed by switching on certain genes)	pluripotent; no embryo; patient's own cells, so no rejection	reprogramming may not be complete; risk of **tumours**; still being developed

3.17 How society decides

1. Scientists **publish** results; other scientists check them by **peer review**.
2. **Ethics committees** and the public (through consultation and debate) weigh benefits against moral objections.
3. **Governments** pass laws, and **regulatory authorities** license, guide and monitor research and treatments (e.g. in the UK, embryos may be used for research only up to **14 days** under licence).
4. Decisions differ between countries, cultures and religions, so some therapies are allowed in one country and banned in another.

EXAM TIP

"Discuss" means **both sides** and a **conclusion**: the medical benefit, the ethical objection (embryo destruction), alternatives (adult or iPS cells), and who decides (regulators, law). Use the details in the question, e.g. "the disease is fatal", "the cells are put into the liver, so gametes aren't affected".

MARK-SCHEME LANGUAGE

- Morula: "**solid** ball" of **totipotent** cells. Blastocyst: "**hollow** ball", outer layer (trophoblast), **inner cell mass** of pluripotent cells, fluid-filled cavity. Both divide by mitosis; both diploid.
- Compare morula and blastocyst: full marks need a **similarity** (both diploid, both made by mitosis) plus differences (totipotent v pluripotent; solid v hollow; the blastocyst has more cells).
- Totipotent: all cell types **including extra-embryonic / placenta** cells. Pluripotent: "most cell types" (not "any"), not placenta.
- Totipotent → pluripotent: "**differential gene expression**"; "some genes switched **off**"; "by **epigenetic** modification"; "proteins made from the active genes permanently modify the cell". "Genes activated" alone is ignored.
- Heart attack: "pluripotent stem cells **differentiate into heart muscle** cells" "to **replace** the dead cells". "Repair" is ignored.
- Society: "stem cells can differentiate into different cells/tissues", "great potential for therapies"; against: "**destruction of an embryo**" is an ethical issue; early embryos "have no nervous system / can't feel pain"; iPS: "no embryo used".

Often asked as: name or circle the morula or blastocyst; describe the blastocyst; compare a morula and a blastocyst; explain the change from totipotent to pluripotent; ethics of a stem-cell therapy (often in an animal or new-drug context).

COMMON MISTAKE

Not being able to name the **morula** (three quarters failed in Oct 2019). "Blastocyte" isn't credited.
Swapping totipotent and pluripotent; saying the blastocyst is totipotent.
Describing the differences between totipotent and pluripotent when the question asks **how** a cell changes (half the candidates scored 0, Jun 2019).
Saying dead heart cells are "repaired". They are **replaced**.

BEYOND THE SPEC, BUT EXAMINED

- **Trophoblast**, **inner cell mass** and the **blastocoel** (cavity) by name (WBI12 Jun 2023, Jan 2026).
- **Induced pluripotent stem cells** (hiPSC), made by switching on genes such as *sox2* (WBI12 Jan 2023, Jun 2025).
- Plant **meristem** cells are totipotent: a small piece of tissue grown on agar with nutrients and hormones forms a whole new plant, genetically identical to the parent (tissue culture, WBI12 Oct 2025).

SPEC 3.18

Cell specialisation by differential gene expression

WHAT THIS SPEC POINT ASKS

Understand how cells become specialised because different genes are expressed: active genes make mRNA, the mRNA is translated into proteins, and the proteins control cell processes or decide cell structure, in animals and plants.

3.18 Same genes, different genes switched on

- Every body cell in an organism contains the **same genes** (the same DNA), because all came from the zygote by mitosis.

- What differs is **which genes are active**. Some genes (for respiration enzymes, membrane proteins) are on in every cell; others are on only in certain cells.

3.18 The chain

1. A signal (e.g. a chemical from nearby cells, a hormone) reaches the cell.
2. Some genes are **switched on** and others **switched off** (by **transcription factors** binding to DNA, and by **epigenetic** changes, 3.20).
3. Only the active genes are **transcribed**, producing **active mRNA**.
4. The mRNA leaves the nucleus and is **translated** on ribosomes into **proteins**.
5. These proteins (enzymes, structural proteins, receptors) **change the cell's structure and function**, so the cell becomes specialised. Changes build up over time and are usually **permanent**.

3.18 Transcription factors

- Proteins that **bind to specific DNA sequences**, usually the **promoter** next to a gene.
- **Activators** help **RNA polymerase** bind, so the gene is transcribed. **Repressors** stop RNA polymerase binding, so it isn't.

ANIMAL EXAMPLE: A B CELL IN THE PANCREAS

The **insulin gene** is switched on in β cells (and off in every other cell type).
Insulin gene transcribed → insulin mRNA → translated on the rER → insulin made, packaged by the Golgi and secreted (3.4).
The cell also has lots of rER, Golgi and secretory vesicles. A skin cell has the insulin gene too, but it is switched off.

PLANT EXAMPLE: A MERISTEM CELL BECOMING SCLERENCHYMA OR XYLEM

Genes for the **enzymes that make cellulose and lignin** are switched on.
mRNA transcribed → enzymes made → extra cellulose laid down and the walls **lignified** (secondary thickening).
The cell contents die, leaving a strong, hollow, dead fibre or vessel (Topic 4).

MARK-SCHEME LANGUAGE

- The chain: "**differential gene expression**"; "gene X switched on in these cells and off in others due to epigenetic modification / DNA methylation / histone modification"; "**transcription** of active genes / **active mRNA** produced"; "**translation**, producing proteins"; "proteins **determine / change** the cell's structure or function", with a specific change for the cell named.
- Name the gene and protein in context: insulin, arginase, muscle proteins, sclerenchyma (cellulose, lignin, thickened walls).
- Gene activity graphs: "gene T decreases and gene U increases"; "gene T switched off because its product isn't needed". Don't say "switched on" if the gene was already active.
- A mutation that switches a silent gene back on: the gene "is not switched off"; transcription and translation follow; a structural change results (e.g. teeth in a chick).

Often asked as: explain how a named cell becomes specialised (sperm, muscle, sclerenchyma, a liver cell); interpret gene-activity graphs; explain how a mutation reactivates a silent gene.

COMMON MISTAKE

"Specialised cells contain different genes." They contain the **same** genes; **different genes are expressed**.
Saying genes are "switched on/off" without saying **how**.
Mixing up transcription and translation, or not saying where translation happens.
Stopping at "proteins are made" without linking them to the **structure or function** of the cell named.
Writing about totipotent and pluripotent cells in a plant question.

BEYOND THE SPEC, BUT EXAMINED

- Damage switches on a gene in regenerating tissue: chemical changes to DNA or histones loosen the DNA, so **RNA polymerase / transcription factors** can bind the promoter (WBI12 Jun 2025).
- Plant tissue culture: totipotent cells, mitosis, genetically identical cells, then differential gene expression to form roots and leaves (WBI12 Oct 2025).

SPEC 3.19 One gene, more than one protein

WHAT THIS SPEC POINT ASKS

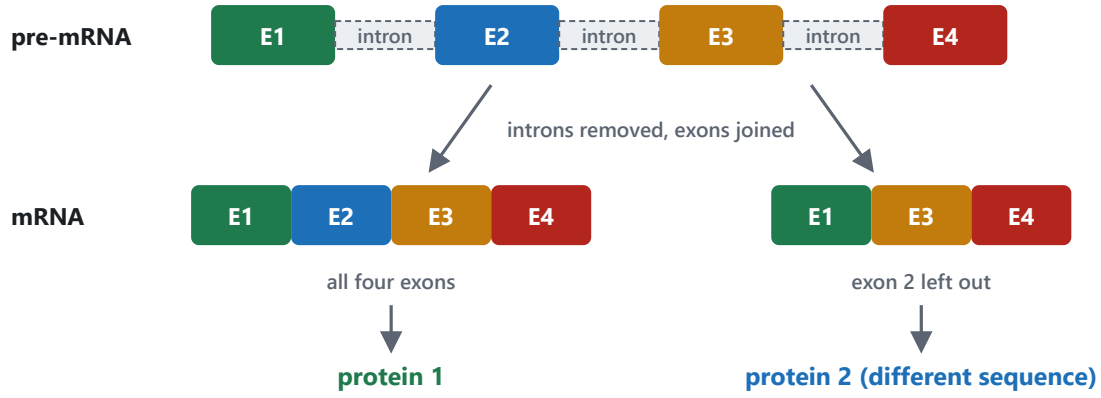
Understand how one gene can code for more than one protein because of changes made to mRNA after transcription.

3.19 Introns, exons and splicing

- In eukaryotes, a gene contains **exons** (base sequences that code for amino acids) separated by **introns** (non-coding sequences).
- Transcription copies the **whole** gene, introns included, making **pre-mRNA** in the nucleus.
- Before it leaves the nucleus, the pre-mRNA is **spliced**: **all introns are removed** and the **exons are joined** together, making **mature (active) mRNA**. Splicing is done by enzyme complexes called **spliceosomes**.
- Prokaryotic genes have no introns, so their mRNA isn't spliced.

3.19 Alternative splicing

- The exons of one pre-mRNA can be joined in **different combinations**: some exons are **left out**, or the **order** changes.
- Each combination gives a different mRNA → a different **sequence of amino acids** (primary structure) → a different folding and shape → a **different protein**.
- So one gene can make several proteins, and humans make far more kinds of protein than they have genes.



One gene, two proteins. Every intron is always removed; which exons are kept (or their order) can vary.

	Pre-mRNA	Mature (active) mRNA
Similarities	both single-stranded RNA, with bases A, U, C, G, ribose and phosphate; each exon has the same base sequence in both	
Contains	**introns and exons**	**exons only**
Length	longer	shorter
Where	in the nucleus only	leaves through nuclear pores to the ribosomes
Exons	all of them, in the gene's order	may be **fewer** or in a **different order**

EXAM TIP

If the question gives a diagram with exons labelled (e.g. P, Q, R, S), quote **two different exon combinations** from it, e.g. "PQS and PRS". Most candidates don't, and lose the mark.

MARK-SCHEME LANGUAGE

- "**Post-transcriptional** changes / (pre-)mRNA **splicing**" (alternative splicing accepted).
- "**All introns** removed (by enzymes / spliceosomes)". "Some introns" is **rejected**.
- "Removal of some **exons** / rearrangement of exons / different permutations of exons", with two different orders from the diagram.
- "Different **amino acid sequence** / primary structure / polypeptide", so different folding and shape. "Different proteins produced" alone is ignored.
- Pre-mRNA v mRNA: similarities (single-stranded; A, U, C, G; ribose, phosphate) and differences (introns + exons v exons only; fewer or reordered exons; mRNA shorter).

Often asked as: explain how one gene gives two proteins (3–5 marks, often with an exon diagram from an unfamiliar organism); compare pre-mRNA and active mRNA.

COMMON MISTAKE

Giving an answer about **mutation**, **differentiation** or **epigenetics**: none of these is post-transcriptional, and they earn nothing here.
Swapping introns and exons. Describing transcription and translation when only the change to the mRNA is asked.
Saying only that exons are rearranged: examiners want **removal** of exons as well.

BEYOND THE SPEC, BUT EXAMINED

- The words **intron, exon, spliceosome** and **alternative splicing** are all used in mark schemes (WBI12 Oct 2020, Jun 2023).
- In fruit flies one gene, *Dsx*, is spliced into two different proteins, one leading to female development and one to male (WBI12 Jun 2024).

SPEC 3.20 Genotype, environment and epigenetics

WHAT THIS SPEC POINT ASKS

(i) Understand that phenotype results from the interaction of genotype and environment. (ii) Know how epigenetic modification (DNA methylation and histone modification) can switch genes on or off. (iii) Understand how epigenetic changes can be passed on when cells divide.

3.20 Phenotype = genotype + environment

- **Genotype**: the alleles an organism has. **Phenotype**: its observable characteristics.
- The genotype sets the **potential**; the environment affects how much of it is reached.

Example	Genotype	Environment
Himalayan rabbit / Siamese cat coat	allele for a **temperature-sensitive** enzyme that makes dark pigment	enzyme is inactive at core body temperature but active in cooler **extremities**, so only ears, nose, paws and tail are dark
Human height	many genes (3.21)	diet, disease during childhood
Plant height and colour	genes for growth, chlorophyll	light (etiolation in the dark), **magnesium** or nitrate in the soil (chlorosis)
Identical twins	same genotype	grow apart in weight, disease and behaviour as their environments (and epigenetic marks) differ

Variation caused by the environment alone is **not** passed on in the DNA sequence.

3.20 Epigenetics

Epigenetic modification: chemical changes to DNA or to the histone proteins it is wrapped round, which **switch genes on or off without changing the base sequence**. Diet, stress, smoking, toxins and age can all change epigenetic marks.

DNA in the nucleus is wound round **histones**. How tightly it is wound decides whether **RNA polymerase** and **transcription factors** can reach a gene.

	DNA methylation	Histone acetylation	Histone methylation / deacetylation
What's added	**methyl ($-CH_3$) groups** to **cytosine** bases (at CG sites), often in a gene's **promoter**	**acetyl groups** to amino acids (lysine) in histones	methyl groups added to histones, or acetyl groups removed
Effect on chromatin	promoter blocked; DNA wound more tightly	histones lose positive charge, **bind DNA less tightly**: open chromatin (euchromatin)	DNA wound **more tightly**: condensed chromatin (heterochromatin)
Transcription factors / RNA polymerase	**can't bind**	**can bind**	can't bind
Gene	**switched off** (silenced)	**switched on**	**switched off**

So: methylated DNA → no transcription → no mRNA → no protein.

3.20 Passing epigenetic marks on

- When DNA replicates, each new double helix has one old strand (with its methyl groups) and one new, unmethylated strand.
- A **methyltransferase enzyme** recognises the half-methylated sites and adds methyl groups to the matching cytosines on the new strand. Histone marks are rebuilt too.
- So after mitosis, both daughter cells have the **same pattern of active and silent genes**. That's how a liver cell's daughters stay liver cells.
- Most marks are erased when gametes form, but a few survive into the next generation. Children of mothers starved in the Dutch famine of 1944–45 showed altered methylation and health decades later.

MARK-SCHEME LANGUAGE

- DNA methylation: "**methyl / CH_3 groups** added to **cytosine**" (CpG, "in the promoter region" accepted); "gene **switched off / silenced**"; "RNA polymerase / transcription factors **cannot bind**"; "no transcription / translation"; "the protein isn't produced".
- The enzyme: "binds a methyl group to DNA", "to cytosine". Histones, amino acids or cysteine are rejected.
- Histone modification: "methylation / acetylation (or phosphorylation) of **histones**". Methylation: "histones closer, DNA more **tightly** wrapped (heterochromatin), gene off". Acetylation: "histones further apart, DNA **less tightly** wrapped (euchromatin), gene on". "Loose / tight histones" is ignored.
- Passing on: "marks are **copied when DNA replicates**", "daughter cells have the same methylation"; if the enzyme is missing, the modification is diluted at each division until it disappears.
- Tumour suppressor gene methylated → gene silenced → no protein to stop the cell cycle → uncontrolled mitosis → tumour. A drug that lowers methylation reverses this.
- Coat colour (6-mark): multiple alleles; a **temperature-sensitive enzyme**; pigment only in **cooler extremities**. Level 3 needs genotype → enzyme → phenotype **plus** the environment.

Often asked as: describe one epigenetic modification; explain how methylation silences a gene (often a tumour suppressor); histone diagrams; how marks pass to daughter cells; 6-mark questions on epigenetic drugs with graphs, or on gene–environment interaction.

COMMON MISTAKE

"Epigenetics changes the DNA sequence." It changes **gene activity**, not the base sequence (a mutation changes the sequence).
Methylating **DNA** in a histone question, or acetylating DNA. Saying methylation "switches genes **on**".
Saying "the **gene** stops cell division": the **protein** does.
Not linking the modification to the gene being switched off; "balanced diet" or "different exposure" with no detail.

BEYOND THE SPEC, BUT EXAMINED

- **CpG sites**, the enzyme **DNMT** (DNA methyltransferase), and the terms **euchromatin** / **heterochromatin** (WBI12 Jan 2025, Oct 2025).
- Other histone modifications, such as phosphorylation, are accepted alongside acetylation and methylation (WBI12 Oct 2019, Jun 2023).
- **Demethylating drugs** reactivate silenced tumour suppressor genes; evaluate them with error bars and the mitotic index (WBI12 Jan 2026, 6-mark).
- In enzyme-transfer experiments, a mark kept by the enzyme passes to every daughter cell; without the enzyme it is halved at each division (WBI12 Oct 2021).

SPEC 3.21 Multiple alleles, polygenic inheritance and continuous variation

WHAT THIS SPEC POINT ASKS

Understand how some phenotypes are affected by multiple alleles of one gene, or by many genes (polygenic inheritance), plus the environment, and how polygenic inheritance produces continuous variation.

3.21 Multiple alleles: one gene, more than two alleles

- A gene can have **three or more alleles** in a population, though each individual still has only **two** (one on each homologous chromosome).
- Each allele has a **different base sequence**, so codes for a slightly different mRNA and protein (or none).
- **ABO blood groups**: one gene, three alleles. I^A and I^B are **codominant**; I^O is **recessive** to both.

Genotype	Phenotype (blood group)
I^AI^A or I^AI^O	A
I^BI^B or I^BI^O	B
I^AI^B	AB (both antigens made)
I^OI^O	O

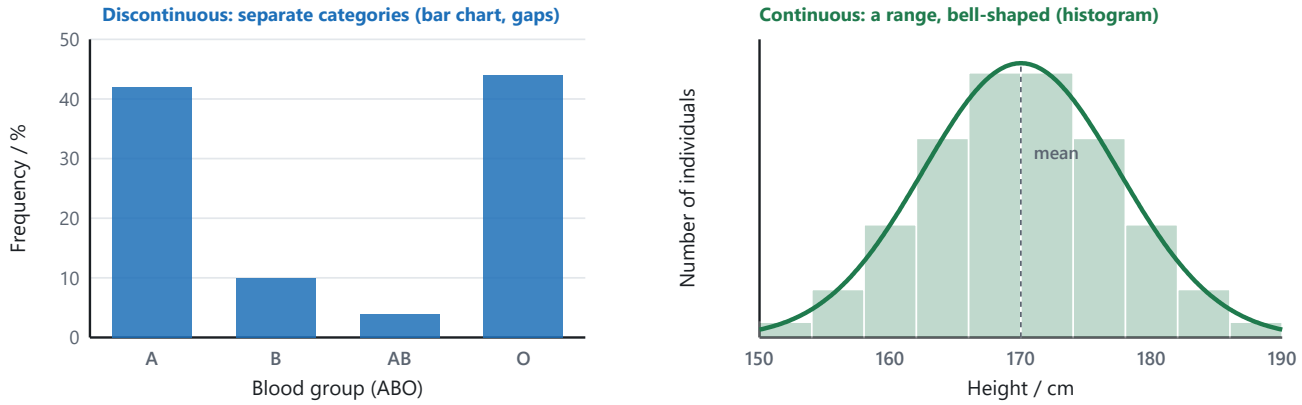
- **Rabbit coat colour**: four alleles in a dominance series, C (full colour) > c^{ch} (chinchilla) > c^h (Himalayan) > c (albino). The Himalayan pattern also depends on temperature (3.20).
- Multiple alleles give a few **distinct categories**: **discontinuous** variation.

WORKED EXAMPLE: A MULTIPLE-ALLELE CROSS

Parents: I^AI^O (group A) × I^BI^O (group B).
Gametes: I^A or I^O; I^B or I^O.
Offspring: I^AI^B (AB), I^AI^O (A), I^BI^O (B), I^OI^O (O): **1 : 1 : 1 : 1**. Two group A and B parents can have a child of **any** blood group.

3.21 Polygenic inheritance: many genes, one characteristic

- **Polygenic**: a characteristic controlled by **many genes at different loci**, each with a small, **additive** effect. Examples: height, skin colour, wheat seed colour, milk yield.
- The many genotypes give many small steps of phenotype. The **environment** then blurs the steps, giving a smooth range: **continuous variation**, usually a **bell-shaped (normal) distribution**. Most individuals are near the **mean**; few are at the extremes.



Left: discontinuous variation (one gene, multiple alleles): distinct groups, drawn as a bar chart. Right: continuous variation (many genes plus environment): a range of values with a bell-shaped distribution, drawn as a histogram. Example data.

WORKED EXAMPLE: TWO GENES WITH ADDITIVE EFFECTS

Wheat seed colour: each **dominant** allele (A or B) adds red pigment; recessive alleles (a, b) add none. Cross AaBb × AaBb. Each parent makes 4 gametes (AB, Ab, aB, ab), giving 16 combinations.

Number of dominant alleles	**0**	**1**	**2**	**3**	**4**
Fraction of offspring	1/16	4/16	6/16	4/16	1/16

So **1 : 4 : 6 : 4 : 1**: the middle shade is most common and the extremes are rare. With three genes (AaBbCc × AaBbCc) it becomes 1 : 6 : 15 : 20 : 15 : 6 : 1 out of 64, closer still to a smooth curve.

	Discontinuous	**Continuous**
Genes	one gene (sometimes with multiple alleles)	**many genes** (polygenic)
Environment	little or no effect	**large** effect
Values	distinct categories, no in-betweens	a **range** between two extremes
Graph	**bar chart** (gaps between bars)	**histogram** or smooth curve, bell-shaped
Examples	ABO blood group, sex, seed shape	height, mass, skin colour, leaf length

MARK-SCHEME LANGUAGE

- Polygenic: "many / multiple **genes** contribute to the **same characteristic**"; "genes at **different loci**". "Alleles" alone isn't enough, and writing "characteristic**s**" lost the mark.
- Sketch: a **bell-shaped** curve starting and ending at zero; x-axis = the characteristic (e.g. height / cm), y-axis = **frequency / number of individuals**. "Population" or "variation" as axis labels is ignored.
- Wheat colour: "more recessive alleles = lighter colour"; "independent assortment and crossing over → gamete variation"; "**random fertilisation**"; "low probability of inheriting 0 or all recessive alleles", so intermediate colours are most common.
- Multiple alleles: "different alleles have different **base sequences** / code for different mRNA", so "a different polypeptide / protein / pigment"; heterozygotes may show both alleles or an intermediate phenotype. "Polygenic" is rejected here.
- Predictions from a rule: follow every step and keep the **unit** (e.g. (1.59 + 1.76 − 0.13) ÷ 2 = 1.61 m).

Often asked as: define polygenic inheritance; sketch the distribution; explain variation in wheat or skin colour (including a 6-mark); explain multiple alleles at one locus.

COMMON MISTAKE

"Polygenic means many alleles for one gene." That is **multiple alleles**. Polygenic means **many genes**. Using "population" or "variation" as an axis label; drawing Punnett squares when a distribution is asked. Writing about crossing over, linkage or epigenetics when the question is about **multiple alleles**. Describing a table instead of explaining it.

BEYOND THE SPEC, BUT EXAMINED

- The probability of the extremes: in AaBbCc × AaBbCc, the chance of inheriting all six dominant alleles is $(½)^6$ = **1 in 64** (WBI12 Oct 2022).
- A normal distribution of bean mass or leaf width is continuous variation (WBI12 Jan 2020).
- Two people with the same polygenic genotype can differ because of **environment** (UV exposure, diet) or **epigenetic** marks (WBI12 Jan 2022, 6-mark).

Topic 4 – Plant Structure and Function, Biodiversity and Conservation

SPEC 4.1 Plant cell structure and how it compares with animal cells

WHAT THIS SPEC POINT ASKS

(i) Know the structure and ultrastructure of a plant cell (cell wall, chloroplast, amyloplast, vacuole, tonoplast, plasmodesmata, pits, middle lamella) and compare it with an animal cell. (ii) Know what each of these structures does.

4.1 What plant and animal cells share

Both are eukaryotic cells, so both have a **nucleus** with a **nucleolus**, **rough and smooth ER**, a **Golgi apparatus**, **mitochondria**, **80S ribosomes** in the cytoplasm and a **cell surface membrane** (Topic 3). Plant cells add the structures below.

4.1 The extra structures in plant cells

Structure	What it is	What it does
Cell wall	Layer outside the cell surface membrane, made mainly of **cellulose** microfibrils in a matrix of other polysaccharides (pectins, hemicelluloses). Freely permeable.	Gives **support** and a fixed shape. Resists the outward pressure of the vacuole, so the cell becomes **turgid** instead of bursting.
Middle lamella	The thin layer between the walls of two neighbouring cells, made mainly of **calcium pectate**. It forms first, when a dividing cell lays down its new wall.	**Sticks neighbouring cells together**.
Plasmodesmata (singular plasmodesma)	Narrow channels through the walls of neighbouring cells, lined with cell surface membrane, so the cytoplasm of one cell is continuous with the next.	Let substances (water, ions, sucrose, signalling molecules) pass from cell to cell: **transport and communication** between cells.
Pits	Thin areas of the wall where **no secondary thickening** (and no lignin) was laid down. They line up in pairs across neighbouring cells.	Let water and mineral ions move **sideways** (laterally), e.g. from one xylem vessel to the next or out to living cells.
Chloroplast	Lens-shaped organelle a few μm long with a **double membrane** (envelope). Inside: flattened sacs called **thylakoids**, stacked into **grana** and joined by membranes (lamellae), surrounded by a fluid **stroma** that holds enzymes, starch grains, circular DNA and 70S ribosomes. Chlorophyll is in the thylakoid membranes.	Site of **photosynthesis**: converts **light energy into chemical energy** (glucose).
Amyloplast	Colourless plastid with a double membrane, packed with **starch grains**; no thylakoids or chlorophyll. Common in storage organs (potato tubers, roots, seeds).	**Makes and stores starch**, and hydrolyses it back to glucose when the cell needs it.
Vacuole	Large, permanent sac filled with **cell sap**: water with dissolved sugars, mineral ions, pigments, waste and enzymes.	Water enters by osmosis, so the vacuole pushes the cytoplasm against the wall: **turgor**, which supports the plant. Also **stores** named substances (pigments, sugars, ions) and isolates waste or harmful chemicals.
Tonoplast	The **single, partially permeable membrane** around the vacuole, with carrier and channel proteins.	Controls what enters and leaves the vacuole.

4.1 Plant cell vs animal cell

Feature	Plant cell	Animal cell
Cell wall	**cellulose** wall outside the membrane	none
Middle lamella, plasmodesmata, pits	present	absent
Chloroplasts	in green tissues (not in root cells)	none
Carbohydrate store	**starch** grains, often in **amyloplasts**	**glycogen** granules
Vacuole	**large, permanent**, with a tonoplast	small, temporary vacuoles or vesicles only
Centrioles	absent (in flowering plants)	present
Lysosomes	rare (the vacuole does much of this job)	present
Cilia / flagella	absent (in flowering plants)	on some cells
Shape	fairly regular, fixed by the wall	variable
Both have	nucleus, nucleolus, rER, sER, Golgi apparatus, mitochondria, 80S ribosomes, cell surface membrane	

EXAM TIP

For "compare" questions, write paired sentences that name both cells: "a plant cell has a cellulose cell wall **whereas** an animal cell has none". A list of features with no cell named isn't credited.

MARK-SCHEME LANGUAGE

- Animal cells only: "centrioles", "glycogen granules", "cilia", "flagella"; lysosomes accepted. Vesicles and temporary vacuoles are ignored.
- Chloroplast: "(site of) photosynthesis" / "converts light energy to chemical energy". "Makes food" is ignored.
- Starch / amyloplast: "store of {glucose / energy}". "Produces energy" and "stores carbohydrates" are ignored.
- Vacuole: "provides support (turgor)", "stores" + a **named** substance (water, sugars, mineral ions, pigments, enzymes). "Cell sap" alone and "gives the cell shape" are ignored.
- Plasmodesmata: "connect cells / allow communication between cells", "signalling substances pass through the cytoplasm (symplast)", or named molecules passing through.
- Pits: "allow (lateral) movement of water / mineral ions" between adjoining cells or out of the xylem.
- Walls with secondary thickening compared with walls without: "thicker / more layers", "contain lignin", "stronger / less flexible", "waterproof / impermeable".
- Drawings: amyloplasts and chloroplasts drawn **in the cytoplasm** and **smaller than the nucleus**; the tonoplast drawn as a **single** membrane, with the label line touching the membrane, not the middle of the vacuole.

Often asked as: MCQ on which structure stores starch (very common), name two structures found in animal but not plant cells, state the function of a named structure, draw and label amyloplast / chloroplast / tonoplast, compare a plant cell with a fungal cell.

COMMON MISTAKE

Saying plant cells have no Golgi apparatus: they do (WBI12 Oct 2024 report).
Putting chloroplasts in a **root** cell. Roots are underground, so their cells have none.
Writing "cell sap" or "nutrients" as the vacuole's contents, or "nutrients" for what passes through plasmodesmata. Name the substances.
Mixing up **pits** (thin areas of the wall) with **plasmodesmata** (cytoplasmic channels through the wall).
Calling the vacuole the tonoplast. The tonoplast is the membrane around it.

BEYOND THE SPEC, BUT EXAMINED

- **Fungal cells** compared with plant cells: fungal walls are made of **chitin**, not cellulose; fungi store **glycogen**, not starch; they have no chloroplasts and no plasmodesmata (WBI12 Jan 2021).
- How a plant cell **gets bigger** after division: it takes up water into the vacuole and makes more cytoplasm, organelles and proteins, plus new membrane and wall. Describing division instead loses the marks (WBI12 Oct 2021).
- Sphere and hemisphere volume in plant contexts ($4/3\pi r^3$, halved for a bowl shape) (WBI12 Jun 2022).

SPEC 4.2

Recognising plant organelles in electron micrographs

WHAT THIS SPEC POINT ASKS

Identify the plant cell structures from 4.1 in electron microscope images.

4.2 What each structure looks like in a TEM

Structure	How to spot it
Cell wall	Thick, fairly even grey band around the outside of the cell. Between two cells you may see a thin darker line down the middle: the **middle lamella**. The cell surface membrane sits tight against the inside of the wall, so it is hard to see.
Plasmodesmata	Thin dark channels running straight across the wall, joining the cytoplasm of two cells.
Pits	Gaps or thin places in a thick (lignified) wall, often in pairs on either side of the wall.
Chloroplast	Large oval with a double membrane. Inside: **stacks of thin parallel lines** (grana) joined by single lines, in a grainy stroma. Often has pale starch grains and small dark lipid droplets. Bigger than a mitochondrion.
Amyloplast	Double membrane, filled with one or more large **starch grains** (sometimes showing rings). **No thylakoids.**
Vacuole	Large pale area that looks empty, bounded by one thin membrane (the **tonoplast**). Often pushes the cytoplasm into a thin layer at the edge.
Mitochondrion (to compare)	Smaller, double membrane, inner membrane folded into **cristae**, not stacked discs.

EXAM TIP

Chloroplast or mitochondrion? Look inside. **Stacked thylakoids** = chloroplast. **Folds of the inner membrane** (cristae) = mitochondrion. Chloroplast or amyloplast? Starch grains with **no** thylakoid stacks = amyloplast.
A large pale vacuole, a cell wall or chloroplasts tells you the image is a plant cell.

WORKED EXAMPLE: ACTUAL SIZE FROM A MICROGRAPH

A chloroplast measures 42 mm long on a micrograph at × 6000.
Convert to µm: 42 mm × 1000 = 42 000 µm.
Actual size = image size ÷ magnification = 42 000 ÷ 6000 = **7.0 µm**.
With a scale bar instead: measure the bar in mm, convert to µm, and divide by the value written on the bar to get the magnification first.

MARK-SCHEME LANGUAGE

- Amyloplast: spelling is marked leniently, but "starch grain" is **wrong** when the organelle is asked for.
- Plasmodesma: if you name pits, the cell wall or the middle lamella instead, you can still get the function mark (ECF) when the function matches.
- Vacuole: a label line to the tonoplast is accepted.

Often asked as: name the structure labelled X (amyloplast in cassava, plasmodesmata between two cells, a chloroplast, pits in a xylem wall, a mesophyll vacuole), then give its function; magnification or actual-size calculations from the image.

COMMON MISTAKE

Not recognising plasmodesmata: a significant minority missed them (WBI12 Oct 2020 report).
In magnification sums, measuring the specimen when the question wants the scale bar, or forgetting to convert mm to µm (× 1000).

BEYOND THE SPEC, BUT EXAMINED

Recognising a **pit** as a gap in a lignified xylem wall in a micrograph, and saying it allows sideways movement of water (WBI12 Oct 2023).

SPEC 4.3

Starch and cellulose

WHAT THIS SPEC POINT ASKS

Know the structure and function of starch and cellulose, including how hydrogen bonds between β-glucose molecules form cellulose microfibrils.

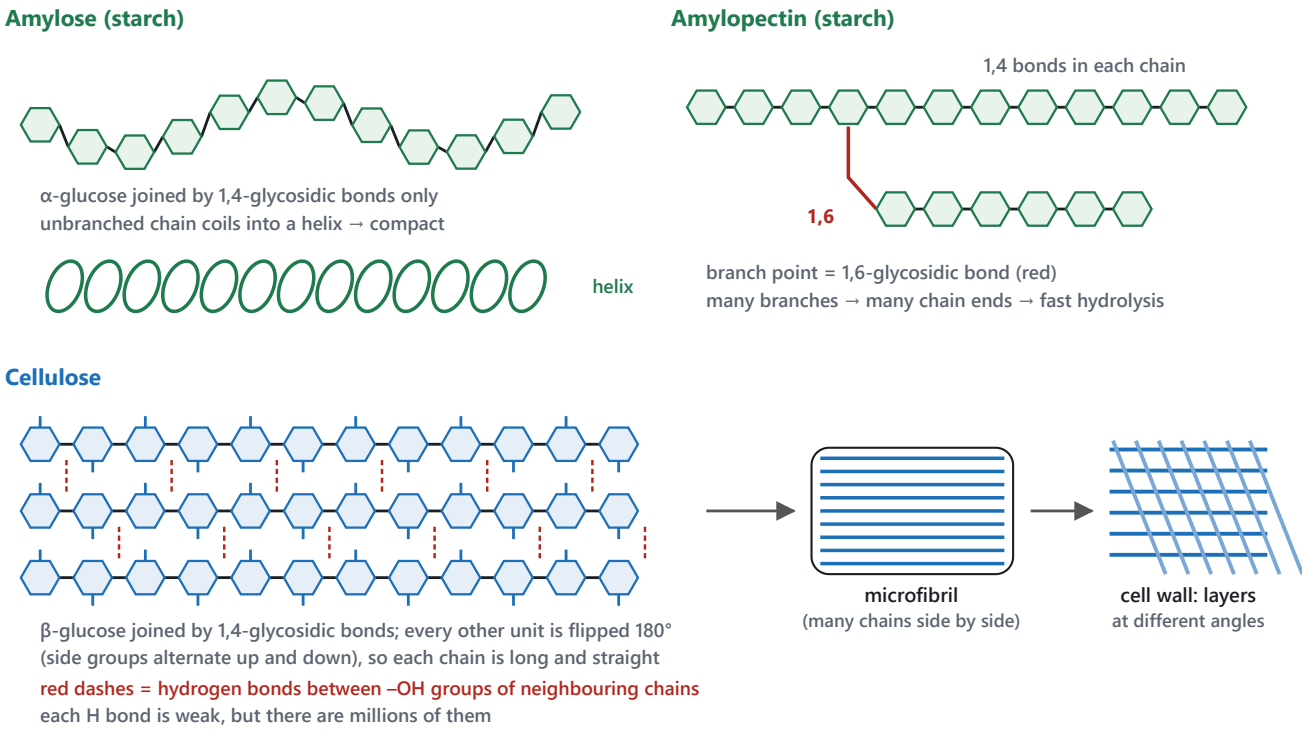
4.3 Starch: the plant's energy store

- Starch is a mixture of two polymers of **α-glucose** (Unit 1, 1.2):
  - **Amylose**: **1,4-glycosidic bonds** only, so an unbranched chain that coils into a **helix**. Very compact.
  - **Amylopectin**: 1,4 bonds **and 1,6-glycosidic bonds** at the branch points, so **branched**.
- It is stored as grains in **amyloplasts** and in chloroplasts.
- Why it is a good store, as feature → consequence pairs:
  - Made of **glucose** → hydrolysed to glucose for **respiration**.

- **Compact** (coiled, branched) → a lot of glucose stored in a small space.
- **Insoluble** → **doesn't affect the water potential** of the cell, so no osmotic water uptake; and it can't diffuse out.
- **Large** → can't cross the membrane, so it stays in the cell.
- **Branched** (amylopectin) → many chain ends where enzymes can work at once → **rapid hydrolysis** to glucose.

4.3 Cellulose: the structural polysaccharide

- Cellulose is a polymer of **β-glucose**. In β-glucose the –OH on carbon 1 points **up** (above the ring); in α-glucose it points down.
- To join carbon 1 of one β-glucose to carbon 4 of the next by a **1,4-glycosidic bond**, **every other molecule is turned over (rotated 180°)**.
- That flipping makes each chain **long, straight and unbranched**, with –OH groups sticking out on both sides.
- The –OH groups of neighbouring parallel chains form **hydrogen bonds** with each other (δ+ H of one –OH attracted to the δ− O of another). One hydrogen bond is weak, but there are huge numbers of them, so together they hold the chains firmly side by side.
- Bundles of many chains held together like this form a **microfibril**. Microfibrils are very strong in tension (high **tensile strength**), insoluble, and hard to break down.
- In the cell wall, microfibrils lie in a matrix of other polysaccharides and are laid down in layers (4.4).



Starch (α-glucose) coils or branches; cellulose (β-glucose) forms straight chains held side by side by hydrogen bonds, which bundle into microfibrils.

4.3 Starch vs cellulose vs glycogen

	Starch (amylose + amylopectin)	Cellulose	Glycogen
Monomer	α-glucose	**β-glucose**	α-glucose
Glycosidic bonds	1,4 (amylose); 1,4 and 1,6 (amylopectin)	**1,4 only**	1,4 and 1,6
Branching	amylose none; amylopectin branched	**none**	**highly** branched
Shape	helix (amylose); branched (amylopectin)	**straight** chains; alternate units flipped 180°	compact, branched granule
Hydrogen bonds	within the amylose helix	**between neighbouring chains** → microfibrils	within the molecule
Soluble?	no	no	no
Job	**energy store** in plants	**structural**: cell walls	**energy store** in animals and fungi
Where	amyloplasts, chloroplasts	plant cell walls	liver and muscle cells
Broken down by	amylase (1,4 bonds) and other enzymes	**cellulase** (most animals don't make it)	enzymes in liver and muscle cells

EXAM TIP

"Compare starch and cellulose" needs at least one **similarity** as well as differences, written as pairs:

- Both are **polysaccharides / polymers of glucose** joined by **(1,4) glycosidic bonds**.
- Starch is made of **α**-glucose **whereas** cellulose is made of **β**-glucose.
- Starch has 1,4 **and** 1,6 bonds (branched amylopectin) **whereas** cellulose has 1,4 bonds only (unbranched).

MARK-SCHEME LANGUAGE

- Starch for storage: "polymer of / contains glucose" (glucose is the energy source); "branched / 1,6 bonds, **therefore** can be rapidly hydrolysed"; "compact, so more {glucose / energy} stored"; "insoluble, therefore does not affect osmosis". "Easy to break down" is ignored.
- Starch: α-glucose; amylose and amylopectin; 1,4 **and** 1,6 bonds. β is rejected.
- Cellulose: "β-glucose (alternate molecules inverted)", "1,4 glycosidic bonds only", "hydrogen bonds between (adjacent) cellulose molecules", "forms microfibrils".
- How cellulose makes a wall strong: "hydrogen bonds between adjacent chains", "{layers / sheets} of microfibrils", "microfibrils at different angles" (criss-cross or mesh accepted).
- Compare: full marks need a similarity. Separate paragraphs on each molecule are **not pieced together**.

Often asked as: MCQ on starch or cellulose statements; identify a cellulose chain or a branched starch from drawings; compare cellulose and starch (3 marks); explain how starch suits energy storage; explain how cellulose and microfibrils strengthen the wall, using a micrograph.

COMMON MISTAKE

Confusing a **cellulose molecule** (one chain) with a **microfibril** (many chains hydrogen-bonded together) (WBI12 Oct 2020 report).
Writing "easier to break down" instead of "**faster hydrolysis**" (WBI12 Oct 2021 report).
Giving "insoluble" when the question asks why a store holds a lot of **energy**. That needs "polymer of glucose" and "compact" (WBI12 Jun 2024).
Saying animals can't digest cellulose because it has 1,4 bonds. Starch has 1,4 bonds too. The difference is **β**-1,4 bonds, which need **cellulase**.
Writing about pectin, hemicellulose or lignin when the question asks only about cellulose and microfibrils.

BEYOND THE SPEC, BUT EXAMINED

- Recognising **β-glucose** in a structural formula (–OH on carbon 1 above the ring; alternate units inverted in a chain) (WBI12 Jan 2020).
- Cellulose is in the walls of xylem, phloem **and** sclerenchyma (WBI12 Oct 2020).
- Picking out the **branched** (amylopectin) drawing and naming the 1,6 bond (WBI12 Jun 2023).

SPEC 4.4

Microfibrils, secondary thickening and the strength of plant fibres

WHAT THIS SPEC POINT ASKS

Explain how the way cellulose microfibrils are arranged, plus secondary thickening, gives xylem vessels and sclerenchyma fibres the physical properties that make plant fibres useful to humans.

4.4 How a cell wall is built

- **Primary wall**: laid down first, while the cell is still growing. Thin and flexible; microfibrils form a loose **mesh** in a matrix of pectins and hemicelluloses, so the wall can stretch.
- **Secondary wall**: added **inside** the primary wall once the cell stops growing (only in some cells, e.g. xylem vessels and sclerenchyma fibres). Microfibrils are laid down in several **layers**, and the microfibrils in each layer run at a **different angle** from the layer next to it.
- **Secondary thickening** = this extra secondary wall, usually impregnated with **lignin**.
- **Lignin** is a complex polymer that fills the spaces between microfibrils. It makes the wall **rigid** (resists compression and stops the cell collapsing) and **waterproof** (impermeable). The cell contents die, leaving a hollow tube.
- Where lignin is missing, **pits** remain, so water can still move sideways.
- In xylem, lignin is often laid down as **rings, spirals** or a net. This lets young xylem stretch as the stem grows and the stem bend without the tube collapsing.

4.4 Why this gives strength

Feature	Physical property it gives
Many **hydrogen bonds** between cellulose chains	each microfibril has high **tensile strength** (hard to pull apart)
Layers of microfibrils at different angles (criss-cross)	strong **in many directions**, not just along one line
Lignin between microfibrils	**rigid**, resists compression, **waterproof**; walls don't collapse under the tension in the xylem
Long cells overlapping end to end	long, strong **fibres** that can be extracted
Gaps between microfibrils (and pits)	water and ions can still pass through the wall

4.4 How humans use plant fibres

Fibre	What it is	Uses
Cotton	seed hairs: almost pure cellulose, no lignin	cloth (textiles)
Flax (linen), **hemp**, **jute**	**sclerenchyma fibres** from the stem	cloth, rope, sacking, fibre composites
Sisal	fibres from the leaves	rope, twine, matting
Wood	mostly xylem and fibres	timber, paper (pulped wood)

Fibres are separated from the rest of the stem by **retting**: microorganisms (or chemicals) break down the pectin of the middle lamella, while the cellulose and lignin of the fibres resist.

MARK-SCHEME LANGUAGE

- Xylem wall properties: "cellulose molecules organised into microfibrils"; "layers of microfibrils at different angles / a mesh"; "{hydrogen bonds / lignin / secondary thickening} give {strength / support}"; "microfibrils held together by {pectin / pectate}"; gaps let water and ions through.
- Lignin: "makes the wall impermeable to water (waterproof)", "provides strength", laid down as rings or spirals.
- Xylem for **support**: "walls have secondary thickening / lignin" + strength. For **transport**: "hollow tubes with no {end walls / cytoplasm}" + "no obstruction to the water column".
- Climbing plant v sunflower: the climber "is supported by the pole" and "needs to be **flexible** to bend round it", so needs less lignin; the sunflower must hold itself up.

Often asked as: explain the physical properties of a xylem vessel wall; role of secondary thickening in sclerenchyma; why a fibre such as jute is strong; interpret cellulose and lignin data (bamboo with age).

COMMON MISTAKE

Leaving out "lignified walls / secondary thickening" in support answers (WBI12 Jan 2020 report).
Writing "hollow" without saying "no end walls / no cytoplasm".
Giving **functions** when **physical properties** are asked (WBI12 Oct 2023), or treating lignin as something that carries water rather than giving strength and waterproofing (WBI12 Jan 2024).
Saying cellulose fibres resist **compression**. Cellulose gives **tensile** strength; lignin resists compression.
Saying lignin strengthens the cell **membrane**. It is in the **wall** (WBI12 Oct 2019).

BEYOND THE SPEC, BUT EXAMINED

- **Bamboo**: as the stem ages, the % cellulose falls and the % lignin rises; ratio cellulose : lignin about 2 : 1 at one year (WBI12 Jun 2025).
- **Jute** has high tensile strength because of its high cellulose and lignin content, and because its fibres are twisted or woven (WBI12 Oct 2025).
- **Tracheids** (long xylem cells with tapering ends) are accepted as support cells like sclerenchyma (WBI12 Jun 2021).
- **Capillary action** in the narrow xylem lumen helps water rise (WBI12 Jan 2020).

SPEC 4.5

Sclerenchyma, xylem and phloem

WHAT THIS SPEC POINT ASKS

Know how sclerenchyma fibres (support), xylem vessels (support, and transport of water and mineral ions) and phloem (translocation of organic solutes) are alike and different in structure, where they sit in the stem, and what they do.

4.5 Xylem vessels

- Formed from a column of cells joined end to end. Their **end walls break down** and their contents die, leaving a **continuous hollow tube** with **no cytoplasm**.
- Walls are **lignified** (secondary thickening in rings, spirals or a net) → support, waterproof, don't collapse under tension.
- **Pits** in the walls let water move sideways between vessels.
- Carry **water and mineral ions** from the roots **up** to the leaves (one direction) in the transpiration stream; also **support** the plant.

4.5 Phloem

- **Sieve tube elements**: **living** cells joined end to end. The end walls are perforated to form **sieve plates**. Each has **no nucleus**, very few organelles and only a thin layer of cytoplasm, so sap can flow through.
- Walls are **cellulose only, not lignified**.
- **Companion cells** lie beside each sieve tube element, linked to it by **plasmodesmata**. They have a nucleus, dense cytoplasm and **many mitochondria**: they supply the ATP for **active loading of sucrose** and keep the sieve tube element alive.
- Phloem carries **organic solutes**, mainly **sucrose** and **amino acids**, from **sources** (e.g. leaves) to **sinks** (growing points, roots, storage organs, fruits). This is **translocation**, and it goes **both up and down** the plant.

4.5 Sclerenchyma fibres

- Long, narrow cells with **tapering, closed ends** (end walls), overlapping in bundles.
- Very thick **lignified secondary walls**, a small empty lumen, some simple pits. **Dead** when mature.
- **Support only**: they don't transport anything.

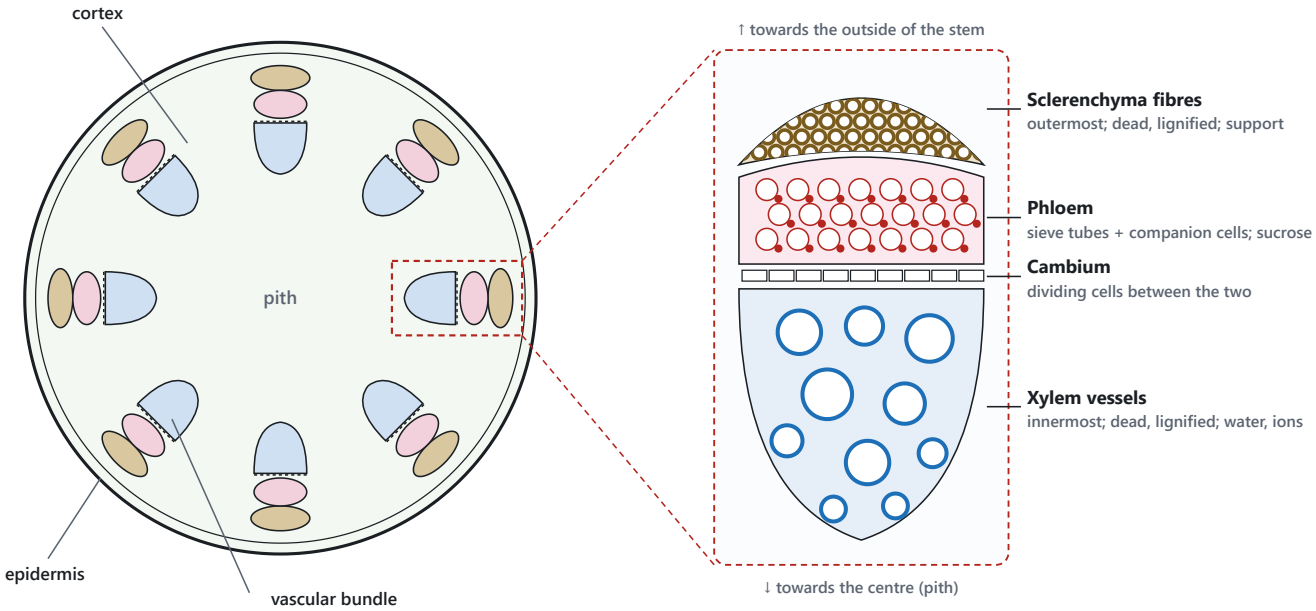
4.5 Comparison table

	Sclerenchyma fibre	Xylem vessel	Phloem sieve tube
Living or dead	dead	dead	**living** (with companion cells)
Contents	none (hollow)	none (hollow)	thin cytoplasm, **no nucleus**
End walls	**present** (tapered, closed ends)	**absent** (continuous tube)	**sieve plates** (perforated end walls)
Lignin / secondary thickening	yes, very thick	yes (rings, spirals, net)	**no**
Wall material	cellulose + lignin	cellulose + lignin	cellulose
Links to other cells	pits	**pits**	**plasmodesmata** (to companion cells)
Function	**support**	**water and mineral ions** (upwards only) + support	**sucrose, amino acids** (both directions)
Position in a stem	**outermost** of the three, a cap on the outside of each vascular bundle	**inner** side of the vascular bundle, towards the centre	**outer** side of the vascular bundle, between xylem and sclerenchyma

Similarities to remember: all three are long, narrow cells with **cellulose** walls. Xylem vessels and sclerenchyma fibres are both **dead, hollow, lignified** and give **support**, and both have **pits**. Xylem and phloem are both **tubes for transport** in the vascular bundles, and neither a mature xylem vessel nor a sieve tube element has a **nucleus**.

4.5 Where they are

- **Stem**: vascular bundles in a **ring** near the outside. In each bundle, from outside to inside: **sclerenchyma** → **phloem** → **cambium** → **xylem**. Pith in the centre.
- **Root**: xylem in the **centre**, often star-shaped, with phloem in the gaps between its arms (all inside the endodermis). A central core resists pulling forces as the root anchors the plant.
- **Leaf**: in each vein and the midrib, **xylem towards the upper surface** and **phloem towards the lower surface**.



Stem in transverse section. In each vascular bundle the sclerenchyma is outermost, then phloem, cambium and, nearest the centre, xylem.

MARK-SCHEME LANGUAGE

- Phloem: "{transport / translocation} of {sucrose / amino acids}" "from {source / leaves} to {sink / site of use or storage}". "Glucose", "nutrients" and "sugar" are **ignored**.
- Xylem: "transport of water and mineral ions" and support. Sclerenchyma: "support" ("strength" and "rigidity" alone are ignored).
- Xylem v sclerenchyma: similarities "both contain {cellulose / pits / lignin / dead cells}", "both hollow / no cytoplasm". Difference: "sclerenchyma has end walls (closed, pointed ends) **whereas** xylem does not".
- Phloem v xylem (structure): similarities "both have cellulose walls", "both tubular", "both have no nucleus". Differences: "phloem has sieve plates", "phloem contains cytoplasm, so not hollow", "phloem has no lignin / secondary thickening", "phloem has plasmodesmata whereas xylem has pits"; living v dead accepted.
- Phloem adapted: "continuous tube / sieve plates let the solution flow", "plasmodesmata", "companion cells provide ATP for active loading of sucrose".
- Root v stem: "in the stem xylem and phloem are in separate vascular bundles arranged in a ring; in the root they are in a central group". Use comparative language and say which tissue.

Often asked as: label a stem section (sclerenchyma, phloem, cambium, xylem); compare xylem with sclerenchyma or with phloem; describe tissue positions in root, stem and leaf; how phloem structure suits its job; dye taken up by a celery stalk; a parasitic plant or a millipede feeding from the vascular tissue.

COMMON MISTAKE

Swapping **sclerenchyma** and **phloem** when labelling, or letting a label line touch the wrong tissue (WBI12 Jan 2023, Jan 2020 reports).
Thinking both xylem and phloem are lignified, or that xylem has end walls or sieve plates (WBI12 Oct 2019, Jun 2022 reports).
Writing "glucose" or "nutrients" for what phloem carries. It is **sucrose** (and amino acids).
Giving functions when the question asks for **structure** (WBI12 Jun 2022 report).

BEYOND THE SPEC, BUT EXAMINED

- Phloem moves sap **both up and down**; xylem moves water **upwards** only (WBI12 Jan 2021).
- **Calcium ions** travel up the plant in the **xylem** (WBI12 Jan 2021).
- A parasitic plant grows into the host's **phloem** and absorbs **sucrose** by diffusion, facilitated diffusion or active transport (**not** osmosis) (WBI12 Jun 2023).
- A dye placed in water travels up a celery stalk in the **xylem**, so only the xylem is stained (WBI12 Jun 2023).
- Recognising **parenchyma** (thin-walled living packing cells) in a stem section (WBI12 Jan 2021).

SPEC 4.6

Core Practical 7: plan diagrams and cell drawings of plant tissues

WHAT THIS SPEC POINT ASKS

Use a light microscope to observe transverse sections of roots, stems and leaves; draw and label plan diagrams of them; draw and label cells of plant tissues; and identify sclerenchyma fibres, phloem, sieve tubes and xylem vessels and where they are.

4.6 Plan diagram or cell drawing?

	Plan diagram (low power)	Cell drawing (high power)
Shows	**outlines of tissues** only: where each tissue starts and ends	a **few individual cells** of one tissue, with their walls (and nucleus etc. if visible)
Never shows	individual cells	the whole section
Use it to	show the **position** of xylem, phloem, sclerenchyma	show the **structure** of e.g. xylem vessels or sieve tubes

CORE PRACTICAL 7: METHOD IN BRIEF

1. Wet a razor blade or scalpel and cut **very thin** transverse sections of a stem (or root) on a white tile, cutting away from your fingers. For a leaf, hold it between two pieces of polystyrene or carrot and shave thin slices.
2. Move the sections with a soft paintbrush into water in a watch glass. Choose the **thinnest** one: a thin partial section is better than a thick whole one (thick sections let little light through and cells overlap).
3. Put it on a slide and add a drop or two of stain, e.g. **toluidine blue O** for 1–2 minutes, then rinse or blot off the excess.
4. Lower a **coverslip** at an angle from one side to avoid air bubbles.
5. Focus on **low power** first and draw a **plan diagram** of the tissue layout. Then switch to **high power** and draw a few cells of one tissue.
6. Use an **eyepiece graticule** calibrated with a stage micrometer to measure sizes, and add a magnification or scale.
   Prepared permanent slides are an acceptable alternative.

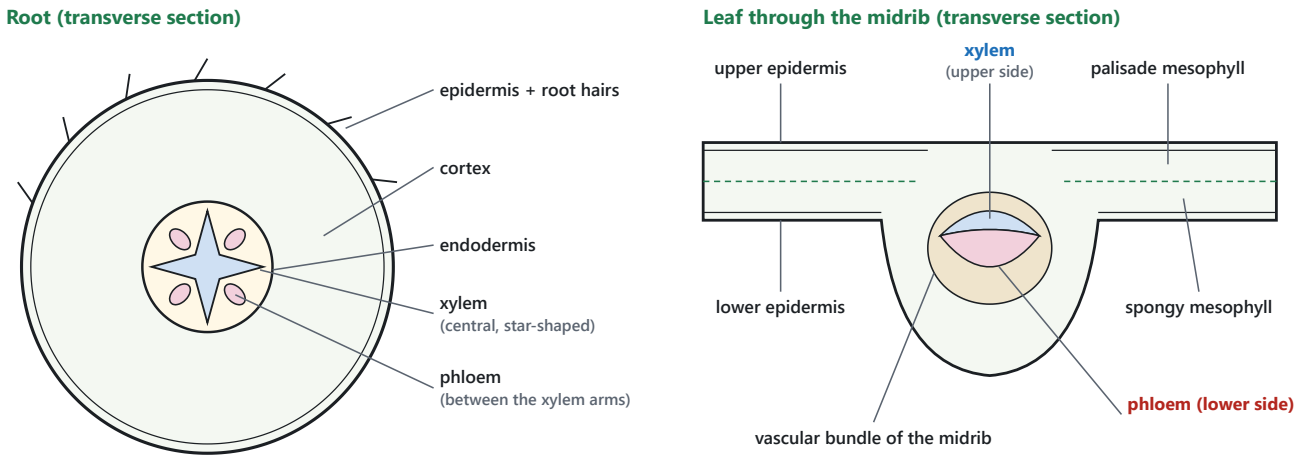
4.6 Stains and what they show

- **Toluidine blue O** stains **lignified** walls (xylem, sclerenchyma) **blue-green** and unlignified walls (phloem, parenchyma) **pink-purple**. Staining adds contrast, because thin unstained sections absorb very little light.

- (Other stains you may meet: **phloroglucinol** with acid stains lignin pink-red; **iodine** stains starch grains blue-black.)

4.6 Identifying the tissues

- **Xylem vessels**: the **largest**, rounded, empty cells with thick (lignified) walls, on the inner side of each bundle in a stem and in the centre of a root.
- **Sclerenchyma fibres**: small cells with **very thick** walls and tiny lumens, in a cap on the outside of each stem bundle.
- **Phloem**: a patch of **small, thin-walled** cells between the xylem and the sclerenchyma; in high power you may see **sieve plates** and small, densely stained **companion cells**.



Root: xylem forms a central star with phloem between its arms. Leaf: in the midrib and veins, xylem is on the upper side and phloem on the lower side.

4.6 Drawing rules (both kinds of drawing)

- Sharp **pencil**, **clear continuous lines**, **no shading** or colouring.
- Draw **large** (use at least half the space) and keep the **proportions** right.
- Plan diagrams: tissue boundaries only, **no cells**.
- Label lines drawn with a **ruler**, ending **exactly on** the structure, not crossing each other, no arrowheads. Labels outside the drawing.
- Add a **title** and the **magnification** (or a scale bar).
- Draw what you **see**, not what the textbook shows.

4.6 Safety

- Razor blades and scalpels: cut **away** from your fingers onto a tile; hand back or put in a sharps container.
- Toluidine blue stains skin and clothing: gloves, wipe up spills.
- Never point a microscope mirror at direct sunlight.

WORKED EXAMPLE: CALIBRATING A GRATICULE

Eyepiece graticule: 40 divisions line up with 0.1 mm (100 µm) on the stage micrometer.
One division = 100 ÷ 40 = **2.5 µm** at this objective.
A xylem vessel spans 18 divisions: diameter = 18 × 2.5 = **45 µm**.
Recalibrate if you change the objective lens.

MARK-SCHEME LANGUAGE

- Label lines must not touch the cambium line or the wrong tissue (WBI12 Jan 2020 report).
- A labelled diagram of the stem is accepted as part of an answer comparing root and stem arrangements.

Often asked as: the full plan diagram has not been set in WBI12 papers from 2019 to 2026, but its skills have: labelling xylem, phloem and sclerenchyma on a stem micrograph, comparing tissue arrangement in a root and a stem, and a stained celery section.

COMMON MISTAKE

Drawing individual cells in a **plan** diagram, or shading.
Few candidates knew where the tissues are in a root or a stem (WBI12 Oct 2021 report): learn the stem order (sclerenchyma, phloem, cambium, xylem) and the central root core.
Air bubbles mistaken for structures: lower the coverslip at an angle.

SPEC 4.7 Plant fibres, starch and sustainability

WHAT THIS SPEC POINT ASKS

Explain how using plant fibres and starch can help sustainability, including plant-based products that replace oil-based plastics.

4.7 What "sustainable" means

A resource is used **sustainably** if we can meet today's needs **without** using it up, so it is still **available to future generations**. Plants are **renewable**: more can be grown. Crude oil is **non-renewable** (finite): once used, it is gone.

4.7 Plant products that can replace oil-based ones

- **Plant fibres** (hemp, jute, flax, sisal, cotton) for rope, cloth, sacks and building materials, instead of nylon and polyester made from oil.
- **Bioplastics** made from **starch** (e.g. from maize, potatoes, cassava) or **cellulose** (wood pulp), instead of oil-based plastics such as polythene and PVC. Examples: thermoplastic starch, PLA (polylactic acid, made by fermenting sugars from maize starch or sugarcane), cellulose films.
- **Bioethanol** made by fermenting sugars from starch or sugarcane, used as fuel instead of petrol.
- Waste plant material (sugarcane bagasse, wheat bran, seed husks, avocado seeds) can be turned into plates, cutlery or packaging.

4.7 Why plant-based products are more sustainable

Plant-based	Oil-based
Renewable: more crops can be grown each year	**Non-renewable**: oil will run out
Biodegradable: broken down by decomposers (bacteria and fungi)	persist for hundreds of years in landfill and the sea
Roughly **carbon neutral**: the CO_2 released when they rot or burn was absorbed by photosynthesis as the crop grew	extracting and burning oil adds CO_2 that was locked away, **increasing atmospheric CO_2**
Less energy needed to make many of them	energy-intensive refining

4.7 The drawbacks (needed for "discuss" or "evaluate")

- Crops for fibres, plastics and fuel compete with **food crops** for land, water and fertiliser.
- Growing them needs land (risk of **deforestation**, monocultures that lower biodiversity), fertilisers (**eutrophication**) and pesticides, and machinery that burns fuel, so they are not fully carbon neutral.
- Bioplastics may be weaker, less water-resistant or more expensive. Many only break down in industrial compost, not in soil or the sea.
- Plant ropes are often weaker than synthetic ones and can rot.

EXAM TIP

Every "more sustainable" point needs a reason: "renewable, **so** more can be grown and it won't run out for future generations"; "biodegradable, **so** decomposers break it down"; "carbon neutral, **so** no net increase in atmospheric CO_2". The word "sustainable" on its own scores nothing.

MARK-SCHEME LANGUAGE

- Sustainable: "renewable / more can be grown / will not run out / available to future generations"; "biodegradable / broken down by decomposers"; "carbon neutral".
- Oil-based not sustainable: "non-renewable (finite)", "takes a long time to decompose", "adds to CO_2 / not carbon neutral".
- **Ignored**: "recyclable", "less pollution" with no detail, "sustainable" with no explanation, "more easily broken down" with no detail. "Not recyclable" is not credited for oil-based plastics.
- Packaging with antimicrobial plant extracts (e.g. oregano): "antimicrobial property reduces bacteria on the food", so a longer shelf life and fewer food-poisoning cases.

Often asked as: explain why a named plant product (hemp, wheat bran, avocado seed, a calabash bowl, wood packaging) is more sustainable than an oil-based one; a 6-mark comparison of bioethanol crops; percentage calculations from product data.

COMMON MISTAKE

Giving "more sustainable" with no reason (WBI12 Jun 2023 report), or leaving out the **future generations** idea (WBI12 Oct 2019 report).
Writing "less pollution" or "recyclable". Neither earns a mark.
In a 6-mark evaluation, listing pros and cons without linking them to sustainability (capped at Level 1).

BEYOND THE SPEC, BUT EXAMINED

- **Bioethanol crops** 6-mark: compare water, fertiliser and pesticide needs and CO_2 output of crops such as hemp, rapeseed and soybean; consider the farmer, the population and the ecosystem; reach a supported conclusion (WBI12 Oct 2019).
- **Starch bioplastic** strength: more starch → higher tensile strength (more hydrogen bonds between starch molecules); more **glycerol** (a plasticiser) → lower tensile strength (WBI12 Jan 2026).
- Percentage of a percentage: e.g. 14% of 48 mg, then 43% of that answer, to 2 s.f. (WBI12 Oct 2020).

SPEC 4.8 Water and inorganic ions in plants

WHAT THIS SPEC POINT ASKS

Explain why plants need water and the inorganic ions nitrate, calcium and magnesium.

Roots take in water by osmosis and mineral ions mostly by **active transport** (using ATP) from the soil. Both travel up the plant in the **xylem**.

4.8 Water

- **Reactant in photosynthesis** (split in the light-dependent stage) and in **hydrolysis** reactions (e.g. starch → glucose).
- **Solvent**: substances dissolve in it so they can react and be **transported** (mineral ions in the xylem, sucrose in the phloem).
- **Turgor**: water entering vacuoles by osmosis makes cells **turgid**, which **supports** non-woody plants and leaves. Without enough water, cells become flaccid and the plant **wilts**. Turgor also opens stomata (guard cells).
- **Cooling**: evaporation of water from leaves (transpiration) cools them.

4.8 Inorganic ions

Ion	Used for	Deficiency symptoms
Nitrate (NO_3^-)	source of **nitrogen** to make **amino acids** → **proteins** (enzymes, membrane proteins), **nucleotides** → **DNA, RNA**, **ATP**, NADP, and **chlorophyll**	stunted growth; leaves turn yellow (**chlorosis**), starting with the **older** leaves
Calcium (Ca^{2+})	forms **calcium pectate** in the **middle lamella**, which holds neighbouring cells together, and in the wall matrix; also needed for membrane function	growing tips and **young** leaves die back or are deformed; weak walls; soft fruit
Magnesium (Mg^{2+})	part of every **chlorophyll** molecule, so needed for light absorption and photosynthesis; also activates some enzymes (ATP works bound to Mg^{2+})	yellowing between the veins, starting with **older** leaves; less photosynthesis, so slower growth

Older leaves show nitrogen and magnesium shortage first because the plant moves these ions out of old leaves into new growth. Calcium can't be moved once it is in a wall, so shortage shows in the **youngest** tissue.

EXAM TIP

Follow the chain to the end: "magnesium → **chlorophyll** → light absorbed → **photosynthesis** → glucose made → growth". "Magnesium is needed for photosynthesis" stops too early, and "calcium pectate" with no job scores nothing. Say what it is **for**: "calcium pectate **holds cells together** in the middle lamella".

RECOMMENDED PRACTICAL: PLANT MINERAL DEFICIENCIES

1. Grow seedlings of the same species, age and size in **culture solutions**: one complete (all ions, the **control**) and others each lacking one ion (e.g. no nitrate, no magnesium, no calcium).
2. Keep everything else the same: volume of solution, light, temperature, aeration of the solution, time.
3. Cover containers so that algae don't grow, and top up with distilled water.
4. After a few weeks, record **leaf colour**, number of leaves, shoot length, and **dry mass** (dry at 80–100 °C to constant mass) or chlorophyll content.
5. Use several plants per solution and compare means (with error bars or standard deviation).

MARK-SCHEME LANGUAGE

- Calcium: "calcium pectate to {hold cells together / form the middle lamella / increase cell wall stability}". "Makes cell walls" and "pectate" alone are ignored. Higher-level answers add that calcium pectate holds microfibrils together in the wall matrix.
- Magnesium: "to make chlorophyll so {photosynthesis can occur / light energy is absorbed / glucose is made}". "Plant growth" and "chloroplasts" are ignored. Mg^{2+} activating enzymes is accepted.
- Nitrate: "to make {amino acids / proteins / DNA / RNA / nucleic acids / chlorophyll}", naming the compound **and** its role (e.g. "proteins for enzymes").
- Water: "for {hydrolysis / photosynthesis}", "as a solvent / to transport named substances", "for turgidity". "Osmotic effects" alone is ignored.
- Data questions: state the trend with both variables, compare with the **control**, and say **which** error bars overlap (no significant difference) or don't (significant difference).

Often asked as: explain why a plant needs Ca^{2+}, Mg^{2+} or NO_3^-; why carnivorous plants catch insects (nitrogen for amino acids and nucleic acids); calcium and fruit firmness; 6-mark questions on water stress, eggshell calcium fertiliser or hydroponic magnesium.

COMMON MISTAKE

Describing deficiency symptoms when the question asks for the ion's **role** (WBI12 Jun 2019 report).
Stopping at "amino acids" or "proteins" for nitrate with no further role (WBI12 Oct 2019 report).
Writing "food" instead of glucose, or "calcium" with no link to calcium pectate.
Not comparing with the control group in data questions (WBI12 Jan 2025 report).

BEYOND THE SPEC, BUT EXAMINED

- Fewer Ca^{2+} ions (e.g. a chemical such as EDTA or verapamil that reduces uptake) → less calcium pectate → weaker walls and a weaker middle lamella (WBI12 Jan 2025).
- Ions enter roots by **active transport** (WBI12 Jun 2025).
- Water stress 6-mark: less water → stomata close → less CO_2 in → less photosynthesis; smaller roots take up fewer ions; less cellulose → thinner, weaker walls; lower growth and yield (WBI12 Jun 2024).
- Nitrogen-fixing bacteria in root nodules (e.g. alfalfa) supply the plant with nitrogen compounds (WBI12 Jan 2026).
- Percentage difference: follow the formula the question gives (one paper used (A − B) ÷ (A + B) × 100) (WBI12 Jun 2025).

SPEC 4.9 Core Practical 8: tensile strength of plant fibres

WHAT THIS SPEC POINT ASKS

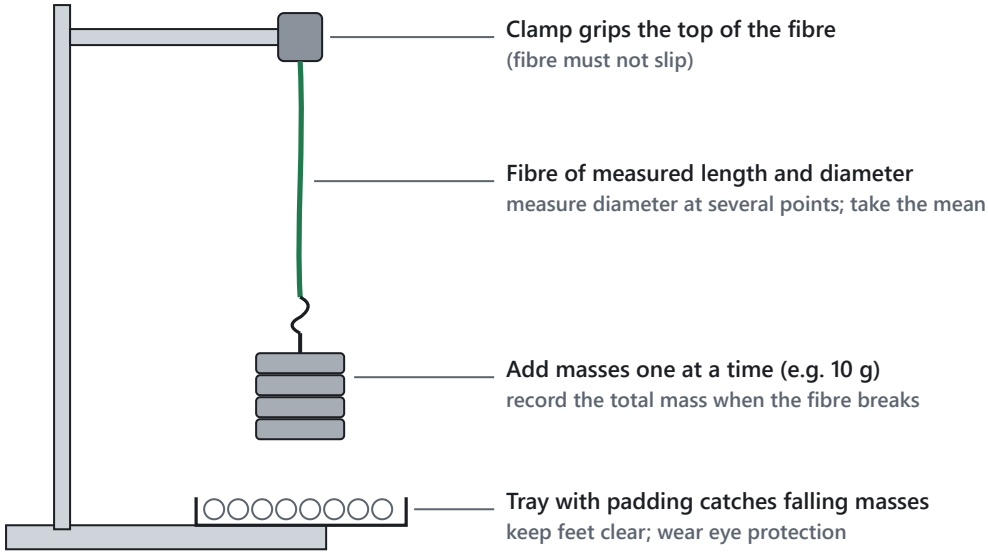
Measure the tensile strength of plant fibres.

4.9 Principle

Tensile strength is the **stretching force a material can withstand before it breaks, per unit cross-sectional area**:

$$\text{tensile strength} = \text{breaking force (N)} \div \text{cross-sectional area (m}^2\text{)}$$

Units: N m^{-2} (= Pa), or N mm^{-2}. A hanging mass gives a force: **force (N) = mass (kg) × 9.81**. Dividing by area lets you compare thick and thin fibres fairly.



Set-up for measuring breaking load. Masses are added one at a time until the fibre snaps.

CORE PRACTICAL 8: METHOD IN BRIEF

1. Extract fibres (e.g. from celery, nettle, flax or banana stems) by scraping or by **retting** (soaking so microbes break down the soft tissue). Choose fibres with no visible damage.
2. Cut each fibre to the **same length** (e.g. 10 cm). Measure its **diameter** at several points with a micrometer or a microscope with a graticule, and take the mean.
3. Clamp the top of the fibre to a retort stand (or tie it round a bar), and hang a mass hanger from the bottom. Put a **tray with padding** underneath.
4. Add masses **one at a time** (e.g. 10 g), waiting a few seconds after each, until the fibre **breaks**. Record the total mass.
5. Repeat with at least three fibres of each type and calculate the mean breaking mass.
6. Convert to force (× 9.81, with mass in kg) and divide by the cross-sectional area (πr^2) to get tensile strength.

4.9 Variables

- **Independent**: type of fibre (or plant species, age, treatment).
- **Dependent**: breaking mass / force, then tensile strength.
- **Control**: fibre length, diameter (or correct for it with area), the part of the plant it came from, its age and how fresh or dry it is, the size of each mass added, the time between additions.

4.9 Safety

Wear **eye protection** (a snapping fibre can whip up). Keep feet clear of falling masses and use the padded tray. Take care with knives or scalpels when extracting fibres.

WORKED EXAMPLE: TENSILE STRENGTH FROM BREAKING MASS

A fibre breaks when the total mass is 450 g. Its mean diameter is 0.80 mm.
Force = 0.450 kg × 9.81 = **4.41 N**.
Radius = 0.80 ÷ 2 = 0.40 mm = 4.0×10^{-4} m.
Area = $\pi r^2 = \pi \times (4.0 \times 10^{-4})^2$ = **5.03×10^{-7} m²**.
Tensile strength = $4.41 \div 5.03 \times 10^{-7}$ = **8.8×10^{6} N m^{-2}** (2 s.f.).
Use the **radius**, not the diameter, and convert mm to m **before** squaring.

4.9 Errors and improvements

- Fibre slips in the clamp or breaks **at** the clamp → wrap the end or tie it round a smooth bar; ignore results where it breaks at the clamp.
- Mass steps too large → the true breaking load lies between two values; use **smaller masses** near the expected break.
- Masses added with a jerk add extra force → lower them gently.
- Fibres vary in diameter and have weak spots → measure diameter at several points, use more fibres, calculate means.
- Fibres dry out over time → test them all fresh, or all at the same time after extraction.

COMMON MISTAKE

Reporting the **breaking mass** as "tensile strength". It is only a breaking load until you convert to force and divide by area. Longer celery strings may seem "stronger" simply because they were thicker. Assuming celery "strings" are xylem. They are mainly strands of thick-walled supporting cells (collenchyma) running alongside the vascular bundles, so a dye test that stains the xylem doesn't show that the whole string is xylem.

MARK-SCHEME LANGUAGE

Not examined as a practical in WBI12 papers from 2019 to 2026. Tensile strength appears in explanations instead: why a fibre is strong (cellulose, hydrogen bonds, microfibrils at different angles, lignin), bioplastic strength data (starch and glycerol), and water stress reducing cellulose and therefore wall strength.

Often asked as: link tensile strength to microfibril structure; interpret strength data with error bars.

SPEC 4.10 Conditions bacteria need to grow

WHAT THIS SPEC POINT ASKS

Know what conditions bacteria need for growth.

Bacteria reproduce by **binary fission**; in good conditions some divide every 20–30 minutes. Each condition below matters because it affects the enzymes and the processes the cell needs to grow and divide.

Condition	Why it is needed
Nutrients	a **carbon source** (e.g. glucose) for **respiration** and to build molecules; a **nitrogen source** (amino acids, nitrate) to make **proteins** and nucleic acids; mineral ions (e.g. phosphate for DNA and ATP). Agar or broth provides these.
Water	solvent for reactions and transport; needed for **hydrolysis** reactions; bacteria in a dry place stop growing.
Suitable temperature	an **optimum** temperature for their **enzymes**. Too cold → slow reactions (little kinetic energy); too hot → enzymes **denature**.
Suitable pH	each bacterium's enzymes have an optimum pH; far from it the active sites change shape (ionic and hydrogen bonds disrupted).
Oxygen (some species)	**obligate aerobes** need O_2 for **aerobic respiration**. **Obligate anaerobes** are killed or inhibited by O_2. **Facultative anaerobes** grow with or without it.

- Photosynthetic bacteria (e.g. cyanobacteria) need **light**, CO_2 and mineral ions instead of a supply of glucose.
- In school labs cultures are grown at **25 °C** or below, **not 37 °C**, so that bacteria which grow well at body temperature (likely human pathogens) are not encouraged.

MARK-SCHEME LANGUAGE

- Each condition needs a reason: "oxygen for (aerobic) respiration" (or "no oxygen" for obligate anaerobes); "glucose for respiration / amino acids for protein synthesis"; "optimum temperature for {enzyme / metabolic} reactions" (or "so enzymes are not denatured"); "optimum pH for … reactions"; "water for {hydrolysis / as a solvent}".
- Photosynthetic bacterium: mineral ions (nitrate, phosphate), pH, light. Glucose, temperature, "nutrients" and "moisture" are ignored there.
- Oxygen graph: "as oxygen increases the growth rate of A decreases whereas B increases"; "B is an (obligate) aerobe"; "B levels off because another factor is limiting"; "high oxygen inhibits A (an anaerobe)".

Often asked as: explain the conditions needed (4 marks); describe and explain growth-rate graphs against oxygen, salt (NaCl) concentration or temperature; bacterial survival on wooden and plastic chopping boards.

COMMON MISTAKE

Listing conditions with no reasons: the commonest reason marks are lost (WBI12 Oct 2020, Jan 2021, Jun 2023 reports).
Saying high temperature "denatures the bacteria". It denatures their **enzymes** (WBI12 Jan 2021 report).
Giving **plant** growth conditions, or glucose for a photosynthetic bacterium.
Describing each bar separately instead of giving the overall trend first (WBI12 Oct 2022 report).

BEYOND THE SPEC, BUT EXAMINED

- **Thermophiles** have heat-stable enzymes with a high optimum temperature; **psychrophiles** grow best in the cold (WBI12 Jan 2023).
- High **salt** lowers the water potential outside the cell, so water leaves the bacterium by osmosis and growth slows (WBI12 Jun 2022).

SPEC 4.11 Plants as sources of antimicrobial and other medicinal substances

WHAT THIS SPEC POINT ASKS

Know that substances made by plants can have antimicrobial and other therapeutic (medicinal) properties.

4.11 Why plants make these chemicals

Plants can't move away from pathogens or herbivores. Many make chemicals that **kill or slow the growth of microorganisms** (antimicrobials: antibacterial and antifungal) or deter animals (e.g. bitter **tannins**). Humans can extract, purify and use these, or copy their structure.

4.11 Examples

Plant	Substance	Use
Willow bark	salicylic acid → modified to **aspirin** (acetylsalicylic acid)	pain relief, reducing fever and inflammation, preventing blood clots
Cinchona bark	**quinine**	treating malaria
Foxglove (*Digitalis*)	**digitalis** (digoxin)	heart failure and dropsy (4.13)
Pacific **yew**	**paclitaxel** (taxol)	some cancers
Sweet wormwood	**artemisinin**	malaria
Garlic, mint, tea tree, thyme, oregano	antimicrobial compounds (e.g. allicin in garlic)	tested on bacteria in CP9; antimicrobial packaging

4.11 From plant to medicine

1. Find a plant used in traditional medicine or one that resists disease.
2. **Extract** the active substance and **purify** it, so the **dose** can be measured and controlled.
3. Work out its structure, so it can be **synthesised** in a lab: no need to harvest huge amounts of a wild plant, and chemists can modify it to work better or have fewer side effects.
4. **Test** it for safety and effectiveness (4.13).

This is one reason to **conserve biodiversity**: plants not yet studied, especially endemic ones in threatened habitats, may contain useful drugs.

MARK-SCHEME LANGUAGE

- Zone of inhibition data: "extract {was effective / had antimicrobial properties} against all three bacteria"; "prevented growth of / killed bacteria around the disc"; "higher concentration → larger zone".
- Comparing two extracts: say which is better **against which bacterium**, then whether the SDs or error bars overlap (no significant difference) or not, then a method point (control disc, same volume and concentration of extract).
- Largest possible difference between two zones: upper limit of the bigger (mean + range or SD) minus lower limit of the smaller.
- A gel with no extract used in a wound trial acts as a **control** (like a placebo) for comparison. This is a **control**, not a controlled variable.

Often asked as: interpret zone-of-inhibition data for plant extracts; 6-mark data on ulcer treatments with antimicrobial plant extracts; wound-healing trials; calculate a standard deviation from zone diameters.

COMMON MISTAKE

Calling penicillin a plant product. It comes from a **fungus**.
Confusing a **control** (a group or disc without the active substance) with a **controlled variable** (WBI12 Jun 2023 report).
Saying a standard deviation "proves reliability", or that overlapping SDs show "accuracy". Overlap means **no significant difference** (WBI12 Jun 2023, Jan 2024 reports).

BEYOND THE SPEC, BUT EXAMINED

- **Standard deviation** by hand, $s = \sqrt{(\Sigma(x - \bar{x})^2 \div (n - 1))}$ (formula given). SD shows how spread out the values are about the mean, not a correlation (WBI12 Oct 2024).
- Named plant chemicals in stems: **tannins** (bitter, deter herbivores), **flavonoids** (antibacterial), fucoxanthin from seaweed (WBI12 Jun 2021, Jan 2022).

SPEC 4.12 Core Practical 9: antimicrobial properties of plants and aseptic technique

WHAT THIS SPEC POINT ASKS

Investigate whether plant extracts have antimicrobial properties, using aseptic techniques to handle bacteria safely.

4.12 Principle

Paper discs soaked in a plant extract are placed on agar covered evenly with bacteria. If the extract kills bacteria or stops them growing, it diffuses into the agar and leaves a **clear zone** (zone of inhibition) with no growth around the disc. A bigger zone suggests a more effective antimicrobial.

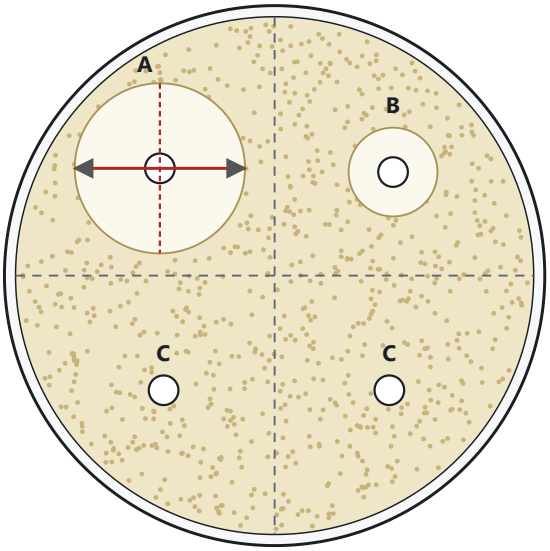
CORE PRACTICAL 9: METHOD IN BRIEF

1. Wash hands and **disinfect the bench** (e.g. 1% Virkon left for 10 minutes). Close windows and doors.
2. **Grind** each plant material (e.g. garlic, mint) with a pestle and mortar in the same volume of **ethanol** (to dissolve the active substances). Leave, then filter or decant the extract.
3. Soak sterile paper discs (all the same size) in each extract for the **same time**; let the ethanol **evaporate** on a sterile surface.

4. Mark the **base** of an agar plate already seeded with a lawn of bacteria (e.g. *Bacillus subtilis*, *Micrococcus luteus*) into quarters.
5. Lift the lid only slightly and use **sterile forceps** (flamed and cooled) to place one disc per quarter. Include a **control disc soaked in ethanol only** (evaporated).
6. **Tape the lid at two points** (not all the way round), **invert** the plate and incubate at **25 °C** for **24–48 hours**.
7. Measure the **diameter** of each clear zone across two directions at right angles, take the mean, and calculate the area. Repeat with more plates and find mean values.

4.12 Aseptic technique and why each step matters

Step	Reason
Disinfect bench; wash hands	kill microbes that could contaminate the culture
Work near a lit **Bunsen burner**	rising hot air carries airborne microbes **away** from the plate
Flame loops and forceps; **flame the neck** of broth bottles when opening and closing	kill microbes on surfaces that touch the culture
Sterile (autoclaved or single-use) plates, agar, pipettes, spreaders	no other microbes present at the start
Lift the lid only slightly and briefly	fewer airborne microbes fall in
Tape the lid at two points, not sealed all round	lets **oxygen** in, so harmful **anaerobic** bacteria don't grow; the plate can't be opened accidentally
Incubate **upside down**	condensation doesn't drip onto the agar and spread colonies
Incubate at **25 °C**, not 37 °C	avoids growing **human pathogens**
Never open plates after incubation; **autoclave** (121 °C) before disposal	protect people from any pathogens that grew



A, B = discs soaked in two plant extracts
C = control discs (solvent only, e.g. ethanol, evaporated)

Clear zone = no bacterial growth (inhibition)
Measure the diameter across two directions at right angles and take the mean (red lines on zone A)

Area of zone = πr^2, where r = mean diameter ÷ 2

Larger zone → more effective antimicrobial
(if concentration and diffusion are the same)

Lawn of bacteria spread evenly over the agar
Quarters marked on the BASE of the dish

Lid taped at two points (not sealed); dish inverted; incubated at 25 °C for 24–48 h

Discs A and B carry two plant extracts; C are solvent-only controls. Measure each clear zone across two diameters and take the mean.

4.12 Variables and controls

- **Independent**: plant extract (or its concentration). **Dependent**: diameter (or area) of the clear zone.
- **Control**: same bacterium and volume of culture, same agar depth, same disc size, same volume and concentration of extract, same soaking time, same incubation temperature and time.
- The **ethanol-only disc** shows the solvent and the disc themselves don't stop growth. An **antibiotic disc** can be added as a positive control for comparison.

WORKED EXAMPLE: AREA OF A CLEAR ZONE

Zone diameters measured across two directions: 18 mm and 20 mm. Mean = **19 mm**.
r = 19 ÷ 2 = 9.5 mm.
Area = πr^2 = π × 9.5^2 = **284 mm^2** (3 s.f.).
If the question asks for the area of clear agar only, subtract the disc: a 6 mm disc has area π × 3^2 = 28.3 mm^2, leaving 283.5 − 28.3 = 255 mm^2.

WORKED EXAMPLE: STANDARD DEVIATION OF ZONE DIAMETERS

Three plates give zone diameters of 7, 9 and 11 mm. Mean $\bar{x}$ = 27 ÷ 3 = **9 mm**.
$(x - \bar{x})$: −2, 0, +2. Squared: 4, 0, 4. $\Sigma(x - \bar{x})^2$ = 8.
s = √(8 ÷ (3 − 1)) = √4 = **2.0 mm**. Remember the **n − 1** and the **square root**.

4.12 Limitations

- Different compounds have different-sized molecules, so they **diffuse** through agar at different rates: a small zone doesn't always mean a weak antimicrobial.
- The concentration of active substance in each extract is unknown and may differ.
- Zones may be irregular (measure two diameters, or use area); discs placed too close overlap.
- An agar plate is not skin or a gut: an extract that works here may be broken down in the body or be toxic.

MARK-SCHEME LANGUAGE

- The method appears in question stems: agar plate spread with bacteria, discs soaked in extract, incubated at 25 °C for 24 h, zones measured.
- Planning answers credit: sterile equipment / aseptic technique; a range of concentrations; discs soaked in extract; a bacterial lawn; incubation (at least **24 hours**); the dependent variable as "the **diameter** of the clear zone" (not just "the clear zone").

Often asked as: aseptic technique itself was not asked in WBI12 papers from 2019 to 2026, but zone-of-inhibition data from this practical appear often (see 4.11).

COMMON MISTAKE

"Aseptic technique means wearing gloves." It means preventing **contamination** of the culture and of people.
Sealing the lid all the way round (encourages anaerobes), or incubating at 37 °C.
Forgetting the solvent-only control disc.

SPEC 4.13 How drug testing developed: Withering to modern trials

WHAT THIS SPEC POINT ASKS

Know how drug testing changed from historical methods to today's protocols, including William Withering's digitalis soup, double-blind trials, placebos and the three phases of clinical trials.

4.13 William Withering and digitalis

- In the 1770s–80s, Withering, an English doctor, heard of a traditional herbal remedy (a "soup" of many plant ingredients) that helped patients with **dropsy** (swelling caused by fluid build-up, often from heart failure).
- He worked out that **foxglove** (*Digitalis purpurea*) was the **active ingredient**.
- Over about ten years he gave foxglove preparations (e.g. dried, powdered leaves) to more than 150 patients, **changing the dose** and **recording the results**. Too little did nothing; too much was toxic (sickness, slowed or irregular heartbeat); the right dose strengthened the heartbeat and removed the fluid.
- Today the drug from foxglove is **digoxin**, still used for some heart conditions.

4.13 What was missing from Withering's method

- No **pre-clinical** tests (cells, tissues, animals) before treating people.
- Tested straight on **ill patients**, not healthy volunteers first.
- No **control group** or **placebo**; not **randomised** or **blind**.
- The amount of active substance in each leaf varied, so doses were not standardised.
- Small numbers, no statistics, no independent checking, no formal consent.

4.13 Modern drug testing

1. **Pre-clinical testing**: computer modelling, then tests on human **cells and tissues** (in vitro), then on **animals**, to see whether the drug works, how toxic it is, and how it is absorbed, broken down and excreted.
2. **Clinical trials** in three phases:

Phase	Who	How many	What it tests
1	**healthy volunteers**	small group (tens)	**safety**, **side effects**, how the body handles the drug; a range of **doses** starting low
2	**patients with the disease**	a few hundred (about 100–500)	whether it **works**; the best **dose**; more on side effects; often against a placebo
3	**patients with the disease**	thousands (about 1000–3000)	whether it is **more effective than the existing treatment** (or a placebo); how common side effects are; usually **randomised, double-blind**

3. Data are reviewed independently between phases. If the results are good, the drug is **licensed**, and its use is still monitored afterwards.

4.13 Placebos and blinding

- A **placebo** looks identical to the drug (same colour, shape, taste) but has **no active ingredient**. It is given to the **control group** so that any improvement caused by the drug can be separated from the **placebo effect** (patients improving because they believe they are being treated).

- When an effective treatment already exists, the control group gets the **existing drug** instead, because withholding treatment would be unethical (e.g. for burns or cancer).
- **Single-blind**: only the patients don't know which treatment they get.
- **Double-blind**: **neither the patients nor the doctors/scientists** know who gets the drug and who gets the placebo (or existing drug) until the end. This removes **bias** from both sides: doctors can't treat or assess patients differently, and patients' expectations can't affect the results.
- Patients are **randomly** assigned to groups, and groups are matched for age, sex, health and severity of disease.

EXAM TIP

Answer in context. If the question names a disease or a drug (plague, burns, headaches), use it in every phase: "Phase 2: a few hundred **patients with burns**...". Generic trial essays are capped at Level 1–2.

MARK-SCHEME LANGUAGE

- Phases: pre-clinical on {human cells / animals}; Phase 1 on **healthy volunteers** (side effects, safety); Phase 2 on a **small group (100–500) of patients** (dose, and whether it works); Phase 3 on a **large group (1000–3000) of patients**, compared with a placebo or the current drug; double-blind; a statistical test; review between phases.
- Phase roles: "to test for side effects", "to find the {safe / minimum / effective} dose", "to see if it is more effective than {existing treatments / a placebo}". "Check for safety" or "see if it works" on their own are ignored. Side effects are **not** credited for cell or animal tests.
- Double-blind: "neither the {patient} nor the {doctor / scientist} knows whether they are receiving the {drug / placebo / existing treatment}".
- Finding a dose: "give a range of doses / increase the dose", "the lowest effective dose / highest dose without serious side effects".
- Withering: the active ingredient was **digitalis** (digoxin accepted). Modern trials know the active ingredient and its dose, test first on healthy volunteers, start with the lowest dose, and use three phases.
- For a burns treatment, a placebo group is ignored: compare with the **current hospital treatment**.

Often asked as: describe a full trial for a named drug (6-mark); roles of phases 1 and 2; compare phases 2 and 3; explain double-blind; compare Withering's method with modern testing; evaluate trial data (sample size, missing data, no information on age or sex, subjective measures such as pain).

COMMON MISTAKE

Describing the three phases when the question asks **how** a double-blind trial works, or describing a **single**-blind trial (WBI12 Oct 2020, Oct 2021 reports).
Wrong group sizes, or putting patients in Phase 1.
Writing "safe" or "patients" unqualified; calling an existing treatment a "placebo".
Saying Withering tested garlic or another plant. He used **foxglove**.

BEYOND THE SPEC, BUT EXAMINED

- Patient numbers per phase (about 100–500 in Phase 2; 1000–3000 in Phase 3) are credited (WBI12 Jun 2019).
- In a trial, the **current drug** is the control in a later stage, not the placebo (WBI12 Oct 2023).
- A plague-treatment 6-mark reached Level 3 only with the disease context, three phases, double-blind, a placebo **and** either a significance test or an in-vitro test on the bacterium (WBI12 Jun 2019).
- Spotting missing data (no results for one drug at some doses) as an evaluation point (WBI12 Jan 2023).

SPEC **4.14**

Classification and the three domains

WHAT THIS SPEC POINT ASKS

(i) Know that classification organises the variety of life by how closely organisms are related, using similarities and differences in phenotype and genotype, built around the idea of a species. (ii) Explain how the scientific community critically evaluates new (molecular) data before accepting new groupings, such as the three domains Archaea, Bacteria and Eukarya.

4.14 Why and how we classify

- **Classification** puts organisms into groups based on how closely they are **related**, so the huge variety of life is easier to study and name. The modern aim is to show **evolutionary relationships** (phylogeny).
- Groups are based on similarities and differences in:
  - **phenotype**: observable features (anatomy, cell structure, biochemistry, behaviour);
  - **genotype**: **DNA base sequences**, rRNA sequences and the **amino acid sequences** of proteins.
- The groups form a **hierarchy**: each group (taxon) sits inside a bigger one, with no overlap:
  domain → kingdom → phylum → class → order → family → genus → species
  (e.g. "**D**on't **K**eep **P**lates **C**lean **O**r **F**amily **G**ets **S**ick").
- Each species has a **binomial** name: *Genus species*, e.g. *Homo sapiens*. The genus has a capital letter, the species name doesn't; print in *italics* (underline when handwritten). Binomials are the same in every language, so scientists everywhere know exactly which organism is meant.

4.14 The species concept

- A **species** is a group of organisms with similar characteristics that can **interbreed** to produce **fertile offspring**.
- Horses and donkeys can mate, but their offspring (mules) are **infertile**, so horses and donkeys are separate species.
- Problems with the definition: it can't be tested for organisms that reproduce **asexually**, for **fossils**, or for populations that never meet; some different species occasionally produce fertile hybrids. So scientists also use phenotype and DNA comparisons.
- **Subspecies**: populations that look different and live apart but **can still interbreed** to give fertile offspring.

4.14 Molecular evidence

- All organisms share some molecules, e.g. **rRNA**, cytochrome c. Compare their sequences:
  - **More similar** sequences (fewer differences in bases or amino acids) → **more closely related** → **shared a more recent common ancestor**.

- Molecular data are more reliable than appearance alone, because unrelated organisms can evolve similar features in similar environments (e.g. sharks and dolphins; the giant panda's "thumb").
- Results are shown as **phylogenetic trees**: species on branches that split closest to the tips share the most recent common ancestor.

4.14 The three domains

In the 1970s **Carl Woese** compared **rRNA** sequences of many prokaryotes and found two very different groups. Prokaryotes had been one kingdom, but the archaea differ from bacteria about as much as either differs from eukaryotes. A new top level, the **domain**, was introduced:

	Bacteria	**Archaea**	**Eukarya**
Nucleus and membrane-bound organelles	no	no	**yes**
Cell wall	contains **peptidoglycan**	**no peptidoglycan** (other polymers)	cellulose (plants), chitin (fungi), none (animals)
Membrane lipids	ester-linked, unbranched fatty acids	**ether-linked, branched** chains	ester-linked, unbranched fatty acids
DNA	circular, no histones	circular, **with histone-like proteins**	linear chromosomes with histones
Ribosomes	70S	70S (but proteins and rRNA more like eukaryotes)	80S in cytoplasm
RNA polymerase	one simple type	several, **similar to eukaryotes**	several, complex
Examples	*E. coli*, *Staphylococcus*	methanogens, extremophiles (hot springs, salt lakes)	plants, animals, fungi, protoctists

Archaea share more molecular features with **eukaryotes** than with bacteria, so Archaea and Eukarya share a more recent common ancestor.

4.14 How the scientific community evaluates new data

1. A scientist publishes the findings (data, methods, conclusions) in a **scientific journal** or presents them at a **conference**.
2. Other experts **peer review** the work before publication, checking that the methods are valid and the conclusions follow from the data.
3. Other scientists **repeat the experiments** or collect **new evidence** (e.g. sequence more genes or more species).
4. If the new evidence supports the idea, it is **accepted** by the community and the classification changes. If not, it is modified or rejected.

Woese's proposal was resisted at first. It was accepted only after other scientists found further evidence (membrane lipids, RNA polymerase, more sequences) that supported it. Classification keeps changing as new evidence arrives.

MARK-SCHEME LANGUAGE

- Species: "(a group of) organisms that are capable of {interbreeding} **and** producing **fertile** offspring". "Inbreeding" is not accepted; similar appearance alone isn't credited.
- Evidence for the three domains: "(evidence from) molecular phylogeny"; "{similarities / differences} in {DNA / RNA / rRNA / proteins / ribosomes / membrane components / cell walls}". "Looking at / analysing DNA" with no detail is ignored.
- Most closely related: "share the **most recent** common ancestor", or "{most similarities / fewest differences} in named molecules (DNA base sequence, amino acid sequence)".
- Evaluation by the community: "**peer review**", "repetition of experiments by other scientists", "{analysis / evaluation} of the data", "published in a scientific journal".
- Deciding that an organism is a new species or type: analyse its **phenotype**, analyse **sequences** of named molecules, and show differences from known species.
- Subspecies, not species: "individuals can still breed together to produce fertile offspring".

Often asked as: name the domain, genus or species; describe the evidence for three domains; read a phylogenetic tree; 6-mark on reclassification by molecular evidence and peer review; how to decide whether populations are different species or subspecies; classify cells from a table of features.

COMMON MISTAKE

Describing the three domains when the question asks for the **evidence** for them (WBI12 Oct 2019 report).
Writing a kingdom ("Animalia") or a group like "birds" or "prokaryotes" when asked for a **domain** (WBI12 Jan 2020 report).
"Share a common ancestor" without "**most recent**" (WBI12 Jan 2020 report).
"DNA analysis" with no detail. Say what is compared: "the **base sequences** of the DNA" (WBI12 Oct 2024, Jan 2025 reports).
Peer review: most say "repeat the experiment" but few say what the community does with the data (evaluates, then accepts or rejects) (WBI12 Oct 2021 report).
Mixing up the old kingdom name "Archaebacteria" with the domain **Archaea**.

BEYOND THE SPEC, BUT EXAMINED

- **Carl Woese** by name and rRNA as his evidence (WBI12 Oct 2019, Oct 2021).
- DNA evidence moved the **giant panda** into the bear family and the red panda away from it: similar adaptations (the extra "thumb") can mislead (WBI12 Jan 2026).
- A 6-mark on reclassifying birds: old groupings used phenotype (beak shape, size, diet); the new tree used molecular evidence (fewer differences = more recent common ancestor); and peer review. Level 3 needed **both** the molecular evidence and the review process (WBI12 Jan 2020).
- **Gel electrophoresis** banding patterns as molecular evidence (WBI12 Oct 2022).

SPEC 4.15

The variety of life and the threat from humans

WHAT THIS SPEC POINT ASKS

Know that the variety of life has become very large over time but is now threatened by human activity.

4.15 How the variety built up

Over about 3.5 billion years, **mutation, natural selection and speciation** (4.20) have produced millions of species, adapted to almost every habitat and niche on Earth. Mass extinctions have happened before, but species are now being lost far faster than the natural background rate, mainly because of **human activity** as the human population and its demand for food, land and resources grows.

4.15 Human threats and how each one reduces biodiversity

Threat	How it reduces biodiversity
Habitat destruction: deforestation, draining wetlands, building, mining	species lose food, shelter and breeding sites; **endemic** species can be lost for ever
Habitat fragmentation (roads, farms splitting forests)	small isolated populations: less **gene flow**, **inbreeding**, loss of genetic diversity, local extinction
Agriculture: monocultures, hedgerow removal	one crop replaces many plant species, so fewer food sources and habitats for animals
Herbicides and pesticides	kill non-target plants and insects (e.g. pollinators) and the animals that depend on them
Pollution: fertiliser run-off, sewage, oil, plastics	**eutrophication** (algal blooms, low oxygen in water), poisoning, changes in salinity or pH
Over-exploitation: overfishing, hunting, poaching, logging	populations fall faster than they can reproduce
Climate change (from burning fossil fuels)	species can't move or adapt fast enough; ranges shift; **coral bleaching** and **ocean acidification**; can upset sex ratios in species whose sex depends on temperature
Invasive species and introduced predators or diseases	outcompete or eat native species that have no defences

EXAM TIP

Always give **cause → effect**: "**deforestation** for farming **destroys the habitat**, so the tapir loses food and shelter, so fewer survive to breed". "Habitat loss" without the human cause, or "deforestation" without the effect, scores nothing.

WORKED EXAMPLE: PERCENTAGE CHANGE IN A POPULATION

A population falls from 7400 to 1850.
% change = (new − original) ÷ **original** × 100 = (1850 − 7400) ÷ 7400 × 100 = **−75%** (a 75% decrease).
Divide by the **original** value, not the new one.

MARK-SCHEME LANGUAGE

- Causes need a reason: "deforestation causing habitat loss"; "less food because of {deforestation / competition / overfishing}"; "{pollution / change in salinity} makes the habitat unsuitable"; "{disease / hunting}"; "a new predator". A cause with no effect scores nothing.
- Herbicide data: "as concentration increases the mean number of insect species decreases", with a figure; "fewer plant species → less food for insects → they leave or die". "Herbicides kill insects" is ignored.
- Biodiversity questions: biodiversity is **not** the same as population size.

Often asked as: explain why a named species declined (crocodile, frog, tapir, baobab, plover, tiger, snow leopard); effect of a herbicide on insect diversity; reintroducing beavers or wolves (6-mark); an invasive plant on an island; percentage change in population.

COMMON MISTAKE

Dividing by the final value in percentage change (WBI12 Jun 2023, Jan 2024 reports).
"Herbicides kill insects" (WBI12 Jun 2019 report). Herbicides kill plants; the insects lose their food and habitat.
Confusing biodiversity with the size of one population (WBI12 Jun 2022 report).

BEYOND THE SPEC, BUT EXAMINED

- **Temperature-dependent sex determination**: in crocodiles nest temperature sets the sex of the young, so global warming can skew the sex ratio (WBI12 Jan 2020).
- Introduced predators (cats, rats) eat eggs and chicks of ground-nesting birds (WBI12 Jan 2025).
- Losing a pollinator (e.g. bats) reduces seed production of the plants it pollinates (WBI12 Oct 2024).
- A keystone predator (wolves) changes a whole ecosystem: fewer elk → more young trees → more beavers and ponds → more species (WBI12 Jun 2022).

SPEC 4.16

Biodiversity and endemism

WHAT THIS SPEC POINT ASKS

Know what biodiversity and endemism mean.

4.16 Biodiversity

Biodiversity is the **variety of living organisms** in an area. It can be measured at three levels:

- **Habitat (ecosystem) diversity**: the range of different habitats in an area (a coral reef has many; a sandy desert few).
- **Species diversity**: the number of different species (**species richness**) **and** how evenly the individuals are spread between them (**species evenness**).
- **Genetic diversity**: the variety of **alleles** within a species or population.

Why high biodiversity matters: a diverse community is more **stable**. If a disease or a change in conditions hits one species, it affects only a small part of the community. A species with high genetic diversity is more likely to include individuals with alleles that let them survive a new disease or a changing climate.

4.16 Endemism

- A species is **endemic** if it is found **naturally in only one geographical area** and nowhere else in the world, e.g. lemurs in **Madagascar**, many species in the Galápagos Islands, the Komodo dragon on a few Indonesian islands.
- Endemism is common on **islands** and in isolated habitats (mountain tops, lakes), because populations there evolved **in isolation**.
- Endemic species are at high risk of **extinction**: if their one area is damaged, there is no population elsewhere.
- Areas with many endemic species and high biodiversity under threat are called **biodiversity hotspots**, and are priorities for conservation.

MARK-SCHEME LANGUAGE

- Endemic: "found **only** in {one area / one geographical location}", naming the place if the question gives one (e.g. "only in Madagascar"). "Only in one **habitat**" is ignored.
- Species richness: "the **number of (different) species** in a {habitat / area / community}". "Number of organisms in a species" is not accepted; "variety of species" is ignored.
- Genetic diversity v species richness: "genetic diversity is **within one species** whereas species richness counts **different** species"; "genetic diversity is about {alleles / genotypes}". "Genes" is ignored.
- Measuring biodiversity: count the species (richness), count individuals of each species, calculate the **index of diversity**; for genetic diversity use the **heterozygosity index**; compare with another habitat or an earlier time.

Often asked as: define endemic (very common, often 1 mark); define species richness; describe how to measure or compare biodiversity (3 marks); difference between genetic diversity and species richness.

COMMON MISTAKE

Defining species richness as "the number of individuals of a species" (that is abundance) (WBI12 Oct 2022, Jan 2024 reports).
Writing "genes" instead of **alleles** for genetic diversity.
Writing "biodiversity index" with no named index and no comparison (WBI12 Oct 2024 report).

BEYOND THE SPEC, BUT EXAMINED

- **Species evenness** (how many individuals each species has) as the second part of species diversity (WBI12 Oct 2024).
- To measure species richness you need the **number of species** and the **size of the area** sampled; a whole national park is not "the habitat" (WBI12 Jun 2022).

SPEC 4.17 Measuring biodiversity: species richness and the heterozygosity index

WHAT THIS SPEC POINT ASKS

Know how to measure biodiversity within a habitat (species richness) and within a species (genetic diversity, using the heterozygosity index: number of heterozygotes ÷ number of individuals in the population).

4.17 Species richness in a habitat

- **Species richness** = the number of different species present. Counting every organism is rarely possible, so **sample**:

  - Plants and slow-moving animals: **quadrats** placed at **random coordinates** (from a random number generator on a grid) so there is no bias; record every species.
  - Along an environmental gradient (e.g. from shore inland): a **transect**, with quadrats at regular intervals.
  - Moving animals: sweep nets, pitfall traps, kick sampling in streams, light traps.
- Take **many samples**, until new samples add no new species, and use the **same method and sampling effort** in each habitat you compare.
- Weakness: richness ignores **how many** of each species there are. Two habitats with 10 species each can be very different if one is dominated by a single species. The **index of diversity** (4.18) fixes this.

4.17 Genetic diversity within a species: the heterozygosity index

- An individual that has **two different alleles** at a gene locus is **heterozygous**. A population with many heterozygotes has many alleles, so more genetic diversity.

$$\text{heterozygosity index (H)} = \text{number of heterozygotes} \div \text{number of individuals in the population}$$

- H runs from **0** (no heterozygotes) to **1** (all heterozygous). It has no units and is **not** multiplied by 100.
- To find it: take DNA samples (e.g. hair, blood, feathers), use **DNA analysis** (sequencing or genotyping) to identify each individual's alleles at a chosen locus, count the heterozygotes, and divide by the total number sampled. Averaging H over many loci gives a picture of the whole genome.
- **Low H** suggests a small, isolated or inbred population (e.g. after a bottleneck or in a small captive group). It has fewer alleles, so it is less able to adapt to new diseases or changing conditions, and more likely to show harmful recessive conditions.

WORKED EXAMPLE: HETEROZYGOSITY INDEX

500 frogs are genotyped at one locus: 118 are heterozygous.
H = 118 ÷ 500 = **0.236**.
Rearranged: a population of 650 has H = 0.42. Number of heterozygotes = 0.42 × 650 = **273**.
A number of organisms must be a whole number. Don't round it away to a decimal that implies part of an animal.

MARK-SCHEME LANGUAGE

- H = "number of heterozygotes ÷ number of individuals in the population". Do **not** multiply by 100.
- How to find it: "analyse the DNA (of the frogs) / identify the number of heterozygotes"; "find the population size"; "divide the number of heterozygotes by the total".
- Genetic diversity: "number of different alleles present" or "heterozygosity index". If you give several methods, only the first is marked. "Allele frequency" and "genes" are ignored.
- Rearranging: answers such as 0.166 × 500 = 83 heterozygotes.
- Keeping alleles in a captive population: find individuals that carry **rare alleles** and use them in breeding with individuals carrying different alleles.

Often asked as: state the heterozygosity index equation; rearrange it to find the number of heterozygotes (very common); describe how to find H for a population; use H in a conservation decision.

COMMON MISTAKE

Writing the index of diversity, Hardy–Weinberg or mitotic index equation instead (WBI12 Jan 2021, Jan 2022 reports).
Dividing heterozygotes by **homozygotes** instead of by the total.
Using H when a Hardy–Weinberg calculation was needed (WBI12 Jun 2024 report).
In a context question (e.g. organisms living in a sloth's fur), describing quadrats and pitfall traps instead of sampling the stated habitat.

BEYOND THE SPEC, BUT EXAMINED

The answer for a number of heterozygotes must be a whole number: e.g. 30 or 30.0 accepted, 30.012 rejected (WBI12 Oct 2024).

SPEC 4.18

Index of diversity (D)

WHAT THIS SPEC POINT ASKS

Use the index of diversity, D = N(N − 1) ÷ Σn(n − 1), to compare biodiversity in different habitats.

4.18 The formula

$$D = N(N - 1) \div \Sigma n(n - 1)$$

- **N** = **total number of individuals** of all species in the sample.
- **n** = number of individuals of **each** species.
- **Σ** = "add up" (do n(n − 1) for every species, then add).
- A **higher D** means **higher biodiversity**. D takes into account **both** the number of species (richness) **and** how evenly individuals are spread between them (evenness).
- The smallest possible value is **1** (only one species). D has no upper limit of 1; with more species spread more evenly it gets bigger.
- A species with only one individual contributes 1 × 0 = 0 to the bottom line.

4.18 Step by step

1. Add up all the individuals: N.
2. Work out N(N − 1).
3. For each species work out n(n − 1). Add a column to the table.
4. Add the column: Σn(n − 1).
5. Divide step 2 by step 4. Round only at the end, to the number of decimal places asked.
6. **Compare** D values and say what they mean.

Sample counts from two fields (same quadrat area and method):

Species	Field 1: n	n(n − 1)	Field 2: n	n(n − 1)
clover	12	132	24	552
buttercup	8	56	3	6
daisy	6	30	2	2
plantain	3	6	1	0
dandelion	1	0	0	0
Total	**N = 30**	**Σ = 224**	**N = 30**	**Σ = 560**

WORKED EXAMPLE: COMPARING TWO FIELDS

Field 1: N(N − 1) = 30 × 29 = 870. D = 870 ÷ 224 = 3.8839 → **3.88** (2 d.p.).
Field 2: N(N − 1) = 870. D = 870 ÷ 560 = 1.5536 → **1.55** (2 d.p.).
Conclusion: Field 1 has the **higher biodiversity** (3.88 > 1.55). It has more species, and its individuals are spread more evenly; Field 2 is dominated by clover.

EXAM TIP

Watch the rounding instruction: **2 d.p.** and **2 s.f.** are different. 1.5536 is 1.55 to 2 d.p. but 1.6 to 2 s.f. Keep the full values of N(N − 1) and Σn(n − 1) and round once, after dividing.
A conclusion needs the comparison **and** the values: "Habitat A has higher biodiversity **because** its D is higher (3.88 compared with 1.55)".

4.18 What a change in D tells you

- D falls after pollution, an invasive species or habitat damage: some species are lost, or one species becomes dominant. Say **why**, e.g. "the invasive weed blocks light, so other plants die, reducing food and habitat for animals".
- Only compare D values from samples collected the **same way** (same area, same method, same time of year), because D depends on how you sample.

MARK-SCHEME LANGUAGE

- Marks: N(N − 1) correct; Σn(n − 1) correct; D to the stated d.p. (ECF from your working).
- Fill in the n(n − 1) column before adding. If a table gives n(n − 1) but not n, work out n.
- Include **every** species in the table: leaving one out lost a mark (WBI12 Oct 2022).
- Conclusion: "habitat 1 has higher biodiversity / a higher index of diversity" **with** the values or a reason. A reason without a conclusion scores 0.

Often asked as: calculate D from a partly completed table (a staple from Jun 2021 onwards); decide which habitat has higher biodiversity and agree or disagree with a statement; interpret a fall in D after an invasive weed or sewage.

COMMON MISTAKE

Using the **number of species** for N (e.g. 7 × 6) instead of the total number of individuals (WBI12 Jun 2024 report).
Wrong rounding or ignoring the d.p. asked (WBI12 Oct 2022, Jan 2025 reports).
Writing the heterozygosity or Hardy–Weinberg equation when asked to name the formula (WBI12 Jun 2022 report).
Stopping after "D decreased" when an explanation is asked (WBI12 Jan 2024 report).

BEYOND THE SPEC, BUT EXAMINED

Sewage pollution: D falls as sensitive species (e.g. mayfly nymphs) disappear and tolerant ones (e.g. bloodworms, sludgeworms) become dominant (WBI12 Oct 2025).

SPEC 4.19 Niche and adaptation

WHAT THIS SPEC POINT ASKS

Know what a niche is, and discuss examples of behavioural, anatomical and physiological adaptations of organisms to their environment.

4.19 Habitat and niche

- **Habitat**: the place where an organism lives (its address).
- **Niche**: the **role** of an organism in its habitat (its job): what it eats, when and where it feeds, what eats it, how it interacts with other species (**biotic** factors) and the range of physical conditions it tolerates, such as temperature, light, pH, water (**abiotic** factors).
- **Two species cannot occupy exactly the same niche in the same habitat** for long: one will outcompete the other (competitive exclusion). Species that share a habitat avoid this by having **different niches**, e.g. birds feeding at different heights in a tree, or wading birds with different beak lengths taking prey from different depths of mud.

4.19 Adaptations

An **adaptation** is an inherited feature that makes an organism better suited to its environment, so it is more likely to **survive and reproduce**. There are three types:

Type	Meaning	Examples
Anatomical (structural)	a **body structure**, inside or out	thick fur or **blubber** (insulation); small ears and a compact body (small SA : V, less heat loss); streamlined shape and webbed feet (swimming); long claws (digging); **camouflage**, **warning colours** and **mimicry**; rolled leaves, hairs and sunken stomata in marram grass (less water loss); spines instead of leaves in cacti
Physiological	how **processes inside the body** work	producing **venom** or **poison**; antifreeze proteins in polar fish; producing very **concentrated urine** (desert mammals); tolerating salt; enzymes that work at high temperatures (thermophiles)
Behavioural	what the organism **does**	**migration**; hibernation; courtship displays; huddling (penguins); basking in the sun (lizards); being active at night in deserts; caring for eggs or young (e.g. carrying eggs in the mouth)

EXAM TIP

For "explain how X is adapted", always link **feature** → **function** → **survival**: "large paws spread the weight **so** the snow leopard doesn't sink into snow, **so** it can chase prey". "It has fur" or "it helps it survive" alone scores nothing. Use details in the photo or the stem.

MARK-SCHEME LANGUAGE

- Niche: "the **role** of an organism in its {environment / habitat / ecosystem}"; both biotic and abiotic parts accepted.
- Classifying: **bright colours / warning colouration = anatomical**; **poison = physiological**; **mimicry = anatomical** (physiological also accepted once); **migration = behavioural**; an extra "thumb" = anatomical; a display = behavioural.
- Typical linked points: "webbed feet to paddle through water"; "long wings to fly long distances / glide"; "thick fur to insulate"; "streamlined shape reduces resistance when swimming"; "blubber for insulation"; "tendrils secure the shoot so it can climb higher, reach more light, photosynthesise more"; "large hoof surface area stops it sinking in sand". "Keep warm" alone is ignored; "camouflage" for bright warning colours is wrong.

Often asked as: define niche; classify given adaptations as anatomical, physiological or behavioural; explain how an animal in a photo (penguin, snow leopard, wombat, mimic octopus) is adapted; suggest the advantage of a feature (tendrils, mouthbrooding); two species sharing a habitat.

COMMON MISTAKE

Calling poison "behavioural" or writing "psychological" for physiological (WBI12 Oct 2019, Jan 2022 reports).
Calling bright colours "camouflage" (WBI12 Oct 2019 report).
Vague "it has wings" with no function, or fewer points than the question asks (WBI12 Oct 2020, Jun 2024 reports).
Confusing **niche** (role) with **habitat** (place).
Lamarckian wording: "the animal grew thick fur because it was cold". Individuals don't change to adapt; those that already had the useful allele survived (4.20).

BEYOND THE SPEC, BUT EXAMINED

- Warning colouration and mimicry are classed as **anatomical**, not behavioural (WBI12 Oct 2019, Oct 2023).
- **Mouthbrooding** (a fish carrying fertilised eggs in its mouth) increases survival of the eggs by reducing predation (WBI12 Jun 2023).
- Beavers and elk compete because they have **similar niches** (both eat young trees) (WBI12 Jun 2022).

SPEC 4.20 Hardy–Weinberg, natural selection and speciation

WHAT THIS SPEC POINT ASKS

(i) Use the Hardy–Weinberg equation to see whether allele frequencies in a population are changing over time. (ii) Know that allele frequencies change through mutation and natural selection. (iii) Explain how reproductive isolation lets populations build up different genetic information, which can lead to new species.

4.20 The Hardy–Weinberg principle

In a population the **allele frequencies** stay the same from one generation to the next (the population is in equilibrium) **as long as**:

- the population is **very large**;
- mating is **random**;

- there is **no mutation**;
- there is **no migration** in or out (no gene flow);
- there is **no natural selection** (all genotypes survive and reproduce equally well).

If measured allele frequencies **change** between generations, one or more of these conditions is not being met: **evolution is happening**.

4.20 The equations (both are given in the exam)

For a gene with two alleles, **A** (dominant) and **a** (recessive):

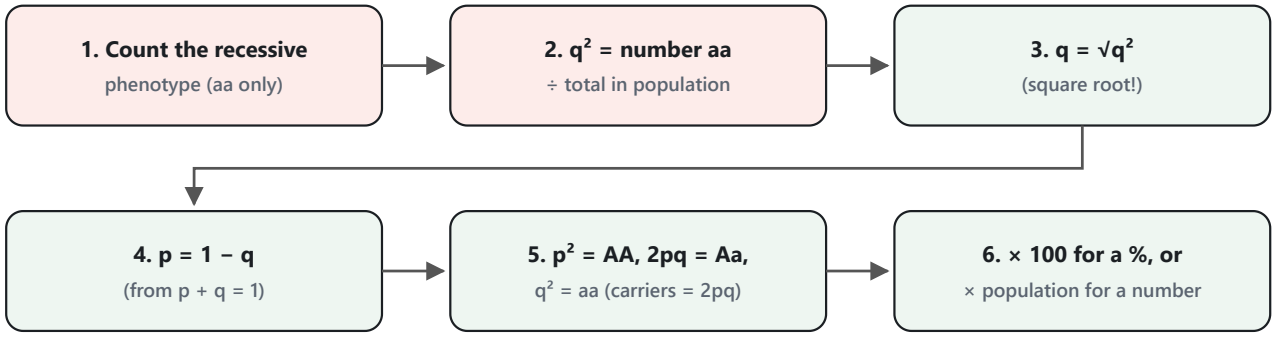
$$p + q = 1$$

$$p^2 + 2pq + q^2 = 1$$

- **p** = frequency of allele A; **q** = frequency of allele a.
- **p^2** = frequency of genotype **AA**; **2pq** = **Aa** (heterozygotes, **carriers**); **q^2** = **aa**.
- Frequencies are decimals between 0 and 1. Multiply by 100 for a percentage, or by the population size for a number of individuals.

4.20 Always start from q^2

The only genotype you can **see** is the homozygous recessive one: an individual with the recessive phenotype must be **aa**. Individuals with the dominant phenotype could be AA or Aa, so you can't count p^2 directly.



The route through a Hardy–Weinberg calculation. Steps 1 and 2 are where most marks are lost.

WORKED EXAMPLE: CARRIER FREQUENCY OF A RECESSIVE CONDITION

A recessive genetic disorder affects 1 in 2500 people. What proportion are carriers?
$q^2 = 1 \div 2500 =$ **0.0004**
$q = \sqrt{0.0004} =$ **0.02**
$p = 1 - 0.02 =$ **0.98**
Carriers = $2pq = 2 \times 0.98 \times 0.02 =$ **0.0392** (3.92%, about 1 person in 26).
In a town of 50 000 people: 0.0392 × 50 000 = **1960 carriers**.

WORKED EXAMPLE: THE DOMINANT-PHENOTYPE TRAP

In 1000 beetles, 640 are black (dominant, B_) and 360 are brown (recessive, bb).
Wrong: $p^2 = 640 \div 1000 = 0.64$. The black beetles include **both** BB and Bb.
Right: $q^2 = 360 \div 1000 = 0.36 \rightarrow q = 0.6 \rightarrow p = 0.4$.
BB = $p^2 = 0.16$ (160 beetles); Bb = $2pq = 2 \times 0.4 \times 0.6 = 0.48$ (480 beetles); bb = 0.36 (360).
Check: 160 + 480 = 640 black beetles. ✓

WORKED EXAMPLE: IS THE ALLELE FREQUENCY CHANGING?

A snail population has yellow shells (recessive, yy). In year 1, 84 of 600 snails are yellow; in year 10, 54 of 600 are.
Year 1: $q^2 = 84 \div 600 = 0.14 \rightarrow q = \sqrt{0.14} =$ **0.37**.
Year 10: $q^2 = 54 \div 600 = 0.09 \rightarrow q = \sqrt{0.09} =$ **0.30**.
The frequency of the yellow allele has **fallen** (0.37 → 0.30), so the population is not in Hardy–Weinberg equilibrium: the allele frequency is changing, e.g. because thrushes see and eat more yellow snails (natural selection).

4.20 How allele frequencies change

- **Mutation** is a random change in DNA that creates **new alleles**. It is the original source of all genetic variation.
- **Natural selection**, step by step:
  1. There is **variation** in the population, caused by **mutation** (and the reshuffling of alleles in meiosis).
  2. A **selection pressure** acts (a predator, disease, climate, competition for food).
  3. Individuals with an **advantageous allele** are more likely to **survive and reproduce**.
  4. They **pass the allele on** to their offspring.
  5. Over many generations the **frequency of that allele increases** in the population (and the disadvantageous allele becomes rarer).
- Example: when a virus spreads through a rabbit population, rabbits with an allele that gives resistance survive and breed, so the resistance allele becomes more common in later generations.

4.20 Reproductive isolation and speciation

Speciation is the formation of new species. It happens when two populations can no longer interbreed to produce fertile offspring.

1. A population is split into two groups that **can't interbreed**: **reproductive isolation**, so **no gene flow** between them.
2. Different **mutations** arise in each group.
3. Each group faces **different selection pressures** (food, climate, predators), so **different alleles** are selected for. Chance (genetic drift) also changes allele frequencies, especially in small populations.
4. Over many generations the two gene pools **accumulate different genetic information**: allele frequencies, then phenotypes, diverge.
5. Eventually, even if they meet again, members of the two groups **can't interbreed to produce fertile offspring**: they are **separate species**.

Ways populations become reproductively isolated:

Type	Cause	Examples
Geographical → allopatric speciation	a physical barrier separates them: sea, river, mountain range, a new island, separate lakes	finches or honeycreepers on different islands; fish in separate crater lakes
Seasonal / temporal	they breed or flower at different times	plants flowering at different times of year
Behavioural	different courtship songs or displays, so they don't recognise each other as mates	birds with different songs
Mechanical	reproductive organs or flowers no longer fit	changes in flower shape and pollinators
Gametic	sperm and egg can't fuse (e.g. egg-binding proteins no longer complementary)	sea urchins releasing gametes into the sea

Isolation **within the same area** (seasonal, behavioural, mechanical, gametic) can lead to **sympatric** speciation.

MARK-SCHEME LANGUAGE

- Hardy–Weinberg route: recessive phenotype → **q^2** as a decimal (e.g. 16 ÷ 100 = 0.16) → $q = \sqrt{q^2}$ → $p = 1 - q$ → p^2, $2pq$, q^2. "q^2 correctly calculated" is a separate mark, and there is no ECF from a wrong q^2 step. Give % or numbers of individuals **as asked**.
- Showing the frequency is changing: "count the number of (recessive phenotype) individuals / DNA analysis to identify the alleles", "use the Hardy–Weinberg equation", "**compare** the allele frequency with previous generation(s)".
- Selection: "mutation(s) occurred"; "(new) allele for the **named feature**"; "selective advantage / more likely to survive and reproduce"; "pass the allele to offspring"; "allele frequency increases". Only 3 marks without the named feature. "Genes" is ignored.
- Speciation: "geographical isolation, so no gene flow"; "different selection pressures"; "mutation → different alleles"; "advantageous alleles passed on"; "**accumulation of different genetic information**"; "can no longer interbreed to produce fertile offspring". Calling populations "species" before they are is rejected.

Often asked as: Hardy–Weinberg calculations (percentages, numbers, carriers); use the equation to show whether a frequency has changed; explain how selection changed a population (poison frogs, island mice, virus-resistant rabbits); 6-mark speciation in context (honeycreepers, cichlids, snow leopard subspecies, sea urchins).

COMMON MISTAKE

Treating the recessive phenotype frequency as q instead of q^2, i.e. not square-rooting: the biggest mark-loser in this topic (WBI12 Jun 2019, Oct 2022, Oct 2023 reports). Only about a third got full marks in Jun 2019.
Taking the **dominant** phenotype as p^2.
Giving 0.49 when the question asks for a **percentage** (49%), or an impossible answer (negative, or more individuals than the population).
"Genes" instead of **alleles**; "advantageous genes" (many reports).
Saying the environment or selection **causes** the mutation, or that individuals change to adapt (WBI12 Oct 2020, Jan 2025 reports). Mutations are random; selection acts on variation that already exists.
Generic natural-selection essays with no reference to the stem stay at Level 1 (WBI12 Jun 2022, Jun 2023 reports). Forgetting the final step: "**cannot produce fertile offspring**, so a new species".

BEYOND THE SPEC, BUT EXAMINED

- Hardy–Weinberg tables where the numbers must add up to the total population (e.g. 400) (WBI12 Oct 2024).
- **Founder effect** and **inbreeding** accepted in speciation answers; **genetic drift** in small populations (WBI12 Oct 2022).
- Gametic isolation: a change in the egg's binding protein (a ZP3-like protein) means sperm no longer bind, so no fertilisation (WBI12 Jun 2025).
- Gene duplication (extra copies of the amylase gene *AMY1*) selected in populations eating high-starch diets (WBI12 Oct 2025).
- Distances from a map scale bar in standard form (WBI12 Jun 2023).

SPEC 4.21

Zoos and seed banks in conservation

WHAT THIS SPEC POINT ASKS

Evaluate how zoos and seed banks conserve endangered species and their genetic diversity, through scientific research, captive breeding, reintroduction and education.

4.21 In situ and ex situ

- **In situ** conservation protects species **in their natural habitat** (national parks, nature reserves, marine protected areas, anti-poaching patrols, controlling invasive species). It is the best option where possible, because it protects the whole ecosystem.
- **Ex situ** conservation keeps species **outside their natural habitat**: **zoos**, botanic gardens, **seed banks**, frozen sperm and embryo stores. It is a back-up when the habitat or wild population is too threatened.

4.21 Zoos

- **Captive breeding programmes** increase the number of individuals of endangered species.
- **Keeping genetic diversity** in a small captive population:
  - **Studbooks** record the ancestry of every animal, so keepers can pair individuals that are **not closely related** and avoid **inbreeding**.
  - **Genetic analysis** of the animals' alleles shows which carry rare alleles.
  - Animals, **sperm or embryos** are **exchanged between zoos** worldwide; **artificial insemination** and **IVF** let distant or reluctant animals breed; **frozen sperm** can be used even after a male has died.
- **Reintroduction** of captive-bred animals to the wild (e.g. the Arabian oryx). It works only if the original cause of decline (habitat loss, hunting) has been removed and the habitat can support them.
- **Scientific research**: animal behaviour, nutrition, reproduction and diseases, which helps both captive and wild populations.
- **Education**: visitors learn about threats to species, which can change behaviour, encourage support for conservation laws and raise money.

4.21 Seed banks

- Seeds are **collected** from **many plants** of a species, from **different populations**, so the store contains as many **different alleles** as possible.
- Seeds are cleaned (to remove microbes), checked (e.g. **X-rayed** to see that an embryo is present), **dried** and stored **frozen** (about −20 °C). Drying and freezing slow metabolism and enzyme activity, prevent germination, and stop fungi and bacteria growing, so seeds stay alive for decades.

- Samples are **germinated at intervals** to check viability; plants are grown from old stock to collect **fresh seed**.
- Stored seeds can be used to **reintroduce** plants to the wild, to restore habitats, for research, and as a source of alleles for crop breeding (e.g. disease resistance from wild relatives of crop plants). Examples: the Millennium Seed Bank (UK), the Svalbard Global Seed Vault (Norway).
- Some seeds (**recalcitrant** seeds, e.g. cocoa, avocado, many tropical trees) die when dried or frozen. They are kept as living plants or by tissue culture, or their embryos are frozen in **liquid nitrogen (−196 °C)**.

4.21 Evaluating zoos and seed banks

	Advantages	Disadvantages
Zoos	protect animals from poaching, predators and disease; vet care and food; increase numbers; studbooks and exchanges keep genetic diversity; reintroduction; research; education and funding	small populations → **inbreeding** and loss of alleles; some species won't breed in captivity; expensive; animals may lose natural behaviour (hunting, avoiding predators), so reintroductions can fail; may carry diseases into wild populations; ethical concerns about keeping animals captive
Seed banks	seeds take **little space**, need **less maintenance and cost** than whole plants; huge numbers stored; seeds survive for a long time when dry and frozen; samples from many populations keep genetic diversity; protected from disease, fire and habitat loss; a source of alleles for crop breeding	seeds lose viability, so need regular germination tests and regrowing; recalcitrant seeds can't be stored this way; a sample may not contain all the alleles in the wild population; stored seeds don't evolve with changing conditions or new pests; plants are not conserved in their habitat

EXAM TIP

"Evaluate" needs **both** sides and a **judgement**: e.g. "Captive breeding increases numbers and, with studbooks, keeps genetic diversity, but reintroduction only succeeds if the habitat is protected, so zoos work best alongside in-situ conservation." Always mention **genetic diversity**: it is the concept the mark scheme anchors on.

MARK-SCHEME LANGUAGE

- Zoo roles: genetic analysis, **stud books**, captive breeding, maintaining genetic diversity / avoiding inbreeding, reintroduction (with detail), education, research, in-situ work.
- Genetic diversity in captive breeding: "analyse individuals' alleles", "use a stud book", "breed individuals with different alleles / from different populations / not closely related", "prevent loss of alleles". "Genes" and "prevent hybrids" are ignored.
- Why seeds, not whole plants: "take up **less space**, so more can be stored"; "need **less maintenance / cost**"; "survive **longer** / can be frozen". Two versions of the same point score only once.
- Seed bank process: collect from **many plants** (different alleles); disinfect or sterilise; X-ray to check viability; **dry**; **freeze below 0 °C** to prevent germination and microbial growth; germinate samples regularly and collect new seeds. "Low temperature" with no reason, "cool", or washing alone are ignored.
- Drying seeds: "prevents germination", "slows enzyme / metabolic reactions", "extends storage time", fewer microbes.
- Education: "teach people not to cut down / hunt X", "promote planting", awareness, donations, influencing policy.

Often asked as: 6-mark evaluation of zoos or a conservation plan (very common); describe how seed banks store seeds; why store seeds not plants; breeding a crop with a seed bank variety for disease resistance; using frozen sperm; wildlife corridors.

COMMON MISTAKE

Describing methods without explaining how they help (Level 1 only) (WBI12 Oct 2020, Jan 2021 reports).
Writing only about increasing **numbers** when the question asks about **genetic diversity** (WBI12 Jun 2024 report).
"Low temperature" instead of "**below 0 °C** / frozen"; washing seeds as the only step (WBI12 Jan 2023, Jan 2025 reports).
"In case of extinction" as the reason for storing seeds rather than plants.
Confusing pollen with seeds, or not knowing what a stud book is (WBI12 Jan 2022 report).

BEYOND THE SPEC, BUT EXAMINED

- **Stud books**, artificial insemination, IVF and exchanging animals between zoos (WBI12 Jan 2022).
- **Frozen sperm** 6-mark: advantages (many males, even dead ones; AI; genetic diversity; little space) and disadvantages (fewer sperm swim after thawing, shorter survival, DNA damage, cost); use the correlation coefficient given (WBI12 Oct 2024).
- **Crocodile eggs** incubated at a chosen temperature to get a useful sex ratio, with eggs collected from many nests for genetic diversity (WBI12 Jan 2020).
- **Wild relatives** of crop plants in seed banks as a source of disease-resistance alleles (bananas and Panama disease; eggplant and bacterial wilt) (WBI12 Jun 2019, Jan 2023).
- **Cryopreservation** of embryos at −196 °C for seeds that can't be dried (WBI12 Oct 2025).
- **Wildlife corridors** link isolated populations, allowing gene flow and outbreeding (WBI12 Oct 2025).
- Seed germination falls with storage time and depends on the container (sealed metal can best, cloth bag worst) (WBI12 Oct 2025).